

FRM 核心考点速通笔记

FOCO 备考速通系列

说明

FRM 考试的主要参考题目是 3 套 PE 以及机经回忆，基于这些题目，我们整理了相应的考点和保姆级的习题详解，希望这份笔记能够帮助大家快速掌握 FRM 考试的核心考点。这份笔记特点如下：

- (1) 在教材整体框架下梳理考点
- (2) 覆盖 PE1、PE2 和 PE3 的 300 道题的考点
- (3) 同时也覆盖近两年的考经考点
- (4) 每个习题都有详细的解析，尽可能做到保姆级地讲解
- (5) 高度浓缩，全是重点
- (6) 单栏排版，80g 护眼纸
- (7) 附有考点目录和 PE 习题索引目录，查找方便
- (8) 可以补充相关要点在上面，形成自己的备考笔记

FRM 备考建议：

1. 聚焦高质量的习题，并结合考经和机经进行复习
2. 注重知识点的理解，而不是死记硬背

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Credit Risk Measurement and Management

信用风险计量和管理

1 信用风险基本概念

考点 1 Fundamentals of Credit risk CR-1

1 基本概念

- 信用风险是由于对方 inability、unwillingness，或 nontimeliness 履行财务义务而导致经济损失的可能性。
- Borrower、obligor、counterparty 和 bond issuer 是接收信贷的一方。
- Lender、creditor 和 obligee 是提供信贷的一方。
- Inability、nontimeliness to Pay
 - 企业的现金流不足以偿还为快速扩张的贷款。
 - 产品或服务过时导致收入不足，无法支付运营和融资成本。
 - 意外事件 (如自然灾害、疫情) 导致企业陷入财务困境。
 - 如果款项未能及时偿还，债权人可能会错失利息收入或产生营运资本融资费用。
- Unwillingness to Pay
 - 合同争议导致的商业纠纷 (Commercial disputes) 是最常见的情况。
 - 如果纠纷进入法律诉讼程序，且贷方在诉讼中获胜 (the lender prevails)，那么债务就可以得到偿还。
 - 不典型的情况——主权国家 (Sovereign state) 不履行其国际债务义务，或强行将其外币债务转换为本国货币。

2 Insolvency、Default 和 Bankruptcy 的区别

- Insolvency (资不抵债)：【Negative equity: Liabilities > 资产 Assets】
Insolvency ≠ Bankruptcy 【把“资不抵债”看作是“破产”的同义词是错误的】
Insolvency ≠ default，只需在下一次支付资金前筹集到资金。
 - Default (违约)：【对手方未能履行合同义务：如未付款】
Default 通常是由于资不抵债或流动性不足 (Illiquidity)。
 - Bankruptcy (破产)：违约实体可以通过申请破产寻求法律保护。
 - 法院会审查违约实体的财务状况，与管理层、债权人以及股东进行谈判，通过出售资产和/或与贷款人重新谈判融资安排，试图让违约实体继续经营。
 - 如果业务无法重组，则企业解散。
- 【破产法第 7 章主要涉及解散 dissolution/清算 liquidation；第 11 章主要涉及重组 restructuring/reorganization】

3 产生信用风险的交易类型

信贷类型 Credit Type	损失来源 Losses Result From	损失类别 Loss Type
Loaned money 贷款	• Nonrepayment 不还 • Slow repayment 慢还 • Dispute 争议 • Enforcement 强制执行	• Face amount 本金 • Interest 利息 • Time value of money 货币时间价值 • Frictional costs 摩擦成本
Lease obligation 租赁义务	• Nonpayment 未付款	• Recovery of asset 回收租赁资产 • Remarketing costs 再营销成本 • Difference in conditions 条件变化损失
Receivables 应收账款	• Nonpayment of goods delivered or service performed 客户未支付已交付的商品或服务	• Face amount
Prepayment for goods or services 商品或服务的预付款	• Nondelivery 供方未交付 • Performance not as contracted 交付不符合合同规定 • Slow delivery 交付延迟 • Dispute/enforcement	• Replacement cost 购买替代品的成本 • Incremental cost 额外的运营成本 • Time value of money • Frictional costs
Deposits 存款	• Nonrepayment	• Face amount • Time value of money
Claim or contingent claim on asset 对资产的索赔/或有索赔	• Nonrepayment/Noncollection • Slow repayment collection • Dispute/enforcement	• Face amount • Time value of money • Frictional costs
Derivative 衍生品	• Default of third party 第三方违约	• Replacement cost (mark-to-market value)

4 存在信用风险的实体

• A. Financial Institutions 金融机构

– Banks 银行

- * 因发放贷款拥有庞大的信贷组合。
- * 信贷风险来自贷款和信贷额度（信贷额度或信用额度）、基于资产的借贷（如回购）以及衍生工具。
- * 通过复杂的风险管理实践和资本市场对冲敞口管理信用风险。

– Asset Managers 资产管理公司

- * 例如贝莱德 (Blackrock) 或先锋集团 (Vanguard Group)
- * 代表个人和机构投资大量资金。
- * 信用风险来源于投资，尤其是债务工具。

– Hedge Fund 对冲基金

- * 投资者有更大的风险偏好，但要求高回报以补偿这种风险。
- * 往往参与风险较高的投资和策略，如不良债务和卖空，从而从信用恶化中获利 (generate profits from credit deterioration)。
- * 一方面承担比较高的信用风险，另一方面通过高信用风险获利。

- Insurance Companies 保险公司
 - * 保险公司在承保活动、投资组合和再保险可收款方面面临信用风险。
- Pension Funds 养老金基金
 - * 基金的一大部分投资于信用风险资产。
- B. Corporations 【主要指非金融公司】
 - 信用风险来源
 - * 应收账款 (accounts receivables) 【应收账款相当于给客户提供短期贷款】
 - * 短期投资和银行存款
 - * 衍生品
 - * 供应商融资 (Vendor financing): 如 IBM 全球融资等公司提供供应商融资, 承担与银行类似的信用风险。
 - * 供应链风险 (supply chain): 关键供应商的破产可能会带来风险。
- C. Individuals 个人
 - 家庭信贷风险
 - 投资活动
 - 存款
 - 个人预付款项 (如预付给承包商的款项) 等

5 管理信贷风险的动机

- Survival 生存

金融机构尤其需要关注信用风险管理, 大额损失可能会导致破产; 非金融企业的信用损失也可能引发破产。
- Profitability 盈利能力

减少损失即增加利润——这对于低利润率的企业非常关键。
- Return on equity 股本回报率

企业需要足够的股本资本和谨慎的风险管理, 在债务和股权之间找到合适的平衡点, 以实现股权回报的最大化。

考点 2 Governance CR-2

实践中, 尽管个别交易可能会使得公司面临大量损失, 但通常不会导致公司破产。公司破产往往是积累了大量的有毒交易。这通常源于风险管理框架薄弱 (poor risk management framework) 和公司治理失效 (corporate governance failure)。本考点主要强调在大型公司中, 良好的信用风险管理框架和公司治理的重要性。

1 Three Lines of Defense 三道防线

- 第一道防线: 负责并管理风险的**业务负责人** (own and manage risk)
- 第二道防线: 监督风险, 建立政策和程序, 对第一道防线进行管理监督 (oversight for the first line) 的**风险管理、合规和法律部门** (enterprise risk management, compliance, and legal)
- 第三道防线: 独立 (provides independent assurance) 管理和监控第一和第二道防线的风险的**内部审计、外部审计和特别审计委员会** (Internal and external auditors and audit committees)

【在操作风险和流动性风险中也有三道防线。大原则是一致的, 主要是第一道防线 (业务负责人) 不同。例如, 在信用风险中第一道防线是信贷客户经理。流动性风险中第一道防线是司库部门】

2 最佳治理体系的四个关键原则：

- A. Guidelines 指导方针：【明确的指导方针，管理关于信用风险交易的审批。】
 - 指导方针是解释交易达成前必须遵守的规则的一系列文件，也叫信贷政策（credit policies）或风险管理标准（risk management standards）等。
 - 特征：
 - * Understandable 易于理解：语言清晰简单，易于理解，避免使用过于法律化的语言
 - * Concise 简明扼要
 - * Precise 准确
 - * Accessible 易于获取
 - Creation and Approval Process 创建和审批流程
 - * 必须由高级管理人员（例如 CRO、CFO）发起，并由董事会批准。
 - Promulgating and Maintaining Guidelines 颁布和维护
 - * 首席风险官办公室负责指导方针的起草、批准、推广和维护。
 - * 指导方针需要定期更新和审查，与业务发展同步。
 - Content of Guidelines 指导方针内容
 - * 指导方针的目的
 - * 定义交易关键参数的方法
 - * 交易审批流程和授权委托
 - * 新产品和市场的处理流程
 - * 审查和更新的流程
 - * 不遵守指导方针的后果
 - Breach（违反）of Guidelines
 - * 违反 Guidelines 是一种严重行为，可能导致立即解雇。
 - * 也可以考虑一些特殊情况（如因外汇波动违反信用额度），设置一些特例。
- B. Skills: 【将权限下放至具备合适技能的委员会和个人】
 - 理想情况下，所有交易都应由公司的高级管理人员批准，但实际工作中不太现实。这就涉及到授权委派问题。
 - 风险管理与业务单位的冲突：
 - * 业务单位的优先事项：赚钱（第一），不亏钱（第二），希望最大自由度。
 - * 风险管理的优先事项：不亏钱（第一），赚钱（第二），需要限制自由度。
 - Delegation of Authority 权力下放的步骤
 - * Step 1: 从风险管理的角度，为每笔交易分配基本参数
 - * Step 2: 基于这些参数委派批准权限。
 - Defining risk parameters 定义风险参数
 - * Amount of exposure 敞口：估计公司可能损失的最大金额。
 - * Credit quality 信用质量：必须制定一个标准总结每个交易对手的信用状况。
 - * Tenor (length) of the exposure 敞口期限：面临潜在损失的期间
 - Delegation of Authority 授权委托的规则：
 - * 交易风险越高，所需的审批级别越高。
 - * 高风险、低信用质量或长期交易，需要高级管理层的关注，必须由具备资格的委员会审批。
 - * 小额、短期且信用质量高的交易可以在较低级别批准。
 - * 简单明了的交易可以由个人批准。



Figure 2.1 Approval process.

fig 1: process

• C. Limits: 【设定适当的风险和交易限制】

- 也叫信贷额度 (credit lines)
- 代表了公司愿意承担的最大风险金额 (最大损失)
- 可对经常使用的交易对手、行业和国家附加限制。
- 因为在某些交易中，最大损失无法提前估计，而是取决于具体市场状况。所以，公司通常会使用不同限额的组合。

• D. Oversight: 【配备具备独立性和资源的合格人员进行监督。】

- Independence 独立性
 - * 风险管理部门不应设在具有盈利中心的业务部门内。
 - * 风险经理的薪酬不应基于业务的盈利能力。
 - * 所有参与风险管理的员工都应该直接向首席风险官 (CRO) 汇报，是业务部门负责人。
 - * 首席风险官直接向首席执行官报告，并拥有接触董事会风险委员会和审计委员会的特权。
- 资格要求 Qualifications: 为了能更有效地执行其职责，风险管理需要具备专业知识和技能。
- Proximity to the Business Unit 与业务单位的接近: 尽管风险管理部门需要保持独立性，但与业务单位的接近能够使风险经理更好地了解业务，并为业务提供支持和建议。
- Open Mind 开放的思维: 一个好的风险经理应该具有开放的思维，能够提出建议以促成交易，而不仅仅是拒绝交易。

3 信贷委员会 Credit Committees

- 最高的审批级别通常是信贷委员会 (credit committee)，由公司最高级别的管理人员组成。
 - 信贷委员会由高级管理人员组成，负责决策重要或高风险交易。
 - 成员应包括组织的关键职能部门，如业务单位、风险管理、税务和会计、合规和法律。
 - 委员会成员应具备足够的专业知识，深刻理解风险。

- 委员会应有一名主席，客观地收集意见并促进讨论。
- 委员会应有明确的章程，详细说明审批流程。
- 一个运作良好的委员会应有批准和拒绝的交易。
- 会议记录应详细准确，并在会议结束后能够迅速下发。

2 违约损失率计算-信用风险建模和分析

考点 1 信用风险建模与评估 CR-5

1 CAMEL 骆驼评级法

- 银行信贷分析师普遍采用 CAMEL 系统来评估银行信贷风险，该系统可被视为银行属性的核对清单，而这些属性被视为评估银行财务业绩的关键。
- 主要关注以下 5 个方面：
 - C: Capital Adequacy 资本充足
 - A: Asset Quality 资产质量
 - M: Management 管理
 - E: Earnings 盈利
 - L: Liquidity 流动性

2 信用风险建模的要素

- Probability of default (PD) 违约概率
 - 反映违约发生的概率。
 - PD 的计算方法有很多种，稍后会介绍多个模型
- Exposure at default (EAD) 风险敞口
 - 反映实际暴露在信用风险中的资产价值。
 - 【如果对方出现违约或无法履行合同，你可能会失去的钱。比如你有 100 元，把钱全部借给你朋友，那么你的预期敞口 Expected Exposure (EE) = 100 元。再比如，你把 20 元存银行（假设这是无风险的），80 元借给你朋友，那么你的 EE = 80 元】
- Loss given default (LGD) 违约损失率
 - 指发生违约时扣除回收的部分所发生的损失率。
- Recovery Rate (RR) 回收率：
 - 在债券或贷款违约时，债权人能够收回的金额，占违约债务总额的百分比。
 - 回收率与违约率负相关。

$$LGD + RR = 1$$

- Expected loss (EL) 预期损失
 - 指在未来一段时间（通常是一年）可能发生的平均损失。也就是损失的期望值。

$$\text{Expected loss (预期损失\%)} = PD \times LGD$$

$$\text{Expected loss (预期损失\$)} = PD \times LGD \times EAD$$

Note

拓展：EL 的推导

一个资产未来有两种可能。一种是违约，违约概率是 PD，如果违约，因为有抵押品等还能回收 RR，损失为 1-RR。另外一种是不违约，收回全部资金，损失为 0。

$$EL = PD * (1 - RR) + (1 - PD) * 0 = PD * LGD$$

- tips: EL 是基于 0 时刻位于未来风险的衡量。EL 本质上是未来的 loss。

3 Predicting Default in Credit Risk Models 违约概率的评估方法

- Judgmental approaches 判断法

- 也称作 qualitative approaches 或 expert systems，是最古老且最简单的信用风险评估方法。
- 完全基于信贷分析师对借款人基本质量和数量特征的专业判断。
- 典型的分析方式是 5C analysis:
 - * 1.character 品格：借款人的个性
 - * 2.capacity 能力：基于现有的收入、支出和其他债务义务，评估借款人偿还贷款的能力。
 - * 3. capital 资本：借款人面临风险的自有资本。
 - * 4. collateral 抵押品：为偿付债务提供的担保品
 - * 5. conditions 条件：描述商业环境状况和贷款特点（如利率）的一般条件。
- 数据来源：
 - * 公开的财务报表 publicly available financial statements：资产负债表 balance sheet、损益表 income statement、现金流量表 cash flow statement。
 - * 贷款申请中提交的商业计划 business plan。
 - * 公司所在行业的数据。
 - * 公司所在地区的数据。
 - * 市场情况和总体经济前景等等。
- 判断法的局限性
 - * 很难验证所获得结果的质量。
 - * 难以根据决策环境和可用信息的变化，更新评估过程的结构和实际结果。
 - * 很难检查可能影响单个借款人信用状况或整个信贷组合风险的情景。
 - * 可能会出现透明性和一致性问题，特别是在缺乏明确的实施和监控的情况下。

- Data-driven empirical models 数据驱动的经验模型

- 数据驱动经验模型根据贷款接受、拒绝、按协议支付和违约情况等历史数据构建模型。
- 适用于有历史数据的企业和消费者贷款。
- 数据来源包括银行内部数据库、信用评级机构和数据提供商。
- 使用统计学、机器学习和运筹学技术等，进行大规模数据集，识别复杂的非线性违约关系。
- 例如，芝麻信用评分；后面的信用评级（credit ratings）
- 优势：
 - * 流程提高了透明度（Transparency）和一致性（consistency）。
 - * 模型的基本结构及其预测能力可以通过实证检验。
 - * 可以进行假设检验、情景分析和压力测试。
 - * 可及时更新并定制化。
- 局限性：

- * 依赖历史数据。
- * 历史数据的静态和滞后性。
- * 不能实时更新，可能会忽略借款人最新状况。
- Financial models 金融模型
金融模型采用规范性方法，基于规范的金融和经济理论建立。
 - Structural models 结构模型
 - * 结构模型假设企业的违约是一个内生过程，由企业的结构特征（如资产价值和债务）决定。例如默顿模型（Merton model）。
 - Reduced form models 简约模型
 - * 简约模型假设违约是随机事件，可能在任何时间发生，由外生随机变量——通常是泊松跳跃（Poisson jump）过程的变化——所驱动。这种模型使用债券和信用衍生品的市场数据作为了解企业信用风险结构的主要信息来源。
- Reduced-form models（简化模型）

考点 2 Credit Scoring and Rating CR-6

- 1. 基本概念
 - 信用评分和信用评级，属于数据驱动的经验模型（Data-driven empirical models）。通过参考与借款人根据商定条款偿还债务的能力和意愿有关的所有可用信息，对借款人（债务人）的信用风险水平进行量化。
 - Credit Scoring 信用评分
 - * 信用评分通常指为每个借款人提供数字信用评分的模型和系统。
 - * 信用评分是衡量借款人信用资质（Creditworthiness）和违约概率（Probability of default）的指标。
 - * 信用评分模型：输出一个反映借款人的信用资质和违约概率的分数。
 - * 信用评分通常仅供金融机构和企业客户内部使用。
 - * 信用评分基于自动分析模型
 - * 例如，芝麻信用分；FICO score。
 - Credit rating 信用评级
 - * 信用评级是以序数（ordinal）定性表示的风险评级，例如 A 级、B 级等。
 - * 每个风险等级都与根据经验估算的违约概率相关联，该违约概率描述了所有被归入该等级的借款人。
 - * 信用评级通常用于公司贷款、债券发行和国家（例如主权信用评级），而且通常是公开的。
 - * 信用评级可能涉及分析性和判断性评估的结合，因此依赖于一个更加复杂和深入的过程。
 - 信用评级和信用评分是有序的，较高的评级意味着较低的信用风险，但评分或评级之间的差异与风险差异不成比例。
 - 信用评分和评级系统的主要优势：
 - * 减少贷款评估过程中的主观性。
 - * 实现分析性的信用风险管理，包括情景分析和压力测试。
 - * 通过对所有客户进行评估的共同框架，实现评估的一致性和透明性。
 - * 降低贷款评估所需的时间和成本。
- 2. External ratings 外部评级

解释说明：一级内容回顾

外部评级指的是外部第三方机构对金融机构的评级。

评级机构：根据特定标准对信用风险提供独立意见的组织。

国际知名的三大评级机构：穆迪 (Moody's)，标普 (Standard and Poor's (S&P)) 和惠誉 (Fitch)

评级机构对债券和货币市场工具使用不同的标准。

- 债券评级被称为长期评级。
- 货币市场工具的评级被称为短期评级。

惠誉的评级类别与标普一样。

- 投资级:BBB-(Baa3) 及以上
- 非投资级（投机级或垃圾债券）：BBB-（Baa3）以下。【注意投资级和非投资级的划分界限】

Category	Moody's	S&P/ Fitch	Description
Investment grades	Aaa	AAA	Highly recommended. Small risk of default. Stable investment.
	Aa	AA	High quality investment. Slightly higher level of long-term risk.
	A	A	High-medium quality investment. Vulnerable to economic changes.
	Baa	BBB	Medium quality. Quite secure at present, with risk problems in long-term period.
Speculative grades	Ba	BB	Medium quality. Not well secured investment
	B	B	Quite secure in present. Likely to default in the future.
	Caa	CCC	Poor quality. High likelihood of default
	Ca	CC	Highly speculative investment. Close to default.
		C	Low quality. High likelihood of default. Still paying at present.
	C	D	In default.

Rating agencies: organizations provide independent opinions on credit risk, based on specified criteria.

- Issue-specific credit ratings: specific obligations or instruments. 【债项评级】
- Issuer credit ratings: an issuer's general creditworthiness. 【发行人评级】

评级需同时考虑定量信息 + 定性信息

被评级公司给与评级公司 fee。这存在利益冲突, this fee arrangement has been criticized.

- 两种评级方式: Through-the-Cycle versus Point-in-Time

不论是外部评级还是接下来的内部评级, 都可以使用这两种评级方式。

* Through-the-Cycle 全周期评级

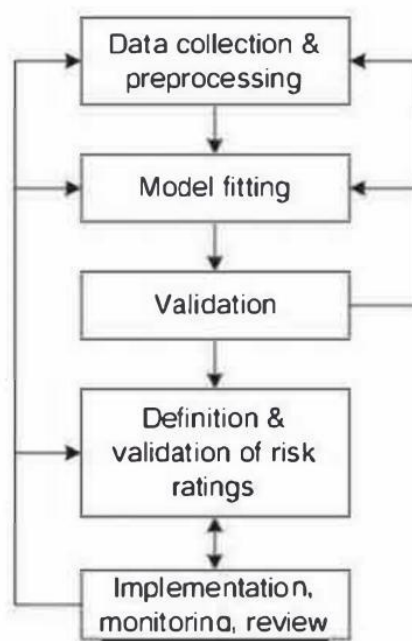
- 周期评级反映一家公司数年内的平均信用度。考虑整体，不太受到整体经济状况起伏的过度影响。
- 评级比较稳定，监管机构倾向于使用 Through-the-Cycle。
- 理论上，Through-the-Cycle 将低估经济周期下行的违约概率，而在经济周期上行时高估违约概率。
- 更新频率较低
- 适用于长期

* Point-in-Time 时间评级

- 时间评级旨在提供对未来违约概率的当前最佳估计。
- point-in-time ratings 具有顺周期性，可能加剧经济波动。
- 评级波动较大，及时更新以反映新信息。
- 适用于短期。

– Development process 信用评分和评级模型的开发过程

信用评分和评级模型的开发是一个复杂和数据密集的过程，既需要分析人员的专业判断能力，也需要技术的先进性。



* a. Data Collection and Pre-processing 数据收集和预处理

数据收集和预处理是信用评分/评级开发过程的首要步骤。

收集违约/不违约的数据。

数据预处理包括去除异常值、根据需要进行数据转换等。

* b. Model Fitting 模型拟合

确定最能描述 (拟合) 训练数据的模型参数 parameters。

比如线性回归模型，需要通过数据，找出参数 beta。

* c. Model Validation 模型验证

从模型拟合过程中得出的模型应使用与模型拟合所用样本不同的样本进行验证。

Validation or holdout sample 验证样本或保留样本

使用样本外 (out-of-sample) 和时间外 (out-of-time) 数据进行回测 (backtesting)，来评估模型的预测性能。

* d. Definition and Validation of Ratings 定义和验证评级

一旦信用评分模型通过验证，就需要把结果一一映射到不同的风险评级类别。

* e. Implementation 实施

- 最后阶段是将结果应用到信用评分/评级系统中，用于实时评估所有新申请贷款的信用风险。
- 经常进行监测和审查，并进行更新。

– 对信用评级机构的主要批评

- * **Lack of transparency and accountability, conflict of interest** 缺乏透明度和问责制度，存在利益冲突：
评级机构通常不公开披露如何进行其风险评级。且评级机构的收入依赖于被评级公司支付的费用。
- * **Promoting Debt Explosion** 促进债务增长：
信用评分和评级提高了金融和信贷机构的风险估计能力，从而使银行和其他信贷机构能够降低其风险溢价和债务融资成本。
- * **Poor Predictive Ability** 预测能力不足：
2007-2008 年全球信贷危机爆发，评级机构未能预测到一些企业破产 (如雷曼兄弟、安然等) 引发了对评级机构评级预测能力的质疑。另外几项实证研究发现，与金融机构内部开发的违约预测模型相比，评级机构的评级预测企业违约的能力较弱。
- * **Pro-cyclicality** 顺周期性：
在经济增长期间，评级过于乐观，在衰退时期过于悲观，因此显示出顺周期性的行为。

• 3. Internal ratings 内部评级

– Internal ratings vs. External ratings

- * 外部评级：通常只适用于已发行公开交易债务的公司。
- * 内部评级：由银行等信贷机构使用，这些机构可以获得所需的历史数据，以开发和校准适合其特定信贷组合的模型。
银行和其他金融机构开发自己的内部评级系统。

– Altman's Z-Score 奥尔特曼 Z 评分模型

- * Altman 于 1968 年开发了后来被称为 Z-Score 的方法。是典型的内部评级模型。
- * 利用一种称为判别分析的统计技术，从五个会计比率中预测违约情况：

$$Z = 1.21X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

X_1 ：周转资金与总资产之比、=

X_2 ：留存收益与总资产之比

X_3 ：息税前利润与总资产之比

X_4 ：股本市值与总负债账面价值之比

X_5 ：销售额与总资产之比

* 判断：

$Z\text{-score} > 3.0$ ，不太可能违约

$2.7 < Z\text{-score} < 3.0$ ，警惕

$1.8 < Z\text{-score} < 2.7$ ，有很大的违约可能性

$Z\text{-score} < 1.8$ ，违约概率很高

- * **【Z-score 只能得出违约与否的大概说明，不能得出违约概率具体是多少】**

- * **【Z-Score 是基于 66 家公开交易的制造业公司样本估算的。该模型在样本外测试时表现相当准确。此外，随着时间的推移，Z-Score 已经发展和扩展，适用于非公开交易的制造业公司和非制造业行业。】**

• 4. Estimating Default Probabilities 估算违约概率

违约概率 (probability of default, PD) 反映违约发生的概率。

PD 的计算方法有很多种，FRM 一级考试主要掌握下面几种方法。

– (1) Unconditional and Conditional default probability 无条件条件和条件违约概率

【注意题目的表达方法，计算不同种类的概率】

* A. Cumulative Default Probability 累计违约概率

- Cumulative PD: 在一定时期内 (例如, 一年内、两年内、三年内) 违约的概率
- Cumulative survival rate 累计生存概率 ($SR_t^{\text{cumulated}}$) = 1 - cumulative PD

* B. Unconditional default probability 无条件违约概率

- 也称 Marginal Default Probability 边际违约概率
- 指在某一特定时间段内, 某一实体 (如公司或债券发行人) 发生违约的概率, 而不考虑其他任何条件或前提。(后面的很多模型都是基于无条件违约概率)

$$PD_k^{\text{Uncond}} = PD_{t+k}^{\text{cumulated}} - PD_t^{\text{cumulated}}$$

* C. Conditional default probability 条件违约概率

- 条件违约概率是指在给定某些条件或前提下, 某一实体在特定时间段内发生违约的概率。
- 例如, 存活 (不违约) 到第 n 年, 其在第 $n+1$ 年违约的概率。

$$PD_{t \rightarrow t+k}^{\text{cond}} = \frac{PD_k^{\text{Uncond}}}{SR_t^{\text{cumulated}}}$$

– (2) Historical Default Probabilities 历史违约概率

根据历史数据计算出的违约概率, 主要指评级机构的各类评级对应的违约概率。

* Rating transitions 评级迁移矩阵 【注意表头。下表 S&P credit rating 由 1981-2000 年数据得到】

Table 9.1 Average Cumulative Issuer-Weighted Default Rates (%), 1981-2020

Rating	Time Horizon (years)							
	1	2	3	4	5	7	10	15
AAA	0.00	0.03	0.13	0.24	0.34	0.51	0.70	0.90
AA	0.02	0.06	0.11	0.21	0.30	0.49	0.70	0.99
A	0.05	0.13	0.22	0.33	0.46	0.76	1.20	1.84
BBB	0.16	0.43	0.75	1.14	1.54	2.27	3.24	4.54
BB	0.63	1.93	3.46	4.99	6.43	8.89	11.64	14.65
B	3.34	7.80	11.75	14.89	17.35	20.99	24.62	28.24
CCC/C	28.30	38.33	43.42	46.36	48.58	50.75	52.76	54.76

* 解读:

例如, 初始评级为 BBB 的债券在第一年的违约概率为 0.16%, 在 2 年内的累计违约概率为 0.43%。

计算特定年份的违约概率: 评级为 BBB 的债券在第二年内 (during the second year) 的无条件违约概率为:

$$0.27\% = 0.43\% - 0.16\%$$

条件违约概率 Conditional PD:

- 计算 CCC/C 评级公司的前两年不违约的条件下, 第三年违约的概率:

$$\text{前两年不违约的概率} = 100\% - 38.33\% = 61.67\%$$

$$\text{第三年违约的概率} = 43.42\% - 38.33\% = 5.09\%$$

$$\text{Conditional PD} = 5.09\% / 61.67\% = 8.25\%$$

* 对于投资级债券, 边际违约概率通常随时间递增。

例如, 初始评级为 A 的债券在第一年、第二年、第三年和第四年违约的概率分别为 0.05%, 0.08% (= 0.13% - 0.05%), 0.09% (= 0.22% - 0.13%), 0.11% (= 0.21% - 0.11%)。

这是因为债券发行人最初被认为信用可靠，时间流逝越久，其财务状况下降的可能性就越大。=

- * 对于投机级债券：边际违约概率可能随时间递减。

例如，最初属于最低类别（CCC/C）的债券在第一年、第二年、第三年和第四年违约的概率分别为 28.3%，10.03%，5.09%，and 2.94%。

这是因为，对于信用评级较差的债券来说，未来一两年可能至关重要。如果发行人能挺过这段时间，其财务状况很可能会有所改善。

- * Default intensity model 违约强度模型

Hazard Rate 风险率，也叫 default intensity 违约强度 (λ)，是短时间内 (Δt) 有条件违约概率的衡量标准 $\lambda * \Delta t$ ：零时间到 t 时间之间没有违约的情况下，时间 t 和 $t + \Delta t$ 之间发生违约的概率。

$$\text{Cumulative PD} = 1 - e^{-\bar{\lambda}t}$$

$$\text{Cumulative survival rate} = e^{-\bar{\lambda}t}$$

$$\text{Unconditional PD} = e^{-\lambda_1 \times t_1} - e^{-\lambda_2 \times t_2}$$

$\bar{\lambda}$ 是时间零点至时间点 t 之间的平均风险率。

例如：假设每年的风险率是恒定的 1.5%。

第一年年底发生违约的概率是 $1 - e^{-0.015 \times 1} = 0.0149$ 。

到第二年年底发生违约的概率是 $1 - e^{-0.015 \times 2} = 0.0296$

在第三年、第四年和第五年结束时发生违约的概率分别为 0.0440，0.0582，和 0.0723

第四年的无条件违约概率为 $0.0582 - 0.0440 = 0.0142$

在没有提前违约的条件下，第四年违约的概率为 $0.0142 / (1 - 0.0440) = 0.0149$

- * Recovery Rates 回收率

table 1: 1983 到 2021 不同公司债券种类的回收率

Class	Average Recovery Rate (%)
1st lien bond	54.6
2nd lien bond	44.3
Senior unsecured bond	38.0
Senior subordinated bond	31.0
Subordinated bond	32.1
Junior subordinated bond	22.3

【债券种类稍微了解即可——公司破产清算时的求偿顺序为：First Lien Loan 优先级最高，回收率因此最高；后面依次递减，Junior Subordinated 回收率最低。】

2007-2008 年危机的教训：抵押贷款回收率与抵押贷款违约率呈负相关。公司债券也存在类似的负相关关系。

【经济疲软时，借出方不仅面临违约率上升的更大风险，还会由于回收率低而遭受更大的损失。】

考点 3 计算违约概率 CR-9

1 信用利差计算违约概率

1. 基本概念

- 粗略地说，信用利差（credit spread）可视为平均损失率。

$$\text{credit spread} \approx \text{EL} (\%) = \text{PD} * \text{LGD} = \text{HazardRate} * (1 - \text{RR})$$

$$\rightarrow \text{Hazard Rate } (\lambda) = \text{credit spread} / (1 - \text{RR})$$

– Recover Rate (RR) 回收率

- 原版书例子: 假设 3 年期、5 年期和 10 年期的 CDS 息差分别为 50、60 和 100 个基点, 预期回收率为 60%。
 - 三年的平均风险率约为 $0.005 / (1 - 0.6) = 0.0125$
 - 五年的平均风险率约为 $0.006 / (1 - 0.6) = 0.015$
 - 十年的平均风险率约为 $0.01 / (1 - 0.6) = 0.025$
 - 第 3 年至第 5 年的平均风险率为 $(5 * 0.015 - 3 * 0.0125) / 2 = 0.01875$
 - 第 5 年至第 10 年的平均风险率为 $(10 * 0.025 - 5 * 0.015) / 5 = 0.035$

2. 信用利差的种类

- A. CDS(Credit credit swap) spread 信用违约互换利差

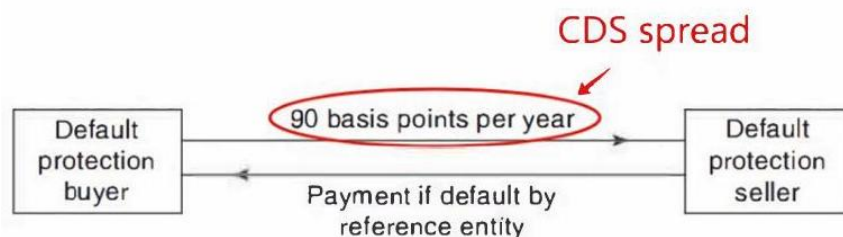


fig 2: 信用违约互换 (Credit default swap)

- CDS spread 是购买信用违约互换时需支付的保险费 premium。
- 反映了特定债务发行人的违约风险。发行人 (reference entity) 违约风险较高, CDS spread 越高, 反之亦然。
- B. Bond yield spread 债券收益率利差
 - 债券收益率利差是指公司债券收益率与类似无风险债券收益率之差。
 - 反映了特定债务发行人的违约风险。发行人违约风险较高, 债券收益率利差越高, 反之亦然。
 - 通常: 债券收益率利差 = $\text{YTM}_{\text{公司债}} - \text{YTM}_{\text{国债}}$
 - CDS-债券基差 (CDS-bond Basis) 是 CDS 利差超出债券收益率利差的部分。

$$\text{CDS} - \text{BondBasis} = \text{CDSSpread} - \text{BondYieldSpread}$$

- 如果是同一个实体 (债券发行人), 风险情况一定, 那么 CDS Spread 应该等于 Bond Yield Spread, 即 CDS-Bond Basis 应接近零, 但实际上往往偏离零。

CDS-Bond Basis 偏离零的原因:

- * Negative direction, 即 $\text{CDS Spread} < \text{Bond Yield Spread}$

CDS 存在交易对手违约风险。【CDS 的卖方违约风险大 (即 CDS 的卖方不赔付) → CDS 不值钱 → CDS Spread ↓】

CDS 的偿付未包括交付债券的累计利息 (interest)。【CDS 少考虑了一部分“好处 accrued interest” → CDS 不值钱 → CDS Spread ↓】

- * Positive direction, 即 $\text{CDS Spread} > \text{Bond Yield Spread}$

There is a cheapest-to-deliver bond option in a CDS. 【这个 option 对于 CDS 是好的 → CDS 更值钱 → CDS Spread ↑】

The restructuring clause (重组条款) in a CDS contract may lead to a payoff when there is no default. 【restructuring clause 对于 CDS 是好的 → CDS 更值钱 → CDS Spread ↑】

重组条款 (Restructuring Clause) 是 CDS 合约中的一项条款, 规定如果参考实体的债务发生某种形式的重组 (如修改偿还条件、延长债务期限或降低利率等), 即使参考实体没有违约, CDS 买方也可以获得赔付。

* 可能 Negative direction, 也可能 Positive direction

- 债券的售价可能与面值相差很大。
高于面值的债券价格往往会产生负基差。
低于面值的债券价格往往会产生正基差

· 【无风险利率的选择不同】

债券收益率利差可能是相对于无风险参考利率计算的, 而这个参考利率并不是市场上使用的利率。

– 若 $CDS-Bond\ Basis \neq 0$, 即 $CDS\ Spread \neq Bond\ Yield\ Spread$

* $CDS\ Spread > Bond\ Yield\ Spread \rightarrow$ 卖出 CDS(收 CDS Spread), 做空债券 (支出 coupon)

* $CDS\ Spread < Bond\ Yield\ Spread \rightarrow$ 买入 CDS(支出 CDS Spread), 做多债券 (收 coupon)

– 市场动荡前后的基差趋势:

【CDS-Bond Basis 偏离零的情况】

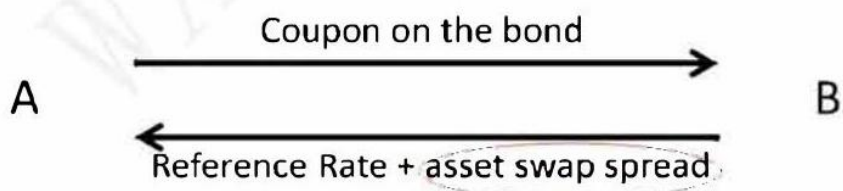
* 次贷危机前: the basis tended to be positive.

* 次贷危机期间: 基差有时非常负, 但由于流动性短缺和其他考虑, 金融机构难以在债券和 CDS 之间进行套利。

* 次贷危机后: CDS-Bond Basis 通常较小且为负。

• C. Asset swap spread

– 资产互换 (Asset swaps) 为信贷市场的交易者提供了一个方便的参考点, 因为它能直接估算出债券收益率超过浮动参考利率的部分。【简单了解】



小结:

- 使用 bond yield spreads 和 asset swap spreads 计算 hazard rate, 需要满足一定的前提条件: 标的债券以接近其面值的价格出售。
- 如果不是 par bond, 需要可以采用其他方式求得 hazard rate。

3. Comparing default probabilities 比较违约概率

- 比较从历史数据 (来自标普) 计算的累计违约概率与从信用利差 (来自美林 Merrill Lynch) 计算的累计违约概率

table 2: 累计违约概率与信用利差对比

Rating	Cumulative 7-Year Default Probabilities (%), 1981-2020	7-Year Credit Spread (bp) 1996-2007
AA	0.51	35.74
AA	0.49	43.67
A	0.76	68.68
BBB	2.27	127.53
BB	8.89	280.28
B	20.99	481.04
CCC/C	50.75	1,103.70

- 例如: A 评级的累计违约 =0.76%，求得 hazard rate

$$\text{CumulativePD} = 1 - e^{-\lambda t} \rightarrow 0.76\% = 1 - e^{-\lambda * 7} \rightarrow \lambda = 0.1090\%$$

- A 评级的 CS=0.6868%(The recovery rate is assumed to be 40%)，求得 hazard rate

$$\lambda = 0.6868\% / (1 - 40\%) = 1.1447\%$$

table 3: 平均七年违约强度 (Hazard Rate)

Rating	Rate from Default Probabilities (%)	Rate Implied by Credit Spreads (%)	Difference
AAA	0.073	0.596	0.523
AA	0.070	0.728	0.658
A	0.109	1.145	1.036
BBB	0.328	2.126	1.798
BB	1.330	4.671	3.341
B	3.366	8.017	4.651
CCC/C	10.118	18.395	8.207
% per annum.			

- 由 CS 计算的违约概率大于历史数据 (期限更长 1981-2020) 得出的违约概率。
- 随着信用质量下降，两种违约率之间的差异也在增加。Reason for the Difference:
 - Corporate bonds are relatively illiquid. 【债券的流动性不好，因此收益率比较高来补偿流动性风险。然而，这只能解释 difference 的一小部分，大约 25 个基点。】
 - the subjective default probabilities 主观违约概率 of bond traders are much higher 【债券交易员对违约概率的主观判断可能要更高】
 - bonds do not default independently of each other. 【债券违约与否并不是独立的 → systematic risk]
 - * Credit contagion 信用传染: 市场经济好，降低了所有公司的违约概率，而市场条件不好，则增加了所有公司的违约概率
 - nonsystematic (or idiosyncratic) risk
 - 每只债券都伴随有非系统性风险，这种风险难以通过持有少数债券来有效分散，需要补偿。【债券一个典型的特征是: 收益少 (只有利息)，但是损失比较大 (不仅利息也可能本金)】

2 股价计算违约概率

- 1. Merton model 莫顿模型
 - 一种结构化模型 (Structural model)。与 BSM 模型有着千丝万缕的关系。【根据公司股票的实时价格得出公司的违约概率】
 - 它取决于公司资产负债表的结构——资产、负债和权益。由于关键变量是公司的资产价值，因此也可称为公司价值模型。
 - 基于以下假设:
 - * a. 债务负债是在 T 时间到期的零息债券。

* b. 违约概率是这一结构模型的内生因素。

* c. 公司的资产可自由交易。

- 逻辑: 【公司资产 $\text{asset} = \text{负债} + \text{股东权益}$ 】

假设某公司的资产价值 V ，同时公司向债权人发行了面值为 $K = 100$ ，期限为 1 年的零息债券，到期一次性偿还本金。一年后债券到期日，债券持有人和股权持有人的价值情况:

一年后	资产价值 (V_1)	债券价值 (B_1)	股东权益 (S_1)
情况 1	$V_1 = 300 > K$	$B_1 = K = 100$	$S_1 = 200 = V_1 - K$
情况 2	$V_1 = 100 = K$	$B_1 = K = 100$	$S_1 = 0 = V_1 - K$
情况 3	$V_1 = 80 < K$	$B_1 = V_1 = 80$	$S_1 = 0$
一般情况	$V_1 = B_1 + S_1$	$B_1 = \text{Min}(K, V_1) = K - \text{Max}(K - V_1, 0)$	$S_1 = \text{Max}(V_1 - K, 0)$

- 结论:

* 对于股东来说: 【持有公司股票，等同于持有一个标的资产是公司资产，执行价格是 K 的欧式看涨期权】

$$\text{股东权益}(E_0) = VN(d_1) - Ke^{-rT}N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{v}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, d_2 = \frac{\ln\left(\frac{v}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Φ : Volatility of assets (assumed constant)

* 对于债权人来说 【千万记得，债务价值不是 put 价值】

$$B_1 = \text{Min}(K, V_1) = K - \text{Max}(K - V_1, 0) = K - \text{put option}$$

* Value of risky debt = value of risk-free debt - value of a put option

持有公司的债务，等同于持有一种无风险债券(面值为 K)，并同时卖出一个欧式看跌期权(标的资产是公司资产，执行价格是 K)

$$B_1 = V_1 - E_1 = V_1 - \text{Max}(V_1 - K, 0) = V_1 - \text{call}$$

持有公司的债务，等同于持有公司资产，同时卖出一个欧式看涨期权

* 风险中性违约概率 Risk-neutral probability of default = 看涨期权不行权的概率 = $1 - N(d_2)$

回顾 BSM: 期权的价值

$$\text{Call} = S_0 N(d_1) - Xe^{-rT}N(d_2)$$

$$\text{Put} = Xe^{-rT}N(-d_2) - S_0N(-d_1)$$

$$N(d_1) + N(-d_1) = 1$$

$$d_1 = \frac{\ln\left(\frac{S_0}{x}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln\left(\frac{S_0}{x}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$N(X)$: Standard normal cumulative distribution function.

$N(d_1)$: Delta of call option

$N(d_2)$: Risk-neutral probability of call exercise 行权概率

* Distance to default (DD) 违约距离

违约距离是指在未来 T 年触发违约所需的，资产价格变化的标准差数量。

$$DD = d_2 = \frac{\ln\left(\frac{V}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \approx \frac{\ln(V) - \ln(K)}{\sigma}$$

违约距离越小，公司更有可能违约。

通过计算 DD，通过查标准正态分布表即可得违约概率。

另外，当有几个备选可投资的公司，无需精确算出每个公司的违约概率，只需要知道几个公司的 DD 大小即可，DD 越大越安全。

* Firm value(V) 和 Volatility of firm value (σ)

计算违约概率时，需要知道资产价值 V 和资产价值的波动率，可以通过下面等式求解。本身的求解过程并不重要 (计算过程很复杂)，结果也不重要。关键掌握这一点——Volatility of firm value 和 the volatility of firm 即使未知，也依然可以利用 Merton model 求出违约概率。即 Volatility of firm value 和 the volatility of firm 未知，不是 Merton model 的缺陷。

联立下面两个方程，可求出 V 和 σ ：

$$\begin{aligned} \text{delta} = N(d_1) &= \frac{\text{期权价格的变化}}{\text{标的资产价格的变化}} = \frac{\sigma_E \times E_0}{\sigma_V \times V_0} \rightarrow \sigma_E \times E_0 \\ &= N(d_1) \times \sigma_V \times V_0 \end{aligned}$$

$$\left. \begin{aligned} E_0 &= VN(d_1) - Ke^{-rT}N(d_2) \\ \sigma_E \times E_0 &= N(d_1) \times \sigma_V \times V_0 \end{aligned} \right\} \rightarrow \text{可得: } V \text{ 和 } \sigma$$

- 2. Merton model 的扩展模型，有很多种形式，例如：什么情况下算违约，即资产价值小于多少算违约：a default occurs whenever the value of the assets falls below a barrier level. 【当资产价值跌破某个障碍水平，认为就会发生违约。】最基本的莫顿模型：barrier level 是债务价值。扩展模型：barrier level 可以是其他设定的情况

Another allows payments on debt instruments to be required at more than one time.

最基本的莫顿模型：假设公司的所有债务只有一个到期时间 (即所有债务一次性到期)。

扩展模型：实际上，公司的债务发生在多个时间点。例如，公司可能有分期偿还的债务或本身有不同到期日的债券。

下面的 Moody's-KMV model and the Kamakura model 也是其中的一个扩展模型

- 3. Moody's-KMV model and the Kamakura model 【原版书这里介绍的甚少】

- 都是基于默顿模型 Merton model 的框架，用于估计公司的违约概率。

- Moody's-KMV 模型：

由 Moody's Analytics 旗下的 KMV 开发，更加专注于利用市场数据和公司财务数据来估计违约概率。它将 Merton 模型的理论应用于实际市场数据中，提供了一种量化的风险评估工具。

- Kamakura 模型：

由 Kamakura 公司开发，同样也是基于公司财务数据和市场数据来计算违约概率的模型。在具体的数学公式和模型参数上有所不同，但目标和方法与 Moody's-KMV 模型类似。

- Moody's KMV and Kamakura provide services to transform Merton's model default probabilities into real-world default probabilities.

- 4. Real-World vs. Risk-Neutral Default Probabilities

- In a risk-neutral world, the expected growth rate of the value of the company's assets is the risk-free rate.

- In the real world, the growth rate of the company's assets is usually higher than this (reflecting a risk premium demanded by the market).

- Risk-Neutral Default Probabilities 风险中性违约概率
- Real-World Default Probabilities 真实世界违约概率
也叫 physical default probabilities

table 4: 风险中性违约概率与现实世界违约概率的比较

	风险中性违约概率	现实世界违约概率
含义	在风险中性世界中计算的公司违约的概率。假设投资者对风险无偏好, 公司资产的预期增长率等于无风险利率。	在现实经济环境中, 公司实际发生违约的概率。考虑了市场对风险的态度, 因此反映了投资者预期的公司资产增长率, 该增长率通常高于无风险利率, 因为包括了风险溢价。
资产的增长率	无风险利率	高于无风险利率 (反映了风险溢价)
数据	基于信用利差	基于历史数据 (例如评级机构)
适用场景	适用于估值, 因为它们更接近无偏的市场定价。	适用于情景分析, 因为更能反映实际经济条件下的违约风险。
模型	Merton 模型	Moody's-KMV 和 Kamakura 模型
风险中性违约概率高于现实世界违约概率。		

考点 4 Retail Credit Risk Management CR-7

1. 基本概念

- 零售银行 retail banking 主要接受消费者和小企业的存款以及向其提供贷款。
- 贷款形式包括:
房屋抵押贷款 Home mortgages、房屋产权信贷额度 Home equity loans(HELOCs)、分期付款贷款 Installment loans(涵盖汽车、信用卡等)、小企业贷款 Small Business Loans (SBL)。
- 主要风险:
信用风险 credit risk
运营风险: 与业务运营相关的日常风险。
业务风险: 与新产品或趋势相关的战略风险, 以及与利率变化时抵押贷款量等措施相关的数量风险。
声誉风险: 银行在客户和监管机构中的声誉。
利率风险: 银行为其资产和负债提供特定利率, 利率随市场变化而变化。
资产估值风险 Asset valuation risk : 与资产、负债和抵押品类估值相关的市场风险
- 零售风险为具有以下特征的同质化组合 homogeneous portfolios:
A large number of small, low-value loans 大量的小额、低价值贷款 With either a consumer or business focus Where the incremental risk of any single exposure is small 增量风险较小

2. Retail credit risk vs. Corporate credit risk

- Retail credit risk 零售信用风险
- 即使单个客户违约, 影响也很小
- Expected loss can be treated like other costs of doing business and can be built into the price charged. 【预期损失可以体现在产品定价中】
- Corporate credit risk 公司信用风险
 - 单一名称的大额风险敞口 Large exposures to single name;
 - Concentrations of exposures.

- 以信贷损失上升到某种意外水平的风险为主。Dominated by the risk that credit losses will rise to some unexpected level.

3.The Dark Side 阴暗面 of Retail Credit Risk

【正常情况下，零售银行风险对于银行来说并不会带来太大影响，但有些时候却会带来很大问题】

4 个原因:

- 并非所有创新的零售信贷产品都能与足够的历史损失数据相关联，从而使其风险评估变得可靠。【缺乏足够的历史损失数据，风险难以评估】
- 在经济环境急剧变化的影响下，特别是在风险因素同时恶化的情况下（即所谓的完美风暴情景），即使是广为人知的零售信贷产品也可能开始出现意想不到的行为。Even well-understood retail credit products might begin to behave in an unexpected fashion under the influence of a sharp change in the economic environment, particularly if risk factors all get worse at the same time (the so-called perfect storm scenario). 【未预料到的经济环境急剧变化】
- 消费者违约（或不违约）的倾向是一个复杂的社会和法律体系的产物，而这个体系又在不断变化。The tendency of consumers to default (or not) is a product of a complex social and legal system that continually changes. 【消费者违约行为的复杂性: 消费者违约行为受到复杂的社会和法律制度的影响，并且这些制度不断变化。】
- Any operational issue that affects the credit assessment of customers can have a systematic effect on the whole consumer portfolio. 【信贷过程中的操作缺陷: 由于信贷评估过程的半自动化结构，可能导致高风险个体获得信贷，从而增加违约风险。】

4. 管理零售信贷风险

- 加强风险管理和监控。
- 强化贷款审查和评估标准。
- 遵守法规和监管要求。
- 通过使用概率违约率、违约暴露和违约损失率等指标，对各种信贷组合的风险进行量化和管理，预先设定适当的风险资本储备。
- 市场监测和趋势分析。

5. Credit Scoring Models 信用评分模型

- 信用评分模型利用统计程序将信贷申请人信息转化为数字评分，用于衡量信用风险或偿还概率。
- 分数越高，风险越低
- Three Types of Models:
 - Credit Bureau Scores 信用局评分
 - * 由信用评分公司开发。例如，Fair Isaac Corporation 开发的 FICO scores
 - * 这种信用评分成本低。
 - * 例如，Fair Isaac 信用局风险评分范围: 在 300 到 850 之间; 次级贷款通常针对评分低于 660 的客户。
 - Pooled Models 汇总模型
 - * 这些模型由外部供应商利用从具有类似信贷组合的众多贷款人那里收集的数据建立。These models are built by outside vendors, using data collected from a wide range of lenders with similar credit portfolios.
 - * 模型根据行业进行调整，但不是针对具体公司的。
 - Custom Models 定制模型
 - * 这些模型通常是内部开发的，使用的数据来自贷款机构自己独特的信贷申请群体。These models are usually developed in-house using data collected from the lender's own unique population of credit applications.
 - * 定制模型可能针对特定的申请人。例如，信用卡和抵押贷款。
- 成本: Credit bureau scores< Pooled Models< Custom models.

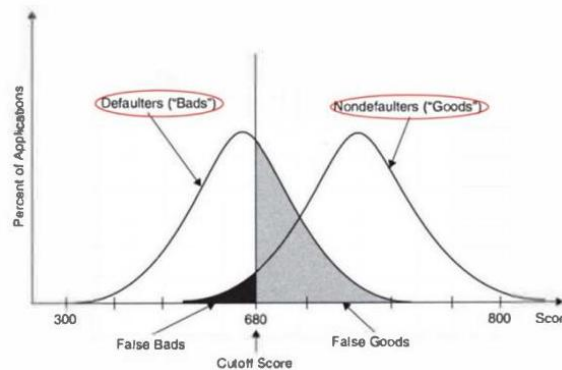
- 抵押贷款信用评分需要考虑的关键变量:

$\text{Risk Score} = f(\text{Doc Type}, \text{Transaction Type}, \text{FICO}, \text{LTV}, \text{DTI}, \text{Occup Type}, \text{Prop Type}, \text{Pmt}, \text{Economic Cycle})$

- FICO score
- Loan-to-value (LTV) ratio 贷款价值比: 抵押贷款额除以相关房产的评估总值。the amount of the mortgage divided by the associated property's total appraised value. 【LTV= 贷款金额/房屋价值, 越低越好】
- 债务收入比 (DTI): 借款人每月债务支付 (抵押贷款、汽车贷款等) 与每月总收入的比率。
- 支付类型 (pmt): 决定抵押贷款的类型 (可调利率、固定利率等)。
- Debt-to-income (DTI) ratio 债务收入比: 每月债务支出 (房贷、汽车等) 与借款人每月总收入的比率。【DTI= 月供/月总收入】
- Payment (pmt) type 支付类型: 固定利率、可变利率等
- Documentation (doc) types 文件类型, 例如资产证明文件、税收文件等。

6. Cutoff Scores 信用评分临界值

- : 截止分数 (Cutoff Scores): 基于主观标准的申请人分数临界值。【什么分数可以发放贷款, 什么分数拒绝发放贷款】



设定 Cutoff Scores 的目的: 避免放款给高风险客户, 同时尽量不错过的低风险客户。

- Scorecard Performance:
 - 判断一些评分模型 Scoring Model 是否有效
 - 在衡量记分卡的性能时, 传统上采用的验证技术是累积准确率曲线 (CAP) 及其汇总统计量, 即准确率 (AR)。When measuring a scorecard's performance, the validation technique traditionally employed is the cumulative accuracy profile (CAP) and its summary statistic, the accuracy ratio (AR).
 - 累计准确性曲线 (cumulative accuracy profile, CAP)
 - * 横坐标 (horizontal axis) 为按得分从最高风险分值到最低风险分值排序的人群。
 - * 纵坐标 (vertical axis) 是根据银行记录得出的实际违约百分比。
 - * 完美模型 (perfect model) 曲线: 代表最完美的情况下, 模型会识别出所有坏客户。此时, CAP 曲线是折现 (由下图红色箭头组成);
 - * 随机模型 (random model) 曲线: 代表模型对好坏客户毫无区分能力, CAP 曲线会是一条斜率为 45 度的线 (下图蓝色箭头);
 - * 实际模型 (actual model) 曲线: 代表实际模型的曲线, 越接近完美曲线代表预测能力越强; 越接近随机曲线, 代表预测能力越弱。

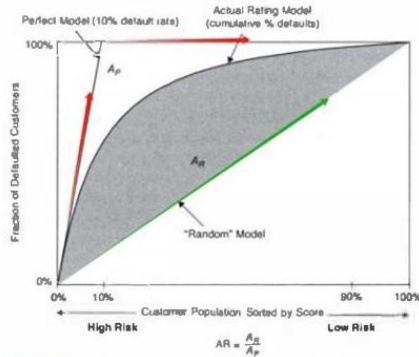


Figure 7.3 Cumulative Accuracy Profile (CAP) and Accuracy Ratio (AR).

- 完美模型 (perfect model) 下的面积记为 A_P ，而实际评分模型 (actual rating model) 下的面积记为 A_R

$$\text{accuracy ratio (AR)} = \frac{A_R}{A_P}$$

AR 越接近 1，说明这个评分模型越准确。

考点 5 Country Risk CR-8

【这是 24 年新增加的内容，和一级有很大的相似度，专有名词有变更、考纲要求稍微有些不同。如果一级学习的比较好，这里可以稍微看一下就行。

FRM 一级在 2020 年未改版之前，关于 Country Risk 这个内容，和现在 FRM 二级的这个知识点选用的是同一个人的同一篇文章，只是版本不同而已 (FRM 二级比较新)。

许多大公司在世界各地都有商业布局。虽然这些公司可能在一个国家注册成立，但对它们来说，评估它们在国外经营相关的风险是很重要的。这些风险统称为国家风险或国别风险 (country risk)。

【本章节的逻辑: country risk 的来源成因 → country risk 的衡量 (几个机构在做这个事情，但有缺陷) → (抓主要矛盾) → 衡量主权债务违约风险 (2 种手段) → 主权信用评级 V.S 信用价差 → 这两种手段各有优缺点】

• 1. Evaluation of Risk

– Economic Growth Life Cycle 经济增长生命周期

- * 评估国家风险的一个重要考虑因素是一个国家将如何应对经济周期。An important consideration in assessing country risk is how a country will respond to economic cycles.
- * 在经济衰退期间，发展中国家国内生产总值的下降幅度往往大于发达国家。During economic downturns, GDP in developing countries tends to decline more than GDP in developed countries.
- * 核心思想: 在成熟市场，国家风险较低；在新兴市场，国家风险较高。

– Political risk

- * 政治风险是指政府、政府决策或政府运作方式的变化对企业或投资的盈利能力产生重大影响的风险。Political risk is the risk that changes in governments, decisions made by governments, or the way governments operate will significantly affect the profitability of a business or an investment.
- * 评估政治风险绝非易事。Assessing political risk is never easy.
 - Democracies 民主制 or authoritarian 独裁制 governments
 - Corruption 腐败
 - Violence 暴力
 - Nationalization 国有化 or expropriation 征用

– Legal Risk

- * Legal risk is the risk of losses due to inadequacies or biases in a country's legal system. Property rights and contract enforcement are important aspects of a legal system.

* 核心思想: 法律体系是否健全有效 -

– Economic Structure

- * It is also important to assess the country's competitive advantages and the extent to which its economy is diversified.
- * 过度依赖单一商品/服务的经济体风险较高, 比如过度依赖石油、或旅游业等。
- * 经济多样化在小型经济体中很难实现。

• 2. 国家风险的衡量

对于国家风险的衡量是否存在一个可靠的综合的风险度量 (比如, 求出一个国家的 VaR 或 ES)。有几个机构在做这个事情:

– Political Risk Service (PRS)

- * Uses 22 measures of political, financial and economic risk to calculate its index. Individual companies can customize the PRS forecasting model to their own projects or exposures.

– 媒体机构: provide country risk scores.

- * Euromoney's score is based on a survey of 400 economist.
- * The Economist develops country risk scores internally based on currency risk, sovereign debt risk, and banking risk.

– The World Bank

- * Provides country risk data measuring corruption (腐败), government effectiveness, political stability, regulatory quality, the rule of law, and accountability.

– 要注意这么几个问题:

- * 1) 几个机构之间的评分无法进行比较。It is difficult to compare these services because they use different scoring methods and consider different attributes of country risk.
- * 2) not every dimension of country risk is necessarily relevant to every individual or corporation.
- * 3) the question of scaling (评分的尺度问题): 如果 A 国家的得分是 B 国的两倍, 并不意味着 A 国的风险是 B 国的两倍。因此, 国家的得分排序 (rankings of countries) 比它们分数本身 (their numerical scores) 更有意义。

• 3. Sovereign Credit Risk

- One measure of a country's risk is the risk it will default on its debt. There are two types of sovereign debt: the type issued in a foreign currency (such as the USD) and the type issued in the country's own currency. 衡量一个国家的风险大小问题, 一个主要的代表就是看其政府债务的履约情况。即衡量其主权债务违约的风险。主权债务 (sovereign debt) 有两种类型: 以外币发行债务和以本国货币发行的债务。

– Foreign Currency Defaults 外币违约

- * 以外币发行的债务
- * 发行国无法通过简单地印更多的钱 (print more money) 来偿还债务。
- * 违约是由金融、经济和政治问题共同造成的, 这些问题在最初发放贷款时基本上是无法预见的。
- * 举例来说, 美国政府可以通过印刷更多的美元来偿还以美元发行的债务 (被称为增加货币供应量, 并可能导致通货膨胀)。然而, 像阿根廷这样的国家在借入美元时就无法用印美元的方式来还钱。
- * 例如: 希腊债务危机: 南美的有些国家在过去 200 年中多次违约。

– Local Currency Defaults 本币违约

- * 一些国家对以本国货币发行的债务以及以外币计价的债务违约。

- * 例如: 巴西 (1990 年) 和俄罗斯 (1998 年) 的违约
- * 穆迪的研究表明, 这两类债务同时违约的国家越来越多。
- * Reasons: 【既然很多国家可以印更多的钞票, 为什么它们还会违约以本国货币计价的债务? 是小可爱吗? 不是】

客观上: 无法印钱 (print money)

比如, 1971 年之前, 要有对应的黄金储备, 才能印对应的钱: 希腊债务危机 (2012 年), 希腊没有印制欧元的权利。

In the decades before 1971, currencies had to be backed by gold reserves.

Members of the European Union use the euro as their domestic currency.

Printing more money debases the currency and leads to inflation

- * 主观上: 印钱 (print money) 带来的危害更大
- 一方面, 印更多的钱会使货币贬值并导致通货膨胀。
- 另一方面, 印钞在短期内可能解决一部分问题, 但长期看危害较大。

– Impact of a Default

- * The modern consequences of a default by a sovereign nation include:
- * A loss of reputation along with an increased difficulty in raising funds for several years
- * A lack of investors willing to buy the debt and equity of corporations based in the country
- * An economic downturn (经济衰退)
- * Political instability as the population loses faith in its leaders
- * A default has a negative impact on GDP growth, has a negative impact on the country's credit rating for many years, can hurt exports, and can make the banking system of the defaulting country more vulnerable. 【违约对 GDP 增长有负面影响, 对该国的信用评级有多年的负面影响, 可能损害出口, 并可能使违约国的银行系统更加脆弱。】

– Factors Determining Sovereign Default Risk

【政府违约的原因与个人和企业违约的原因相似: 在经济繁荣时借贷超过其收入; 在经济衰退时, 无法履行债务义务。】

债务程度 Degree of indebtedness: 看一个主权实体欠多少债, 不仅是对外债务, 还有对内的债务。

评级机构会使用 government debt-to-GDP ratio 来衡量国家的债务水平。

Table 8.3 Debt as % of Gross Domestic Product in 2020

Country	In 2010	In 2020
Venezuela	25.00%	304.13%
Japan	205.69%	254.13%
Greece	147.49%	211.21%
Italy	119.20%	155.82%
Portugal	100.21%	135.19%
United States	95.14%	133.92%
Spain	60.52%	119.92%
Cyprus	56.43%	119.14%
Canada	81.22%	117.46%
France	85.26%	115.08%
Belgium	100.27%	114.14%
Montenegro	45.01%	107.15%
United Kingdom	74.27%	104.47%
Mauritius	57.15%	99.91%
Brazil	62.43%	98.09%
Egypt	69.59%	89.84%
India	66.40%	89.61%
El Salvador	60.04%	89.17%
Croatia	57.68%	88.74%
Saint Vincent and the Grenadines	65.45%	85.00%

Source: IMF

- 养老金／社会服务承诺 **Pensions/Social Service Commitments**: 社会福利好的国家，可能应具有更高的违约风险。
- 政府收入／流入 **Revenues/Inflows to government**: 政府收入通常来自税收，这取决于税法和税基。较大的税基可以增加潜在税收收入，从而用于偿还债务。
- 收入的稳定性 **Stability of revenues**: 债务必须在好坏时期都要偿还。因此，收入流更稳定的国家违约风险较低。
- 政治风险 **Political risk**: 专制政权可能比民主政权更容易违约。此外，中央银行的独立性和权力也会影响违约风险。
- **Implicit backing** 隐性支持 **from other entities**

• 4. Sovereign Credit Ratings

【评级机构通常提供本币评级 (**local currency ratings**) 和外币评级 (**foreign currency ratings**)。】

- 本币评级：对一个国家本币债务违约的可能性的意见。
- 外币评级：对其外币债务的违约可能性的意见。
- 本币评级和外币评级的差异主要基于一个国家的货币政策独立性 (**monetary policy independence**)。
 - * 失去货币政策独立性的国家（例如欧盟国家）其本币和外币评级将趋于一致。
 - * 拥有浮动汇率制度并可以通过稳健的国内市场进行借款的国家，其本币和外币评级差异最大，通常本币评级较高。
- 评级机构的两种方法：
 - * 上调法 (**Notch-up approach**)
 - 外币评级是主权违约风险的关键指标，本币评级在此基础上进行上调。
 - * 下调法 (**Notch-down approach**) :

· 本币评级是主权违约风险的关键指标，外币评级在此基础上进行下调。

– Shortcomings of the sovereign rating systems:

- * Ratings are biased upward.
- * Herd behavior 羊群行为 当一个评级机构上调或下调一个国家的评级时，其他评级机构往往会跟随。
- * Too little, too late. 评级变化不及时。
- * Vicious Cycle 恶性循环
- * Ratings failures 评级失败

• 5. 衡量主权债务风险

【评级机构的主权债务评级可以在一定程度上衡量主权债务风险，有缺陷。可以采用更加基于市场的衡量方式：Sovereign Default Spread、CDS spread】

– (1) Sovereign Default Spread 主权违约利差

- * The credit spread for sovereign debt in a given currency is the excess interest paid over the risk-free rate in that currency.
- * 这一利差反映了市场对该国的信用风险。
- * 例如，2022 年 7 月 1 日，巴西政府发行了一笔 10 年期美元计价债券，市场利率为 6%。与此同时，美国政府 10 年期国债的利率为 3.02%。
 · Sovereign Default Spread = $6\% - 3.02\% = 2.98\%$
- * 主权违约利差可以在一定程度上反映一国偿还债务能力
- * 主权违约利差与评级之间存在着很强的相关性。

Default Spreads on US \$D enominated Bonds-Emerging Markets in July 2022

Country	Moody's Rating	\$ 10-year Bond Rate	US T.Bond Rate	Default Spread
Brazil	Ba2	6.19%	3.02%	3.17%
Chile	A1	4.33%	3.02%	1.31%
Colombia	Baa2	5.58%	3.02%	2.56%
Indonesia	Baa2	4.38%	3.02%	1.36%
Mexico	Baa1	5.07%	3.02%	2.05%
Peru	Baa1	4.76%	3.02%	1.74%
Turkey	B2	10.92%	3.02%	7.90%

* 比较：

	Sovereign Default Spread	Sovereign Credit Ratings
相同点	都可以反映一国偿还债务的能力	
不同点	more granular 颗粒度 than ratings	评级较稳定（评级稳定性是评级机构的目标之一）
	比评级更快地适应新信息	
	不稳定	

– (2) CDS spread (credit default swap spread)

- * CDS spread 也可以作为主权债务的违约风险
- * 在这里 CDS 其参考实体是主权国家政府的债务，而不是公司债券。例如，如果购买了秘鲁政府债券的 CDS，那么秘鲁政府就是这个 CDS 合同中的参考实体。
- * 如果参考实体（某个主权政府）风险大，CDS spread 大；

- * 如果参考实体（某个主权政府）风险小，CDSspread 小；
- * 优点：
 - CDS 利差基于市场交易，反映市场看法，可以比信用评级更好地预测主权风险
- * 缺点：
 - 首先，CDS 利差反映了包括信用风险、市场风险和流动性风险在内的多种风险，单纯依靠 CDS 利差评估信用风险可能会高估风险。
 - 其次，CDS 市场常常存在流动性不足的问题，这可能扭曲其价格。

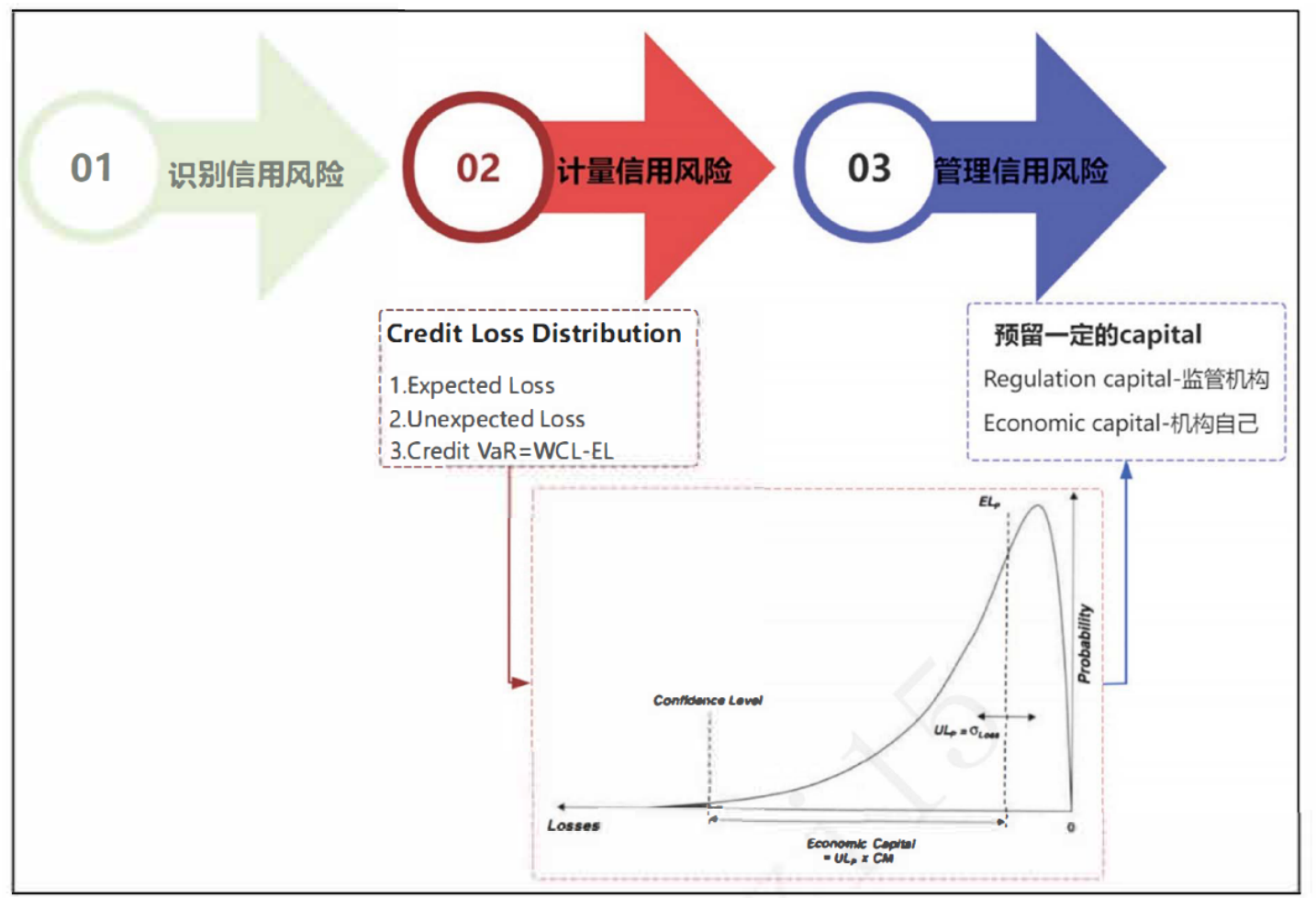
3 信用风险计量及资本-信用 VaR 和资本

考点 1 Credit loss Distrihution CR-4

1 EL、UL、ULC（重要的不能再重要了）

(1) 预期损失 (Expected Loss, EL): 指在未来一段时间 (通常是一年) 可能发生的平均损失。也就是损失的期望值。	
单个资产: $EL(\%) = PD * LGD$ $EL(\$) = PD * LGD * EAD$ 【主要用这个公式】	PD → 信用评级、CS、莫顿模型 计算 PD $LGD = \text{loss rate} = 1 - \text{回收率}$ EAD, 也写作 exposure amount (EA)
组合: $EL_p = \sum_{i=1}^n PD_i \times LGD_i \times EAD_i$ 两资产组合: $EL_p = PD_1 \times LGD_1 \times EAD_1 + PD_2 \times LGD_2 \times EAD_2$	和资产之间的相关性 ρ 无关
(2) 非预期损失 (Unexpected Loss, UL): UL is the standard deviation of credit losses, that is, the standard deviation of actual credit losses around the expected loss average (EL). 切记: 在这个科目中, UL 的概念, 与一级以及二级其他科目中的 Unexpected Loss 概念不同。	
单个资产: $UL = EAD \times \sqrt{PD_i \times \sigma_{LGD}^2 + LGD^2 \times \sigma_{PD}^2}$ Bernoulli 伯努利分布 $\rightarrow \sigma_{PD}^2 = PD \times (1 - PD)$	σ_{LGD}^2 : LGD 的方差 σ_{PD}^2 : PD 的方差
组合: $UL_p = \sum_i \sum_j \rho_{i \square j} \times UL_i \times UL_j$ 两资产组合: $UL_p = \sqrt{UL_1^2 + UL_2^2 + 2 \times \rho_{1 \square 2} \times UL_1 \times UL_2}$ 若单个资产的 UL 相等, 相关系数相等 $\rightarrow UL_n = UL \times \sqrt{n + n \times (n - 1) \times \rho}$	$\rho_{i \square j}$: 资产 i 和 j 的违约相关性
(3) Unexpected Loss Contribution (ULC): ULC 是指单个资产 (如 loan) 对整个 UL 的边际贡献。即衡量一个特定资产在整个组合可能出现的损失中占了多大的份额。【重点掌握两资产】	
$ULC_i = \frac{UL_i \times \sum_j \rho_{i \square j} \times UL_j}{UL_p}$	
两资产: $ULC_1 = \frac{UL_1^2 + \rho_{1 \square 2} \times UL_1 \times UL_2}{UL_p}$; $ULC_2 = \frac{UL_2^2 + \rho_{1 \square 2} \times UL_1 \times UL_2}{UL_p}$; $ULC_1 + ULC_2 = UL_p$	
前提: n 笔 loan; 每笔 loan 规模相同、风险特质相同 $ULC_i = \frac{UL_p}{n} = UL_i \times \sqrt{\frac{1}{n} + \rho \times (1 - \frac{1}{n})}$ 前提: n 比较大时 $ULC_i = UL_i \times \sqrt{\rho}$	the correlation between assets increases, the bank suffers from concentration risk 集中度风险。

2 Credit loss distribution and capital



- Capital multiplier(资本乘数, CM) is dependent on the overall credit quality of the portfolio and the confidence level.
 - 信用质量变差, CM 变大
 - confidence level 置信水平上升, CM 变大
- Credit Loss Distribution 信用损失分布是一个概率分布, 描述在给定的时间段内 (通常为一年) 可能的信用损失及其发生的概率。
 - 信贷风险不是正态分布的, 而是高度倾斜 (highly skewed) 的, 因为上升潜力仅限于最多收到承诺的付款, 在极少数情况下会损失大量资金。【收益有限, 比如利息; 损失比较大, 比如本金】
 - 信用损失的期望—预期损失 (Expected Loss, EL)
 - 信用损失的标准差—非预期损失 (Unexpected Loss, UL)
- 可以用 beta distribution 建模信用风险损失。【关于 beta 分布, 无需了解太深】
 - 尾部拟合工作的最佳方法是将分析 (贝塔分布) 解决方案与蒙特卡罗模拟等数值程序相结合。The tail-fitting exercise is best accomplished by combining the analytical (beta distribution) solution with a numerical procedure such as a Monte Carlo simulation.
 - 在选定的置信水平下, 利用贝塔函数的倒数, 我们可以确定 CM, 即资本乘数。Taking the inverse of the beta function at the chosen confidence level, we can determine CM, the capital multiplier. 给定置信水平, 拟合完分布, 倒求出资本乘数

3 Problems with the Quantification of Credit Risk

- 只考虑违约，没考虑价值变化 This approach assumes that credits are illiquid assets.
- More liquid asset: a value approach would be more suitable.
- 借款人信用质量会变 Multi-period nature of credits: the expected and unexpected changes in the credit quality of the borrowers (and their correlations).
- 只考虑信用风险 All other risk components (such as market and operational risk) are separated.

考点 2 Credit Value at Risk CR-10

1 Vasicek model

- 只考虑违约情况，不考虑评级下调情况
- Vasicek's Gaussian copula model is a way of calculating high percentiles of the distribution of the default rate for a portfolio of loans.
- WCDR(T, X): the Xth percentile of the default rate distribution during a period of length T .
- 也叫 worst case default rate 最差情况违约概率
参数: probability of default in time T; PD; credit correlation(ρ)
- 模型的目标: 找到一个糟糕市场情况下的条件概率
- The Basel Committee sets $X = 99.9$ or regulatory capital in the internal ratings-based approach(IRB, 内部评级法).
- Capital requirement(Credit VaR)=(WCDR- PD) $\times$ EAD $\times$ LGD

$$WCDR(T, X) = N\left(\frac{N^{-1}(PD) - \sqrt{\rho}N^{-1}(X)}{\sqrt{1-\rho}}\right) N\left(\frac{N^{-1}(PD) - \sqrt{\rho}N^{-1}(0.001)}{\sqrt{1-\rho}}\right)$$

WCDR * LGD * EAD is the loss at this 99.9 percentile point.

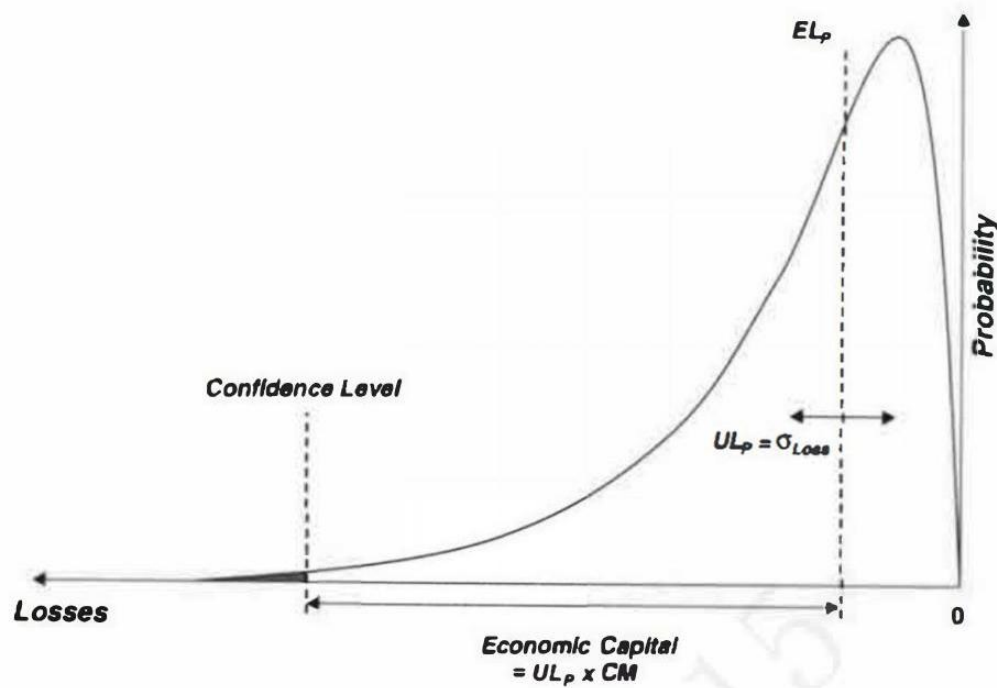
PD * LGD * EAD is the expected loss.

- For a large portfolio with n small loans, the X th percentile of the loss distribution can be approximated as:

组合: n 笔贷款，每笔贷款只是组合的一小部分

$$\sum_{i=1}^n WCDR_i(T, X) \times EAD_i \times LGD_i$$

- 巴塞尔协议二 【在操作风险这个科目会进行更详细的介绍】
IRB 基础法: 银行自行估计:PD; 巴塞尔规定:EAD、LGD 和 MA
IRB 高级法: 全部由银行自行估计
但 ρ 均由巴塞尔规定给出



2 CreditMetrics Model

- 由 JPMorgan 于 1997 年提出
- 考虑了降级和违约的情况 (不仅仅只是违约情况), 通过使用评级转换矩阵 rating transition matrix 进行蒙特卡罗模拟, 最终计算出跨所有对手方 (相关性) 的 Credit Loss Distribution 信用损失分布 $\rightarrow$ VaR。
- 评级可以是银行的内部评级或评级机构提供的评级。
- One-Year Ratings Transition Matrix

Initial Rating	Rating at Year-end							
	AAA	AA	A	BBB	BB	B	CCC/C	D
AAA	89.85	9.35	0.55	0.05	0.11	0.03	0.05	0.00
AA	0.50	90.76	8.08	0.49	0.05	0.06	0.02	0.02
A	0.03	1.67	92.61	5.23	0.27	0.12	0.02	0.05
BBB	0.00	0.10	3.45	91.93	3.78	0.46	0.11	0.17
BB	0.01	0.03	0.12	5.03	86.00	7.51	0.61	0.70
B	0.00	0.02	0.08	0.17	5.18	85.09	5.66	3.81
CCC/C	0.00	0.00	0.12	0.20	0.65	14.72	50.90	33.42
D	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

- 使用 CreditMetrics 计算 Credit VaR

蒙特卡罗模拟: 结合评级转移矩阵; 每次模拟试验中, 确定下一期信用评级。→ 得出评级及对应概率

计算信用风险时, 需要考虑不同信用评级类别的信用利差期限结构 the term structure of credit spreads → 结合无风险利率, 得出折现率

可以直接使用现有市场数据代表信用利差, 或者使用综合指数预测的信用利差。

当考虑多个交易对手, 可以通过高斯 copula 模型来建模违约相关性

- 原版书例题: 【说明, 在这个例题中, 通过蒙特卡罗模拟已得出下一期的评级变化; 关于信用利差的期限结构也已经给出; 只有一个债券, 无需考虑相关性。最后得出 → 信用损失分布】

- Consider a company that owns a single two-year zero-coupon bond with a principal of \$1, 000.

◇ bond's current rating is BB

\) the risk-free rate is 3%

the current credit spread is 200 basis points

Rates are expressed with annual compounding.

The current price of the bond is $1000 / (1.05^2) = \$907.03$.

- during the next month there is

* a 0.6% chance that it will increase to BBB,

* a 98.5% chance that it will stay the same,

* a 0.8% chance that it will decrease to B, and

a 0.1% chance that it will default. If it defaults, the bond will be worth \$400.

对于每个可能的评级类别, 有三个等可能的信用利差 credit spreads 分别为:

* B: 400bps、450bps、500bps

* BB: 160bps、200bps、240bps

* BBB: 80 bps、100bps、120bps

分析: 时间变化为 $23/12 = 1.917$ 年

Default: 概率 0.1%, $\text{loss} = 907.03 - 400 = 507.03$

B 评级: 500bps 情况, 概率 $0.8\%/3 = 0.267\%$,

债券价值 $= 1000 / (1.08^{1.917}) = 862.85$, $\text{loss} = 907.03 - 862.85 = 44.17$

同理可得下表, 显示了所有可能的结果及其概率 → 信用风险损失分布

Rating	Spread (bp)	Probability	Bond Value (\$)	Loss (\$)
Default		0.100%	400.00	507.03
B	500	0.267%	862.85	44.17
B	450	0.267%	870.56	36.47
B	400	0.267%	878.38	28.65
BB	240	32.833%	904.11	2.92
BB	200	32.833%	910.72	-3.70
BB	160	32.833%	917.41	-10.38
BBB	120	0.200%	924.17	-17.14
BBB	100	0.200%	927.58	-20.55
BBB	80	0.200%	931.01	-23.98

表中显示, 如果置信水平高于 99.9%, VaR 为 507.03 美元;

如果置信水平在 99.9% 到 99.633% 之间, VaR 为 44.17 美元; 以此类推。当置信水平为 99% 时, VaR 为 2.92 美元。

3 Credit Risk Plus (CreditRisk+)

- Credit Risk Plus: 瑞士信贷金融产品 (Credit Suisse Financial Products) 在 1997 年提出
- 预测与保险公司的违约模型类似
- 只考虑违约情况, 不考虑评级下调情况
- 模型假设: 不同贷款或债务的违约事件是独立发生的。
- CreditRisk+ 模型中使用多种分布:

(1) 二项分布 Binomial Distribution 和泊松分布 Poisson Distribution 【一级数量分析内容】

对于每笔贷款, 假设其在一定时间内违约的概率为 q , 利用二项分布 Binomial Distribution 可以估算出在给定的 n 笔贷款中违约的数量 m 。

$$P(X = m) = C_n^m q^m (1 - q)^{n-m}$$

$$\Leftrightarrow E(X) = nq$$

$$\Leftrightarrow \text{Var}(X) = np(1 - q)$$

当贷款数目 n 很大且每笔贷款的违约概率 q 很小时，二项分布的概率分布形状会趋向于泊松分布，泊松分布更加简单且方便计算。 $(\lambda = nq)$

$$P(X = m) = \frac{(nq)^m e^{-nq}}{m!}$$

(2) 伽马分布 gamma Distribution 和负二项分布 negative binomial distribution: - the default rate(q) 不确定, default rates vary greatly from year to year.

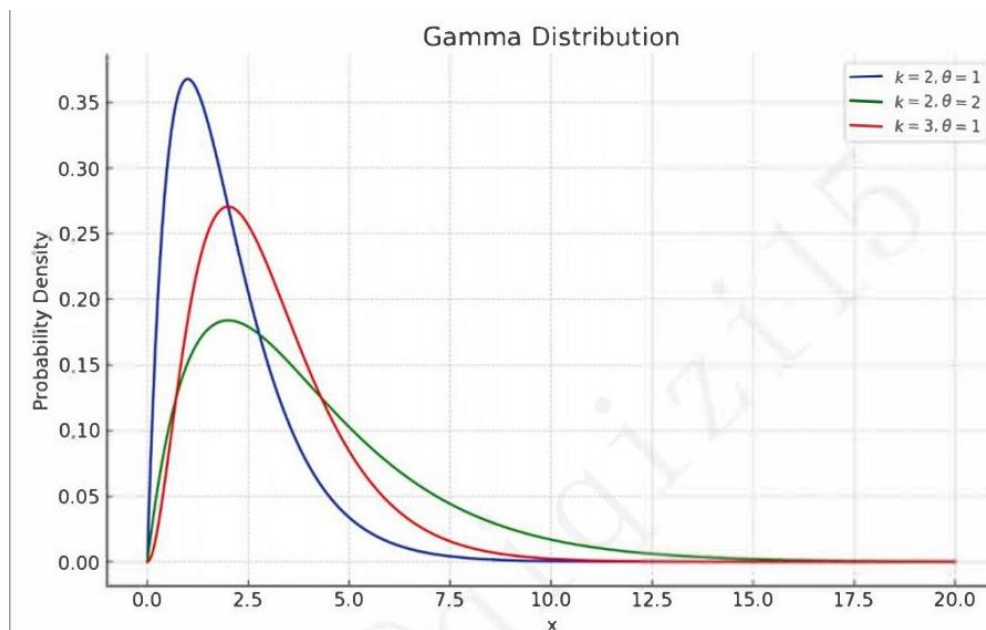
假设 the expected number of defaults(qn), 服从伽马分布 gamma Distribution(mean μ and standard deviation σ), 那么 The Poisson distribution then becomes a negative binomial distribution(负二项分布)

$$\text{Prob}(m \text{ defaults}) = p^m (1 - p)^\alpha \frac{\Gamma(m + \alpha)}{\Gamma(m + 1) \Gamma(\alpha)}$$

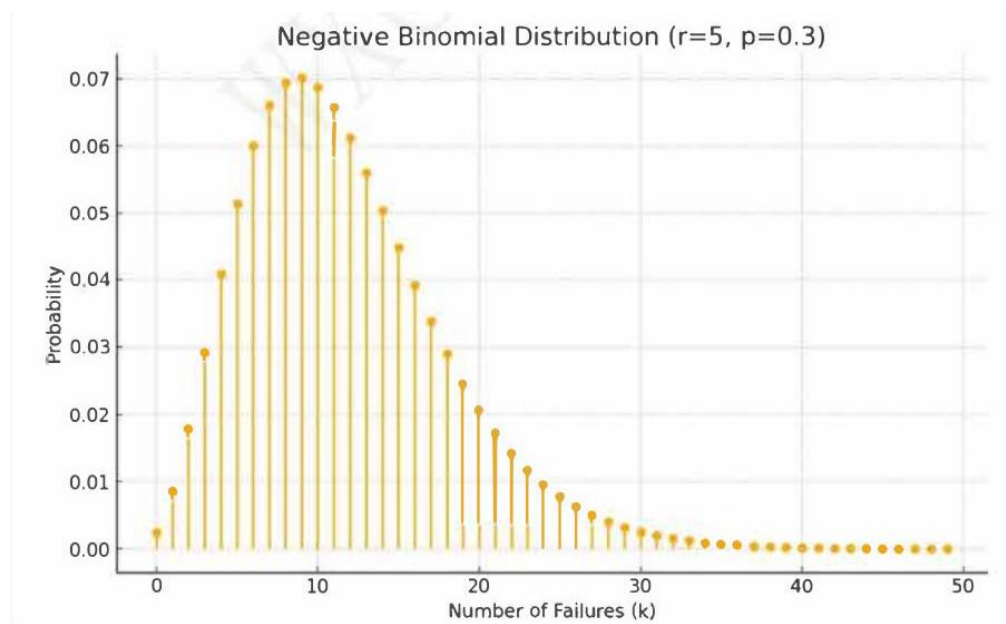
【公式不要求掌握，主要看里面的参数，及对应的性质】

σ [4] the negative binomial distribution = Poisson distribution.

σ (1the likelihood of experiencing a large number of defaults increases. 拓展-gamma 分布图形:



负二项分布图形:



考点 3 Default Correlation and Credit VaR CR-11-12

1 Default correlation 违约相关性

- 衡量多个实体同时违约的趋势 Measures the tendency of multiple entities to default simultaneously
- 一家公司的违约可能引发另一家公司的违约，这被称为信用传染效应 (credit contagion effect)。
- 联合违约概率 joint default probability (π_{12})：在时间范围 T 内两者都违约的概率

$$\rho = \frac{\text{Cov}(X_1 X_2)}{\sigma_{X_1} \sigma_{X_2}} = \frac{\pi_{12} - \pi_1 \pi_2}{\sqrt{\pi_1 (1 - \pi_1)} \sqrt{\pi_2 (1 - \pi_2)}}$$

π_1 ：资产 1 的违约概率

π_2 ：资产 2 的违约概率

拓展推导过程：

一级数量科目，相关系数 $\rho = \frac{\text{Cov}(X_1 X_2)}{\sigma_{X_1} \sigma_{X_2}}$

$E(X_i) = \pi_i$; $E(X_1 X_2) = \pi_{12}$

$\text{Var}(X_i) = E(X_i)^2 - [E(X_i)]^2 = \pi_i (1 - \pi_i) \quad i = 1, 2$

$\text{Cov}(X_1, X_2) = E(X_1 X_2) - E(X_1) E(X_2) = \pi_{12} - \pi_1 \pi_2$

关于 $E(X_1 X_2) = \pi_{12}$ ，见下表： $E(X_1 X_2) = 0^* (1 - \pi_1 - \pi_2 + \pi_{12}) + 0^* (\pi_1 - \pi_{12}) + 0^* (\pi_2 - \pi_{12}) + 1^* \pi_{12} = \pi_{12}$

所以：

$$\rho = \frac{\text{Cov}(X_1 X_2)}{\sigma_{X_1} \sigma_{X_2}} = \frac{\pi_{12} - \pi_1 \pi_2}{\sqrt{\pi_1 (1 - \pi_1)} \sqrt{\pi_2 (1 - \pi_2)}}$$

结果	X_1	X_2	$X_1 X_2$	概率
No default	0	0	0	$1 - \pi_1 - \pi_2 + \pi_{12}$
Firm 1 only defaults	1	0	0	$\pi_1 - \pi_{12}$
Firm 2 only defaults	0	1	0	$\pi_2 - \pi_{12}$
Both firms default	1	1	1	π_{12}

2 Credit VaR

- **Market VaR vs. Credit VaR** Credit VaR = WCL - EL

Worst case loss (WCL): Loss at the confidence level - Credit risk VaR & Market risk VaR

	Credit risk VaR	Market risk VaR
Time horizon	One-year time horizon	One-day time horizon
Tools	More elaborate model 更复杂的模型, such as the Vasicek model	Main tool: historical simulation

- **违约相关系数如何影响 Credit VaR**

- 如果一个信贷组合的违约相关性等于 1, 那么可以把这个组合看成一个资产。
No credit diversification is achieved. 没有分散化效果
- 如果违约相关性等于 0, 那么投资组合中的违约次数就是二项分布随机变量。可以实现显著的信用分散。 If default correlation is equal to 0, then the number of defaults in the portfolio is a binomially distributed random variable. Significant credit diversification may be achieved.

- **Effect of Granularity on Credit VaR**

- “Granular 颗粒度”指的是增加组合中独立信贷的数量, 并减少每个信贷在整个组合中所占的比例。
组合越“granular”, 就意味包含更多的独立小额信贷, 每个信贷占组合的比重越小。
- 粒度是对投资组合集中风险的衡量; Granularity is a measurement of the portfolio's concentration risk;
- 对于给定的违约概率, 信用风险值会随着信用组合的粒度增大而减小。[信用风险值 For a given default probability, Credit VaR decreases as the credit portfolio becomes more granular. [granular] Credit VaR
- 在违约概率较高的情况下, 收敛更为剧烈。The convergence is more drastic with a high default probability. 【违约概率越高, Credit VaR 下降越多】
- 如果违约概率较低, 则更难通过提高投资组合的颗粒度来降低 VaR。违约概率越低, 信用风险值越高 It is harder to reduce VaR by making the portfolio more granular, if the default probability is low. 【违约概率越低, Credit VaR 下降的比较少, 甚至很难下降】 - 其他计量违约相关性的模型 1) Reduced Form Models

3 Single-factor model

(1) Model Assumption

- 公司的 asset return is represented as a function of two random variables:

m : the return on a "market factor" that captures the correlation between default and the general state of the economy.

ε : shock capturing idiosyncratic risk.

- Assume that m and ε are standard normal variables, and are not correlated with one another:

$$\alpha_i = \beta_i \times m + \sqrt{1 - \beta_i^2} \times \varepsilon_i$$

$$\alpha \sim N(0, 1)$$

$$E(\alpha) = \beta E(m) + \sqrt{1 - \beta^2} E(\varepsilon) = 0$$

$$\sigma^2(\alpha) = \beta^2 \sigma^2(m) + (1 - \beta^2) \sigma^2(\varepsilon) + 2\beta\sqrt{1 - \beta^2} \text{cov}(m, \varepsilon) = 1$$

β_i : own correlation between market value.

- 例如，当 $\alpha < -2.33$ 时，公司发生违约。由于 α 服从标准正态分布，所以 $P(\alpha < -2.33) = 1\%$ ，即违约概率为 1%。
- Asset return correlation of the two firms $= \beta_i \beta_j$
- 原版书例题: Suppose firm 1 is "cyclical" and has $\beta_1 = 0.5$, while firm 2 is "defensive" and has $\beta_2 = 0.1$. The asset return correlation of the two firms is then $\beta_1 \beta_2 = 0.5 \times 0.1 = 0.05$.

(2) unconditional PD 与 conditional PD

- 当 m 未知时，通过单因素模型得出的是非条件违约概率 (unconditional PD)。
- 当 m 已知时，即市场因子 m 设定为常数 ($\bar{m}$ ，对市场情况有一定预期)，通过单因素模型得出的是条件违约概率 (conditional PD)。

$$\alpha = \beta \bar{m} + \sqrt{1 - \beta^2} \varepsilon$$

其中 $E(\alpha) = \beta \bar{m}$, $\sigma^2(\alpha) = 1 - \beta^2 \rightarrow \alpha$ 服从均值为 $\beta \bar{m}$ ，方差为 $1 - \beta^2$ 的正态分布。

- 违约概率 $PD = N\left(\frac{k_i - \beta_i \bar{m}}{\sqrt{1 - \beta_i^2}}\right)$ $i = 1, 2, \dots$ 【这其实是标准化过程】

default threshold、债务因子: k

具体得出多少概率，需要查标准正态分布表

- 例子: Given a firm with beta of 0.5, and default threshold of -2.33, the unconditional PD is $N(-2.33) = 1\%$. If the market factor is -0.5, the conditional mean is $= 0.5 \times (-0.5) = -0.25$ and conditional variance is $1 - 0.5^2 = 0.75$, therefore $\text{std} = 0.8660$. The conditional cumulative PD is?

分析:

$$PD = N\left(\frac{-2.33 - (-0.25)}{0.8660}\right) = N(-2.4) = 0.82\%$$

- 性质

the market factor shock happens to be zero	→ unconditional PD = conditional PD
或者 the firm's returns are independent of the state of the economy	
条件方差比无条件方差 1 要小	
The default outcomes for two different firms i and j are independent	

– Impact of correlation parameter

$\beta \rightarrow 1$ (perfect correlation)

$m < k$, nearly all the credits default

$m > k$, almost none default

$\beta \rightarrow 0$ (zero correlation):

If there is no statistical relationship to the market factor, so idiosyncratic risk is nil 不存在的, then the loss rate will very likely be very close to the default probability p .

In less extreme cases, a higher correlation leads to a higher probability of either very few or very many defaults, and a lower probability of intermediate outcomes.

4 交易对手方风险

考点 1 Derivatives CR-14

1 Derivatives market

• Counterparty credit risk

- Derivatives transactions: represent contractual agreements either to make payments or to buy or sell an underlying security at a time or times in the future.
 - * Financial institutions, mainly banks, provide derivative contracts to their end user clients and hedge their risks with one another.
 - * are dominated by a relatively small number of large counterparties(dealers) 许多市场都被数量较少的大型交易方所控制
- Counterparty credit risk: the possibility that a counterparty may not meet its contractual requirements under the contract when they become due.

• OTC vs. Exchange

	Exchange-traded	Over-the-counter (OTC)
Terms of contract	Standardised (maturity, size, strike, etc.)	Flexible and negotiable
Maturity	Standard maturities, typically at most a few months	Negotiable and non-standard, often many years
Liquidity	Very good	Limited and sometimes very poor for non-standard or complex products

- **Clearing**

- **Execution:** buyer and seller enter into a legal obligation to buy/sell securities or another underlying
- **Clearing:** Transaction is managed prior to settlement (margining, cashflow payments, etc.)
- **Settlement:** Transaction is settled via exchange of securities and/or cash and legal obligations are therefore fulfilled.
- Since it requires a certain amount of standardisation, not all OTC derivatives can be centrally cleared.
- The main OTC derivatives are: interest rate (and inflation) derivatives, foreign exchange derivatives, equity derivatives, commodity derivatives, and credit derivatives

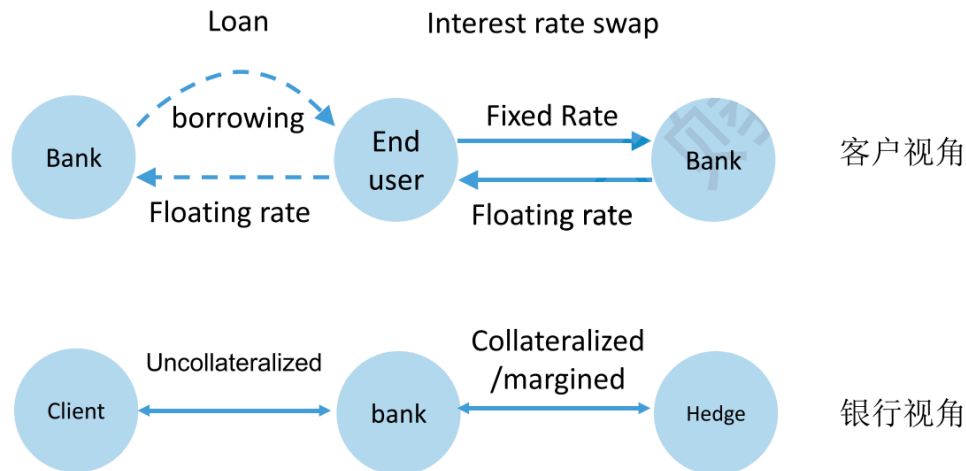
- **Participants**

- **Large player:** large global bank, often known as a dealer.
 - * Have a vast number of derivatives trades on their books and have many clients and other counterparties
 - * Trade across all asset classes and will collateralize positions
 - * Will be a member of most or all exchanges and CCP to facilitate trading on their own account and for their clients.
- **Medium-sized player:** a smaller bank or other financial institution
 - * Smaller number of clients and counterparties
 - * Will be a member of local exchanges and may be a member of a number of global ones
- **End user:** a large corporate, sovereign, or smaller financial institution with derivatives requirements
 - * Overall position will be very directional
 - * Unwilling to commit to margining or posting collateral
- **Third parties:** may offer settlement/margining/collateral management, software, trade compression, and clearing services.

- **Collateral**

- Exchange-traded derivatives: are effectively settled on a daily.
 - * most simple, liquid, and short-dated derivatives. Margin must be posted against losses on a daily basis
- OTC derivatives: may be collateralized
 - * OTC centrally cleared: Daily collateralization in cash (variation margin)
 - * OTC collateralized: bilateral OTC derivatives. Parties post collateral(which may be cash or securities) to one another in order to mitigate the counterparty risk.
 - * OTC uncollateralized: parties do not post collateral (or post less and/or lower quality collateral)
- Banks and end users

- * 头寸具有方向性，盯市价值及现金流波动大 The overall portfolio will typically be directional. The value of the derivatives, or mark-to-market(MTM)volatility will be significant and any associated margin or collateral flows may vary substantially.
- * 一对一对冲敞口而非整体宏观对冲敞口 End users may hedge risks on a one-for-one basis rather than macro basis.
- * The hedging needs may lead to certain imbalances



• ISDA

- The International Swaps and Derivatives Association (ISDA) is a trade organization for OTC derivatives practitioners.
 - * Standard legal documentation to increase efficiency and reduce aspects such as counterparty risk
 - * The ISDA Master Agreement 主协议: the market standard for OTC derivative documentation
- Master agreement
 - * Bilateral framework which contains terms and conditions
 - * Multiple transactions will be covered under a general Master Agreement to form a single legal contract of an indefinite term, covering many or all of the transactions.
 - * Core section 核心部分 + a schedule containing adjustable terms 附表组 (可调整条款)
 - * 合约条款涉及: netting, collateral, termination events, the definition of default, and the close-out process
- 交易对手风险视角，ISDA 主协议具有以下降低风险的特点：
 - * The contractual terms regarding the posting of collateral 有关抵押品的合同条款
 - * Events of default and termination 违约事件和终止
 - * All transactions referenced are combined into a single net obligation 所有参考交易合并为单一净债务
 - * The mechanics around the close-out process are defined 确定平仓机制
- ISDA 主协议涵盖的违约事件有：
 - * Failure to pay or deliver 未付款或未交付
 - * Breach of agreement 违背合约
 - * Credit support default(collateral terms) 信用支持违约
 - * Misrepresentation 错误陈述
 - * Default under the specified transaction 特定交易下的违约
 - * Cross-default(default on another obligation) 交叉违约
 - * Bankruptcy
 - * Merger without assumption 不承担义务的合并

- May lead to the loss for the nondefaulting party:
 - * The total value of the transactions to the nondefaulting party is positive and greater than the collateral (if any) posted by the defaulting party. 非违约方价值为正，但违约方缴纳的抵质押物不足
 - * The total value of the transactions is positive to the defaulting party and the collateral posted by the nondefaulting party is greater than this value. 违约方价值为正，未为违约方缴纳抵质押物超过合约价值

2 CCP and modeling derivatives risk

- **Systemic risk:** A major concern with respect to OTC derivatives is systemic risk, which generally refers to an uncontained crisis which may cause the failure of an entire financial system or market. 系统性风险一般是指可能导致整个金融系统或市场崩溃的无法控制的危机
- **Special purpose vehicle 特殊目的实体** Special purpose vehicle (SPV) or special purpose entity (SPE) is a legal entity(e.g. a company or limited partnership) created typically to isolate a firm from financial risk.
 - To isolate counterparty risk.
 - SPVs aim essentially to change bankruptcy rules
 - * Shift priorities 改变清偿顺序
 - An SPV transforms counterparty risk into legal risk.
 - * A bankruptcy court to combine the SPV assets with those of the originator
- **Derivatives product companies 衍生品公司**
 - DPCs are generally triple-A-rated entities set up by one or more banks as a bankruptcy-remote subsidiary of a major dealer 由一家或多家银行设立的三 A 级实体，作为主要交易商的破产远程子公司
 - * Provides external counterparties with a degree of protection against counterparty risk by protecting against the failure of the DPC parent. 通过防止 DPC 母公司倒闭
 - Maintained a triple-A rating
 - * Minimizing market risk(Mirror trades: matched book)
 - * Support from a parent: If the parent were to default, then the DPC would either pass to another well-capitalized institution or be terminated
 - * Credit risk management and operational guidelines: Restrictions were also imposed on(external) counterparty credit quality and activities(position limits, margin, etc.)
 - However, was such a triple-A entity of a double-A or worse bank really a better counterparty than the bank itself?
 - * The GFC essentially killed the already-declining world of DPCs.
 - * A DPC's fate inextricably with that of its parent. DPCs 的命运与其母公司的命运密不可分地联系在一起
- **Monolines and CDPCs 单一险种公司**
 - Monoline insurance companies: were financial guarantee companies with strong credit ratings that were utilized to provide 'credit wraps', which are financial guarantees.
 - Credit derivative product companies(CDPCs):were an extension of the DPC concept that had business models similar to those of monolines.
 - When the GFC developed through 2007, monolines experienced major problems due to the MTM-based valuation losses on the insurance they had sold.
- **Central clearing of OTC derivatives**
 - From the late 1990s, several major CCPs began to provide clearing and settlement services for OTC derivatives and other non-exchange-traded products (repo).

- * These OTC transactions are still negotiated privately and off-exchange, but are then novated into a CCP on a post-trade basis. 这些场外交易仍然是私下和场外谈判的，但在交易后被更替到中央交易对等中心
- In 2009, the G20 leaders agreed in Pittsburgh to require:
 - * All standardized OTC derivatives to be traded on exchanges or electronic platforms
 - * Mandatory central clearing of standardized OTC derivatives
 - * The reporting of OTC derivatives to trade repositories 向交易资料库报告场外衍生工具
 - * Higher capital requirements for non-centrally cleared OTC derivatives. 提高非集中清算场外衍生品的资本要求
- Central clearing of OTC derivatives- the roles
 - * To sets certain standards and rules for its clearing members 制定一定的标准和规则
 - * To take responsibility for closing out all the positions of a defaulting clearing member 负责平仓违约会员头寸
 - * To maintain financial resources to cover losses in the event of a clearing member default:
 - Variation margin: closely track market movements
 - Initial margin: cover the worst-case liquidation or close-out costs above the variation margin
 - A default fund: mutualize losses in the event of a severe default
 - * To have a documented plan for the very extreme situation when all their financial resources (initial margin and the default fund) are depleted 全部耗尽
 - Additional calls to the default fund
 - Variation margin haircutting
 - Selective tear-up of positions
 - * CCPs provide a number of benefits: Allow netting of all trades executed through CCPs
 - CCPs manage margin requirements from their members to reduce the risk associated with the movement in the value of their underlying portfolios.
 - The CCP absorbs the 'domino effect' by acting as a sort of financial shock absorber 充当金融冲击吸收器来吸收“多米诺骨牌效应”(Auctioning the defaulted members' positions)
- Bilateral margin requirements
 - Whilst clearing is mandatory, it can be avoided by trading contracts that are not sufficiently standardized to be cleared
 - Regulators introduced mandatory margin(collateral) rules for derivatives that would remain bilateral. 双边合约也进入了强制保证金制度
- Derivatives risk modelling
 - VaR
 - * VaR does not give an indication of the possible loss outside the confidence level chosen
 - * VaR is that it is not a coherent risk measure (not behave in a sub-additive fashion)
 - ES: not ignoring completely the impact of large losses; coherent risk measure
 - Models: the financial markets have a somewhat love/hate relationship with mathematical models.
 - * The quantitative modelling of counterparty risk is complex and prone to misinterpretation and misuse.
 - * Counterparty risk involves looking years into the future rather than just a few days, which creates further complexity that is not to be underestimated. 交易对手风险涉及未来数年而非短短几天
 - Correlation and dependency: counterparty risk takes difficulties with correlation to another level.

考点 2 Counterparty Risk CR-15

1 Definition of counterparty risk

- Counterparty credit risk is the risk that the entity with whom one has entered into a financial contract (the counterparty) will fail to fulfil their side of the contractual agreement.
 - Over-the-counter (OTC) derivatives
 - Bilaterally cleared
 - Uncollateralized

2 Lending risk vs. counterparty risk

- Lending risk:
 - Notional amount at risk during lending period is usually known with a degree of certainty.
 - Only one party takes lending risk. (i.e. bondholder).
- Counterparty risk:
 - The value of the contract in the future is uncertain.
 - Since the value of the contract can be positive or negative, counterparty risk is typically bilateral.

3 Settlement, pre-settlement, and margin period of risk

- Pre-settlement risk: a counterparty will default prior to the final settlement of the transaction. (counterparty risk)
 - Long-dated OTC derivatives
- Settlement risk: arises at settlement times due to timing differences between when each party performs its obligations under the contract.
 - Occurs for only a small amount of time(days or even hours)
 - A very large exposure(potentially 100of the notional)
 - Spot contracts 现货合同；FX markets 外汇市场
- Margin period of risk(MPoR): the risk horizon when collateralized/margined
 - Collateralized bilateral OTC derivatives: 10 business days
 - Centrally-cleared OTC derivatives: 5 business days

4 Institutions and transactions

- Large global banks have exposure to all asset classes
- Smaller banks may have more limited exposure
- End users may also have limited exposure
- Counterparty risk faced with CCPs may receive attention over the coming years

5 Mitigations of counterparty risk

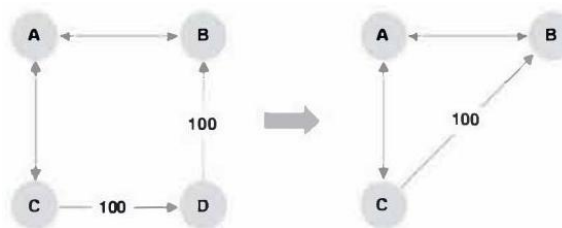
- Netting: Bilateral netting agreements allow cashflows to be offset and,in the event of default
- Collateral: Collateral agreements specify the contractual posting of cash or securities against MTM losses.
- Hedging: hedging counterparty risk with instruments such as credit default swaps(CDSs).
- Central counterparties: CCP act as intermediaries to centralize counterparty risk between market participants.

- Other contractual clauses: resets or additional termination.

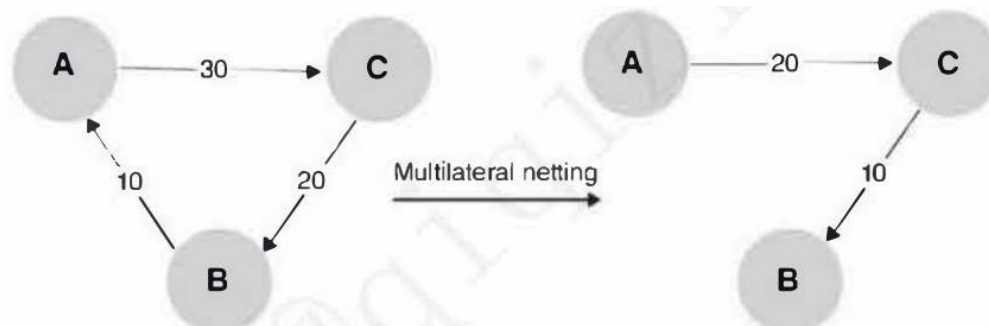
考点 3 Netting, Close-out and Related Aspects CR-16

1 Cash flow netting

- International Swaps and Derivatives Association (ISDA)
 - The market standard for OTC derivative documentation is the ISDA Master Agreement, which was first introduced in 1985 and is now used by the majority of market participants to document their OTC derivative transactions.
 - * Aims to remove legal uncertainties and to provide mechanisms for mitigating counterparty risk.
- Payment netting: involves the netting of different payments between two counterparties that are (generally denominated in the same currency).
 - Reduces the time and costs associated with making payments
 - Reduce operational risk, counterparty and liquidity risk
- Currency netting and CLS
 - Settlement risk is also a major consideration in foreign exchange (FX) markets, where the settlement of a contract involves payment of one currency against receiving the other. (Herstatt risk 赫斯塔特风险)
 - Non-deliverable forward (NDF) transaction: currency exchange is cash-settled based on the difference between the NDF rate and the prevailing spot FX rate applied to the notional.
- Continuous linked settlement (CLS): reduce settlement risk in FX transactions across a range of eligible currencies
 - Payment versus payment (PVP): Only after both currencies have arrived does CLS Bank make the outgoing payment to A and B.
- Clearing rings: multilateral netting before the development of central clearing.
 - Three or more members who would agree to "ring out" offsetting positions.
 - Participants in the ring had to be willing to accept substitutes for their original counterparties.



- Multilateral netting
 - Portfolio compression: achieves multilateral netting benefits via the cooperation of multiple counterparties that partially or entirely eliminate transactions.
 - * Aims to minimize gross notional of positions in the market
 - Portfolio compression:
 - * Preserve the market risk position of participants
 - * Reducing non-market risks such as settlement and counterparty risk.
 - * Reduce operational costs and may also lower systemic risk



- Compression cycle works: an algorithm is run to determine changes to transactions that generate multilateral netting benefits, but keeping each participant's portfolio neutral in terms of value and risk. 用算法来对交易的修改，产生多边净额结算利益，保持每个参与者在价值和风险方面的中性
 - * Participants can specify constraints (internal credit limits of a participant).
 - * Basic portfolio compression by its nature requires standard contracts
- Compression Algorithm
 - * It is not clear what should be minimized.
 - * An obvious choice may be the total notional.
 - * Alternative choices could be to use the squared notional or the total number of positions, which would reduce large exposure and interconnectedness respectively.
- Benefits of cash flow netting
 - Operational costs: By reducing the number of cash flows and transactions, operational costs can be reduced.
 - Legal risks: Portfolio compression may reduce legal risk by contractually simplifying portfolios and not relying on future risk mitigation.
 - Regulatory: some simple regulatory methodologies are partially based on gross notional and will, therefore, appear more beneficial if these values are reduced

2 Value netting

- Close-out netting 终止净额结算/净额交割
 - Close-out netting: comes into force in the event that a counterparty defaults, and aims to allow a timely termination and settlement of the net value of all transactions with that counterparty. 及时终止和结算与违约方的所有交易的净值
 - * Close-out: the right to terminate transactions with the defaulted counterparty and cease any contractual payments
 - * Netting: the right to offset the value across transactions and determine a net balance for the final close-out amount.
 - For the surviving party:
 - * Owes money: makes this payment
 - * Is owed money: makes a bankruptcy claim for that amount.
 - Jump to the bankruptcy queue for all but its net exposure
 - It only depends on mark-to-market(MTM) values at the time of default and not matching cashflows
 - Close-out netting and ISDA
 - * Payment netting reduces settlement risk, close-out netting is relevant to counterparty risk since it reduces pre-settlement risk.
 - * Close-out netting reduces the complexity involved in the close-out of transactions in the event that a counterparty defaults.

- * The ISDA has obtained legal opinions supporting the close-out and netting provisions in their Master Agreements in most relevant jurisdictions.
- * Determining a close-out amount:
 - Firm or indicative quotations for replacement transactions supplied by one or more third parties 由一个或多个第三方提供的替换交易的确定或指示性报价
 - Relevant market data supplied by one or more third parties
 - From internal sources
- Set-off 冲销
 - * Netting across product categories(e.g. OTC derivatives and loans) is not definitely possible as they are documented differently.
 - * 银行既给交易对手提供利率互换服务，同时又提供浮动利率贷款，但内部两个交易是分开的，在违约的情况下两者价值是否可以抵消
 - * Set-off vs. netting: 经济效果上相同，但含义不同
 - Set-off: recognizes the existence of cross-claims
 - Netting: results in a single contractual claim
- The impact of netting
 - 风险降低 Risk reduction:
 - * Close-out netting is risk mitigant for counterparty risk
 - * Has been critical for the growth of the OTC derivatives market.
 - 波动大,可能有系统崩溃风险 Netted positions are inherently more volatile than their underlying gross positions,which can create systemic risk
 - 改变市场参与者对特定交易对手风险增加的看法 If credit exposures were driven by gross positions, then all those trading with the troubled counterparty would have strong incentives to attempt to terminate existing positions and stop any new trading.
 - Netting impact on other creditors
 - * Netting redistributes value to OTC derivatives creditors from other creditors
 - * Netting redistributes value to OTC derivatives creditors from other creditors.
 - Multilateral netting and Bifurcation
 - * Bifurcation 分歧: between cleared and bilateral transactions
 - Market participants executing offsetting transactions since one product may be clearable and the other not.
 - An increase in overall exposure caused by multilateral netting related to central clearing of only a subset of trades
- Additional termination event (ATE)
 - Additional termination event(ATE): is designed to mitigate against counterparty risk by allowing a party to terminate transactions or apply other risk reducing actions when their counterparty's credit quality is deteriorating.
 - 触发事件
 - * Rating downgrade (e.g. below investment grade)
 - * For unrated parties such as hedge funds: market capitalisation, net asset value, or key person departure may be used.
 - 触发后果不一定是直接终止合约
 - * Provide a replacement counterparty
 - * Provide a counterparty offering a third-party guarantee
 - * Terminate the transaction(s) at their current replacement value 按目前的重置价值终止交易

- * Post some sort of margin
- Potential danger of ATE
 - * Risk-reducing benefit: a systematic deterioration in credit quality is much harder to nullify. 系统性恶化很难消除
 - * Weaknesses in credit ratings 评级反应迟钝
 - * Cliff-edge effects: single-notch rating downgrade may cause a dramatic consequence. 下调一个等级，可能会造成巨大的后果
 - * Modelling difficulty: rating transitions probability cannot be implied from market data

考点 4 Margin (Collateral) and Settlement CR-17

1 Margin terms

- Rationale for collateral
 - Counterparties
 - * In the case of a positive value: a party will call for margin
 - * In the case of negative value: a party will be required to post margin themselves
 - Terminology
 - * Collateral: used for OTC derivatives markets and contractual agreements such as ISDA Master Agreement 场外市场用更多
 - * Margin: used for exchange-traded markets to represent a similar concept 场内市场用更多
 - The fundamental idea of OTC derivatives collateralization is that cash or securities are passed (with or without actual ownership changing) from one party to another primarily as a means to reduce counterparty risk.
 - * 场外市场的抵质押物可以是现金、证券甚至是非金融资产
 - * Counterparty risk, in theory, can be completely neutralized as long as a sufficient amount of margin is held against it.
- Margins
 - Variation margin: members who have lost money pay their loss amount to the exchange CCP, while members who have gained receive their gain amount from the exchange CCP. (Reflects current exposure)
 - * MPoR (margin period of risk): effective delay in receiving margin
 - The inherent delay in the process between the time the party last posted margin and the time they were declared in default 最后一次交保证金到违约间的延迟
 - The associated costs in replacing and/or rehedging defaulted transactions. 替换和/或重新安排违约交易的相关费用
 - Initial margin: funds or marketable securities that must be deposited with the CCP in addition to variation margin.
 - * Extra margin that must be posted independently from the value of the underlying portfolio.
 - * For OTC derivatives, use the term 'Independent amount'.
 - * Reflects potential future exposure
 - 99 confidence interval over a 10-day horizon (incorporates a period of significant financial stress).
- OTC derivatives market
 - Bilateral OTC derivatives market: has used margin almost entirely in the form of variation margin, and initial margin (independent amount) has been quite rare. 场外市场，交换变动保证金，比较少交初始保证金
 - * Initial margin is a much more common concept on derivative exchanges and central counterparties (CCPs).
 - Regulation covering margin-posting in bilateral markets is making initial margin more common.

- Uncleared margin requirements (UMR): regulators have started to impose bilateral margin requirements
 - * Reduction of systemic risk: reducing “contagion and spillover effects” 减少传染和溢出效应
 - * Promotion of central clearing: aligning costs (notably initial margin) 双边清算成本和中央清算成本一直，促进中央清算的采用
- Apply to all OTC derivatives (FX swaps and forwards)
- Credit support annex (CSA)
 - Within an ISDA Master Agreement, it is possible to append a Credit Support Annex (CSA) where the parties agree to contractual margin-posting.
 - No CSA: 最典型的是银行和 end user (corporate)
 - Two-way CSA: For two financial counterparties, a two-way CSA is more typical, where both parties agree to post margin.
 - One-way CSA: only one party can receive collateral
 - * A high-quality entity such as a triple-A sovereign or supranational trading with a bank.
 - * Additional risk for the collateral giver
- Margin term
 - Threshold: is the amount below which margin (collateral) is not required, leading to undercollateralization.
 - * If the MTM is above the threshold, only the incremental amount of collateral can be called for.
 - * Threshold=0: collateral would be posted under any circumstance
 - * Threshold < 0: post initial margin
 - * Threshold ∞ : counterparty will not post collateral under any circumstance.
 - Minimum transfer amount (MTA): is the smallest amount of margin that can be transferred.
 - * Represents a balance between risk mitigation versus operational workload. 平衡风险缓释与业务工作量
 - Rounding 取整: A collateral call or return amount may also be rounded to a multiple of a certain size to avoid dealing with awkward quantities.
 - Types of collateral
 - * Partially collateralized: reduction of exposure is imperfect. (Threshold can be seen as approximately capping the exposure.)
 - * Strongly collateralized: assume thresholds are zero and therefore the exposure is reduced significantly.
 - * Overcollateralized: assume initial margin and therefore the exposure is reduced further.
- Credit support amount
 - Within the CSA, two parties can choose a number of key parameters and terms that will define the margin-posting requirements in detail.
 - Credit support amount: the amount of margin that may be requested at a given point in time.
 - * Variation margin is captured and the initial margin is referred to separately as independent amount. 上述金额表达的是变动保证金，初始保证金并不囊括在内
- Margin types and haircuts
 - Haircuts 折减率: acts as a discount to the value of the margin and provides a cushion in the event that the security is sold at a lower price.
 - Extra margin must be posted which acts as an overcollateralization to mitigate against a drop in the value of the asset
 - Cash: no haircuts
 - Defined as fixed percentage amounts

2 Impact of margin

- Valuation agent 估值代理
 - Valuation agent: the party calling for the delivery or return of margin and must handle all calculations.
 - The role of valuation agent:
 - * Calculate the current value, market value of margin (adjusted by haircuts), uncollateralized exposure and credit support amount
 - dispute over a margin call is common.
 - * Margin disputes are not relevant for centrally-cleared transactions since CCP is the valuation agent.
 - * Reconciliations: aim to minimize the change of a dispute by agreeing on valuations and margin requirements across portfolio.
 - Various third parties also offer services around valuation, exchange of margin, dispute management, and portfolio reconciliation.
- Impact of margin on exposure
 - Why margin cannot perfectly mitigate exposure
 - * The presence of a threshold means that a certain amount of exposure cannot be collateralized. 阈值导致无法足值抵押
 - * The delay in receiving margin and parameters such as the minimum transfer amount create a discrete effect 由于 MTA 保证金并非连续的，而是离散的
 - * Initial margin can be thought of as making the threshold negative and can potentially reduce the exposure to zero.
 - Impact on other creditors: margin does not reduce risk, it merely redistributes it. Other creditors will be more exposed
- Rehypothecation and Segregation
 - Rehypothecation 再抵押: the reuse of securities or cash margin by the margin receiver (e.g. in another margin agreement or a repo transaction).
 - * For variation margin: rehypothecation is important in OTC derivatives markets where many parties have multiple hedges and offsetting transactions
 - Segregation 分离: aims to ensure that margin posted is legally protected in the event that the receiving party becomes insolvent.
 - * For initial margin: segregation can avoid there being additional counterparty risk in relation to the initial margin posted.
 - * Whilst safer from the point of view of counterparty risk, segregation does create higher funding costs.
 - Segregated margin can be held:
 - * Held directly by the margin receiver
 - * Held by a third party acting on behalf of one party
 - * Held in tri-party custody where a third party holds the margin
- Potential problems of margin
 - Liquidity risk: in the event that margin has to be liquidated following the default of a counterparty, the non-defaulting party faces transaction costs and market volatility over the liquidation period. 在交易对手违约后必须清算保证金的情况下，非违约方在清算期间将面临交易成本和市场波动。
 - FX risk: margin posted in cash or securities denominated in a currency that does not match the currency or currencies of the portfolio 保证金与有关投资组合计价货币不一致
 - Legal risk: rehypothecation and segregation are subject to possible legal risks

- Operational risk: can be especially significant for the largest banks, which may have thousands of relatively nonstandardized margin agreements with clients, requiring posting and receipt of billions of dollars of margin on a given day. 它们可能与客户签订了数千份相对非标准化的保证金协议，需要在特定的一天缴纳和收取数十亿美元的保证金。
- Funding liquidity risk: increasing collateralization may reduce counterparty risk, but it increases funding liquidity risk.
 - * Potential risk arising from the difficulty to raise required funding in the future, especially when margin needs to be segregated and/or cannot be rehypothecated.

考点 5 Central Clearing CR-18

1 evolution and mechanics

- Evolution of complete clearing
 - Direct clearing
 - Clearing rings: three or more counterparties could “ring out” offsetting positions
 - Complete clearing: a central counterparty (CCP) is placed centrally between all counterparties and assumes all such contractual rights and responsibilities
- What is a CCP?
 - CCP: is a particular financial market infrastructure (FMI) that represents a set of rules and operational arrangements designed to allocate, manage, and reduce counterparty risk in a bilateral market. 是一种特殊的金融市场基础设施 (FMI), 代表了一套旨在分配、管理和降低双边市场中交易对手风险的规则和操作安排。
 - * Can reduce the interconnectedness within financial markets 相互关联性
 - * Can provide more transparency on the positions
 - The true CCP landscape is much more complex than represented
 - * Client clearing: 不能成为 CCP 会员的交易对手需要通过 CCP 会员清算
 - * Bilateral trades: 不是所有衍生品都适合
 - * Multiple CCPs: 全球有许多不同的 CCP
 - Regional: 为本地清算以本币计价的交易；监管
 - Product: 专注某些产品类型
 - Clearing relationships
 - * CCP members: large global banks and some other very large financial institutions
 - Smaller banks, buy-side and other financial firms, and other non-financial end users are unlikely to be direct clearing
 - * Requirement: admission criteria; financial commitment; operational
- Novation
 - Novation: means that the CCP essentially steps in between parties to a transaction and therefore acts as an insurer of counterparty risk in both directions.
 - * Buyer to the seller and seller to the buyer
 - * Matched book:
 - No net market risk
 - Take the counterparty risk
 - Portfolio compression vs. CCP
 - * Portfolio compression: requires actual transactions (or cash flows) to be eliminated
 - * CCP can compress risk, whereas bilateral compression can only compress objective quantities such as cash flows.

2 CCP risk management

- Functions of CCP
 - Multilateral offset: provide netting and/or compression benefits
 - Loss absorbency 损失吸收能力
 - Default management: hedging, auctioning, allocating loss, transferring client trades
- CCP membership requirements
 - Creditworthiness: the probability of default of the clearing member, as assessed by external or internal methods.
 - Liquidity: the ability of the member to meet liquidity requirements, such as the need to meet margin calls at short notice. 可以短时间内满足追加保证金要求
 - Operationality: the ability to adhere to the CCP rules, such as those pertaining to auctions.
- Margin and default funds
 - Variation margin: be transferred on a daily (and sometimes intradaily) and must usually be in cash
 - * Requires timely and reliable price data for all cleared derivatives and, where price data is not directly available, market-standard valuation methods. 可靠的价格数据或市场标准的估值方法
 - Initial margin: requirements may also frequently change with market conditions and must be provided in cash or liquid assets (e.g. treasury bonds).
 - * Cover potential close-out losses in the event of a default by that clearing member.
 - * Is calculated using different scenarios for possible price movements over an assumed close-out period (MPoR or liquidation period), such as five days.
 - Default funds: CCP members all contribute to a CCP
 - * Loss mutualization: losses above the resources contributed by the defaulter are shared between CCP members.

	Higher initial margin	Lower initial margin
Cost	Higher	Lower
Client clearing	Clients pay for their own risk via initial margin, promotes portability	Clients do not pay for their own risk directly, portability difficult
Moral hazard	Lower	Higher

table 5: Comparison of Initial Margin and Default Fund

- CPs are typically required to calibrate the size of the total default fund qualitatively via pre-defined stress tests. CCP 通常需要通过预先确定的压力测试，定性地校准总违约基金的规模
 - * Allocated to clearing members
- Default scenarios and margin period of risk
 - In a default scenario, the CCP neutralizes its exposure through:
 - * Macro-hedging 宏观对冲
 - * Auctions 拍卖
 - Sub-portfolio 拆分成子组合拍卖
 - Surviving CCP members bid the best price 未违约会员还是有很强动力参与拍卖，把不利后果压倒最低
 - * Fire drills: 消防演习，定期进行违约管理程序演习，最大程度提高潜在拍卖效率

* Driving test: 驾驶考试，对新成员进行测试

- The Loss Waterfall

- Skin-in-the-game: equity contribution from the CCP

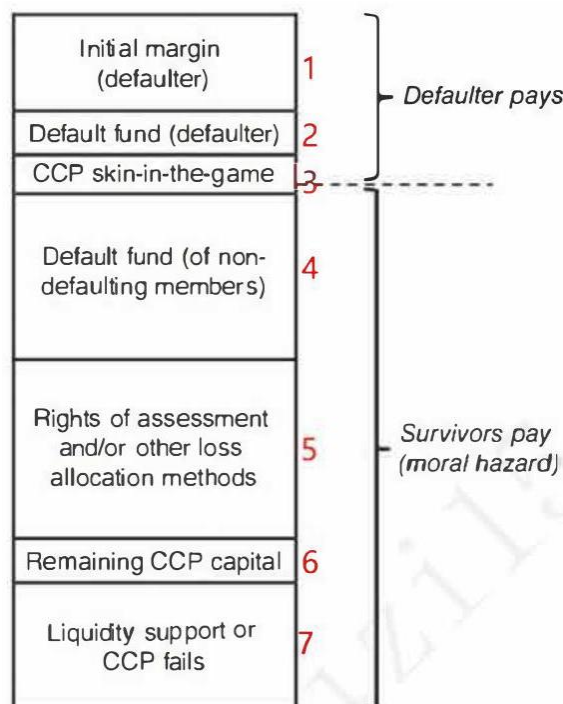


fig 3: Loss Waterfall

- Losses wiping out a significant portion of the default fund

- * Rights of assessment 评估权: an unfunded obligation to contribute additionally to a default fund that has been depleted by losses
 - * Variation margin gains haircutting (VMGH) 变动保证金扣减
 - * Tear-up 强行终止合约，打破原有的 matched book
 - * Forced allocation 把反向交易强行给到未违约会员

- Impact of central clearing

- Advantages

- * Transparency
 - * Multilateral offset
 - * Loss mutualization
 - * Legal and operational efficiency
 - * Liquidity
 - * Default management

- Disadvantages

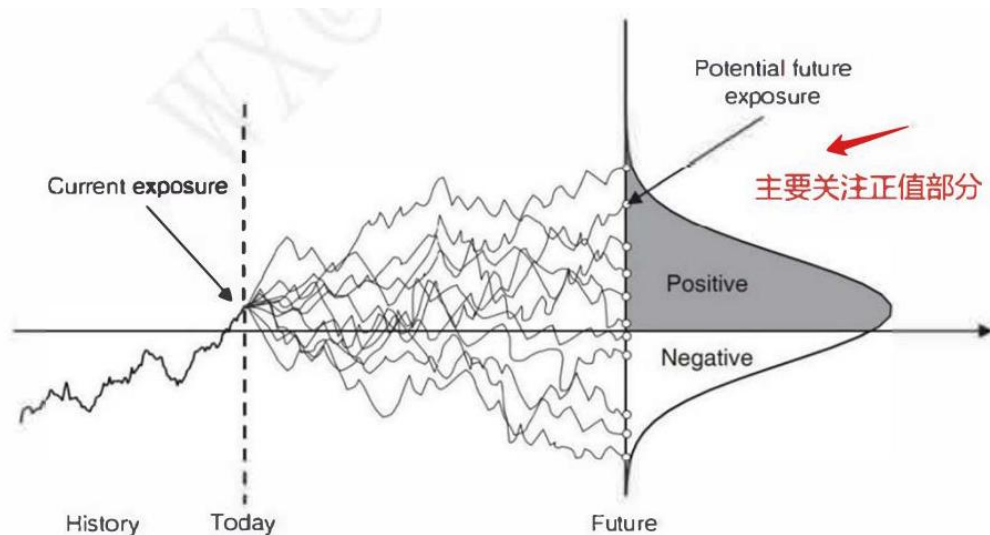
- * Moral hazard
 - * Adverse selection
 - * Bifurcations 清算标准产品的要求可能会在清算交易和非清算交易之间造成分歧
 - * Procyclicality 市场波动或危机的时候提高保证金要求

考点 6 Future Value and Exposure CR-19

1 Metrics for Credit Exposure

- Credit Exposure

- Definition: effective value of the relevant contracts at the default
- Credit Exposure = $\max(\text{value}, 0)$
 - * Loans: the credit exposure is the notional amount and is fixed.
 - * Swap, option and forward: the credit exposure is the value of the asset if it is positive.
 - * Futures and short options: the credit exposure is zero
- Characteristics affects credit exposure:
 - * Volatility
 - * Time horizon
 - * Risk mitigants



- Nature of exposure

- Asymmetric risk profile
 - * Exposure is similar to an option payoff, a key aspect will be volatility (of the value of the relevant contracts and margin)
 - * Options are relatively complex to price 敞口量化也有其复杂性
- VaR vs. quantifying exposure
 - * Time horizon: VaR-short horizon
 - The “ageing” of transaction (cash flows, termination events, exercise decisions, and margin postings)
 - When looking at longer time horizons, the trend (also known as the ‘drift’) of market variables is relevant
 - * Risk mitigants 考虑净额结算和保证金的影响
 - * Application: exposure must be defined for both risk management and pricing (i.e. xVA) 传统方法偏向风控

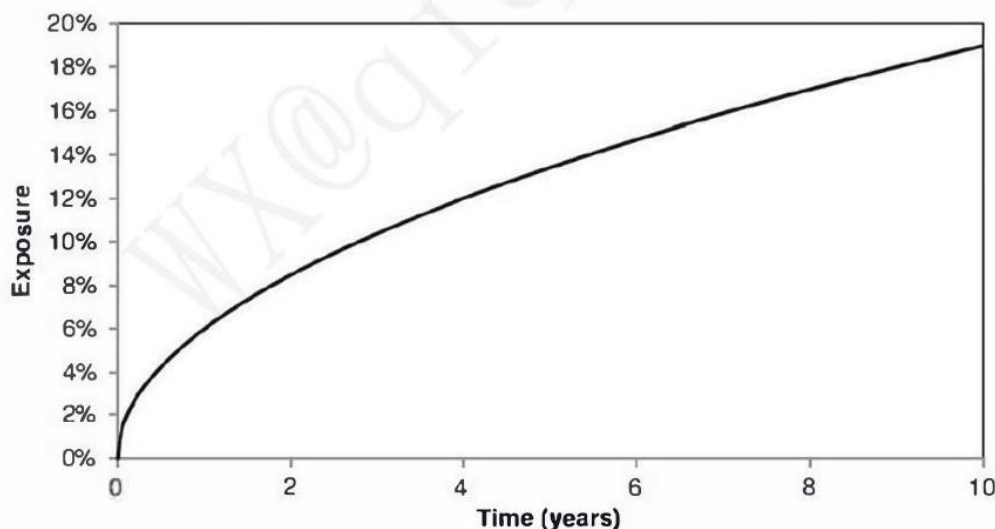
- Metrics for Exposure

- Expected future value (EFV): 对敞口分布求期望
 - * Represents the expected (average) of the future value calculated with some probability distribution.
 - * At time zero: the current value
 - * EFV may vary significantly from current value

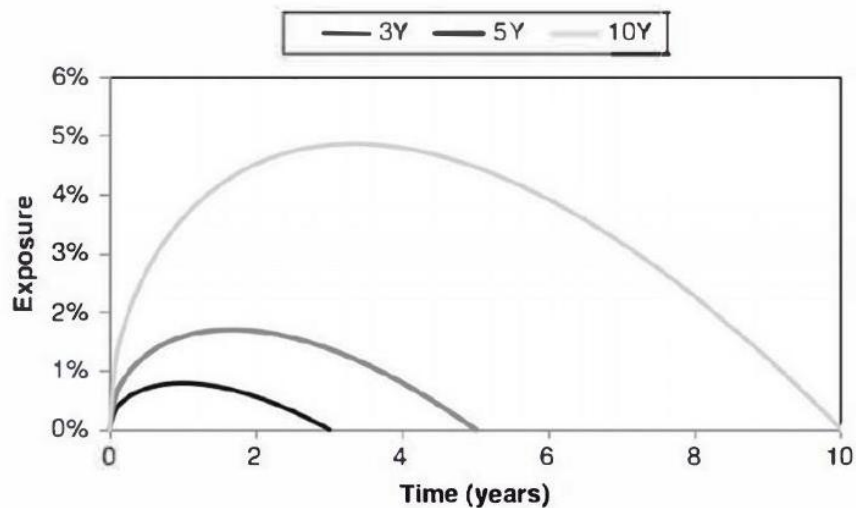
- Cash flow differential 比如利率互换交叉货币互换
- Forward rates: can differ significantly from current spot variables
- Asymmetric margin agreements: one-way CSA
- Potential future exposure(PFE):at a confidence level of x% 未来某个时期出现较坏的风险敞口是多少
 - Is an equivalent metric to VaR
 - The highest (or lowest) exposure calculated to some underlying confidence level
- Expected positive exposure(EPE): the average of all of the positive exposure. 正敞口的平均数 (负的取零)
 - Also referred to as “expected exposure(EE)”
- Expected negative exposure(ENE): the average of all of the negative exposures. 负敞口的平均数 (正的取零)
 - Represents the positive exposure from the counterparty’s point of view.
 - Also referred to as “negative expected exposure(NEE)”
- Average EPE:the average exposure across all time horizons.
 - The weighted average of the EPE across time.
 - Also called a “loan equivalent”: the average amount effectively lent to the counterparty

2 Exposure profile of various securities

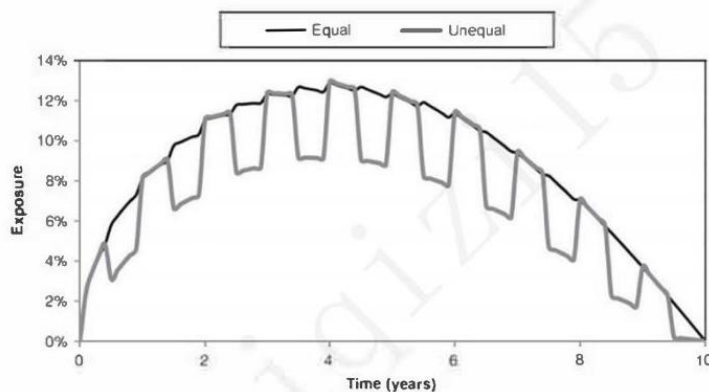
- Loans and Bonds
 - 敞口几乎是确定的，敞口大致等于名义价值
- forward contract 远期合约
 - The exposure is a rather simple increasing function reflecting the fact that, as time passes, there is increasing uncertainty about the value of the final exchange. 【随着时间推移，最终合约价值的不确定性不断增加】



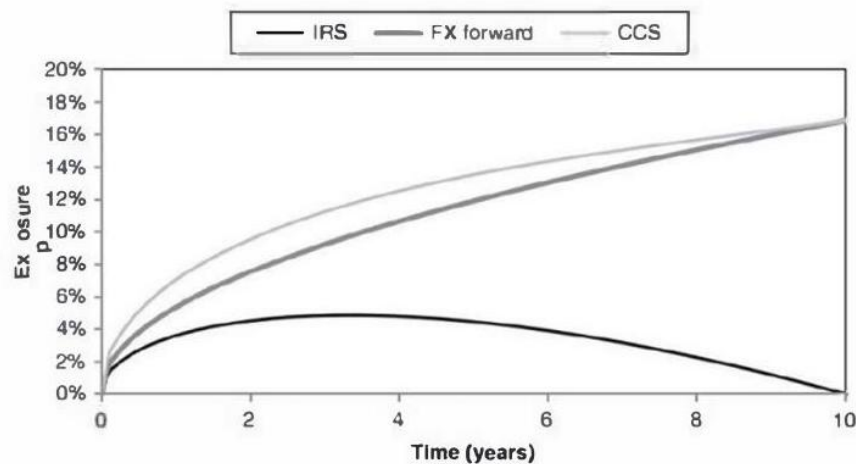
- Interest Rate Swap 利率互换
 - 浮动利率与固定利率的交换。期间换利息，期末换利息不换本金
 - Exposure profile is characterized by a peaked shape.
 - This peaked shape results from the balancing of future uncertainties over payments and the roll-off risk of swap payments over time.



- 另外，有一些非标准化的利率互换可能不是同一个期限进行交割的，比如浮动利率 (每三个月交换一次) 交换固定利率 (每六个月交换一次)，此时现金流是不匹配的。【与普通利率互换敞口形状差不多，只不过是有点锯齿状】

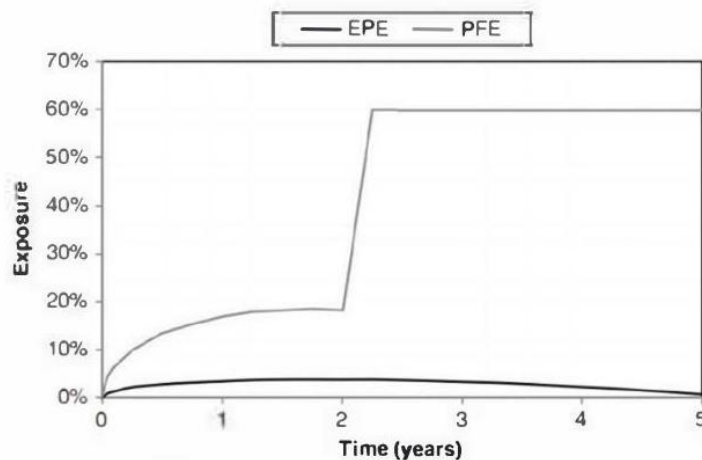


- cross-currency swap 交叉货币互换
 - * 就是一级学习的货币互换。
 - * 与利率互换不同，货币互换期间换利息，期末换利息和本金
 - * Some exposures come from interest rate risk (IR) while the majority results from the uncertainty with the final notional payment of FX rate risk.
 - * 也可以把货币互换看成是利率互换和外汇远期的结合。



– CDS exposure:

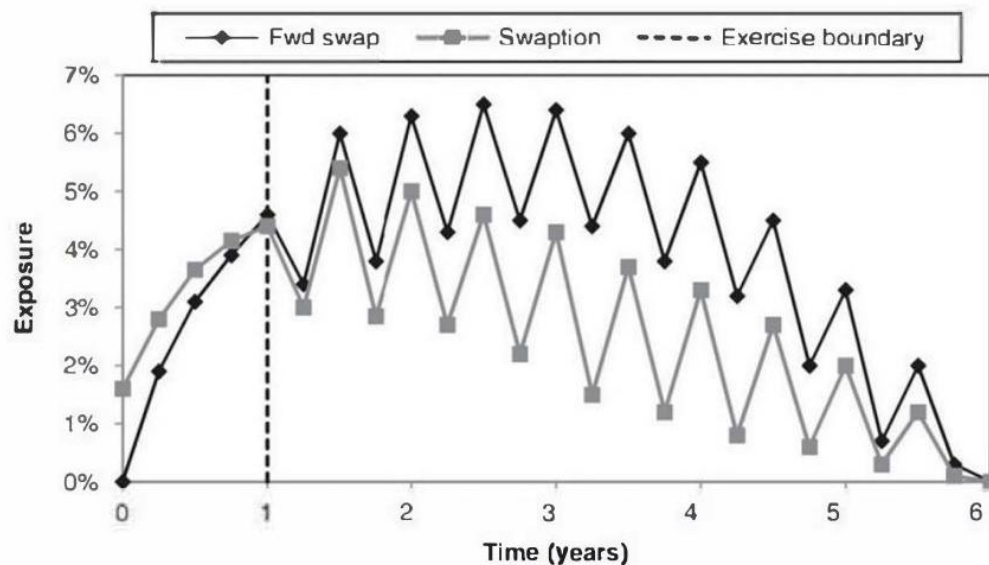
- * EPE: shows a typical swap-like profile.
- * PFE: has a jump due to the default of the reference entity.



– Swaption 和 forward starting swap

- * 补充知识:Swaption 和 forward starting swap 这两个产品，咱们没太关注过。原版书在这里也并没有进行介绍，直接介绍敞口。这里做一个知识补充：
 - Swaption: 赋予持有人在未来某个日期或某段时间，进入一个 swap。是权利，没有义务。
 - Forward Starting Swap 是一种互换，约定在未来某一日期开始生效。从那个日期起，两方将在未来的若干时间段内交换一系列现金流。是权利也是义务，必须要履行合约。
- * Swaption 会根据市场情况选择是否执行，forward starting swap 则是在未来某一特定日期开始执行。
- * 在行权日期之前，swaption 的敞口总是大于 forward starting swap。

- 在行权日期之后，情况会反转，因为在某些情况下，forward starting swap 可能有正值，而 swaption 可能没有被行权 (敞口下降)。



3 The impacts on exposure

- Reducing effect: the exposure profiles are partly offsetting (one is positive whilst the other is negative)
- Exposure
 - Positive exposure: $\max(\text{value} - \text{margin}, 0)$
 - Negative exposure: $\min(\text{value} - \text{margin}, 0)$
- Collateralization is not a perfect form of risk mitigation
 - Granularity effect: margin parameters such as thresholds and minimum transfer amounts
 - Delay in receiving margin: MPoR
 - Potential variation in the value of the margin itself
- Funding, rehypothecation, and segregation
 - Funding benefits and costs: $\text{Funding} = \text{value} - \text{margin}$
 - * Funding cost: term is positive
 - * Funding benefit: term is negative
 - * Posted margin: create a positive funding cost.
 - * Received margin: represent a benefit
 - Various types of margin under certain situations

	降低交易对手风险	提供 funding benefit
Cash that does not need to be segregated	✓	✓
Securities that can be rehypothecated	✓ (前提是 haircut 足够)	✓
Cash or securities that must be segregated or cannot be rehypothecated	✓	×
Counterparty posting own bonds (that can be rehypothecated)	?	✓

- Impact of margin on exposure and funding
 - * Positive exposure = $\max(\text{Value} - \text{VM} - \text{IM}^R, 0)$
 - * Funding = $\text{value} - \text{VM} + \text{IM}^P$
 - * 信用敞口：收到保证金降低敞口，缴纳保证金增加敞口
 - * Funding:
 - 初始保证金：segregation, 无法提供 funding benefit
 - 变动保证金：假设可以使用，可提供 funding benefit

考点 7 CVA CR-20

1 Credit Valuation Adjustment and xVA

- Credit valuation adjustment (CVA): was originally introduced as an adjustment to the risk-free value of a derivative to account for potential default.
 - A dealer calculates CVA for each counterparty with which it has bilaterally cleared OTC derivatives.
 - CVA is one's expected loss from a default by the counterparty. 在无风险价值上做调整 (Risky value = Risk-free value - CVA)
- Motivation and challenge of pricing counterparty risk
 - Motivation: CVA has become a key topic for banks due to the volatility of credit spreads, the associated accounting and capital requirements (Basel III).
 - Challenge: cash flow not fully at risk due to partial offset with opposing cash flows
- Credit limit vs. CVA
 - Credit limit: are a traditional tool to control the amount of risk to a given counterparty over time.
 - * Counterparty risk can be diversified by limiting exposure to any given counterparty, sector, or region.
 - * Characterize the potential future exposure (PFE) over time and compare this with a pre-determined limit.
 - * Effectively favoring short-term exposures
 - * A transaction with low (high) profitability may be accepted (rejected) because the existing credit limit is small (large).
 - CVA: By pricing counterparty risk through CVA, the question becomes whether it is profitable once the counterparty risk component has been "priced in".

- * CVA directly incorporates the credit risk of the counterparty and so defines a minimum revenue that should be achieved.

- Three levels to assessing the counterparty risk of a transaction

Level	Description	Examples
Trade level	Incorporating all characteristics of the trade and associated risk factors such as the precise cash flows	e.g. maturity, currency, CVA
Counterparty level	Incorporating the impact of risk mitigants such as netting and margining	e.g. netting, collateral, CVA
Portfolio level	Consideration of the risk to all counterparties, knowing that only a small fraction may default in a given time period	e.g. large exposure and correlation, Credit limits

table 6: Risk Levels and Descriptions

- CVA: encourages minimizing the number of trading counterparties since this maximizes the benefits of netting
- Credit limits: encourage maximizing the number of trading to encourage smaller exposures and diversification
- CVA and credit limits are typically used together as complementary ways to quantify and manage counterparty risk.

- xVA

- Economic costs of a derivative
 - * Funding
 - * Collateral
 - * Initial margin
 - * Regulatory capital
- Components of xVA Terms

Risk Type	Market Component	Cost Component
Counterparty risk	CVA/DVA	Credit exposure, Default probability
Funding	FVA, MVA	Valuation, Initial margin amount
Collateral	CoVA	Collateral amount, Collateral cost
Capital	KVA	Capital amount, Capital cost

table 7: Risk Components and Their Terms

- Valuation adjustments

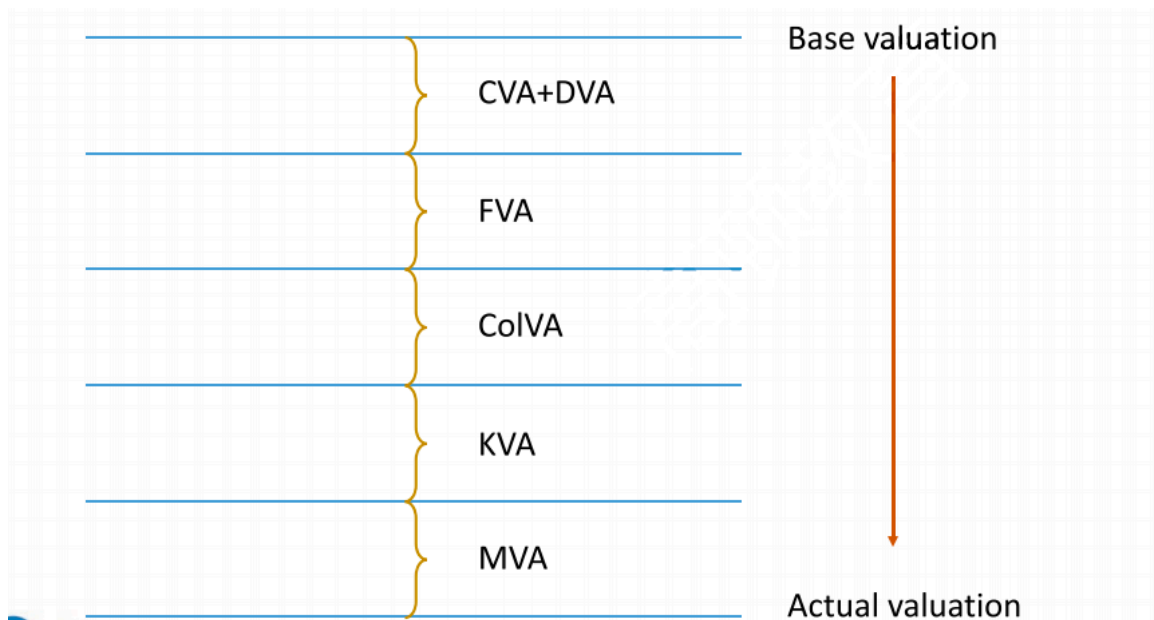


fig 4: Valuation Adjustments

- replacement cost and credit exposure
- Default probability, credit migration, and credit spreads
- Recovery and loss given default
- Funding, collateral, and capital costs

2 CVA and DVA

- Unilateral CVA(UCVA)

$$UCVA \approx -LGD \times \sum_{i=1}^m d(t_i) \times EPE(t_i) \times PD(t_{i-1}, t_i)$$

- $d(t_i)$: 折现因子
- $PD(t_{i-1}, t_i)$: 边际违约概率
- 假定没有 wrong-way risk

- CVA as a spread: adjust a rate paid or received as a running spread (per annum charge)

$$UCVA \approx -averageEPE \times spread$$

- Credit component: the credit spread of the counterparty
- Market component: the exposure, or average EPE

- Bilateral CVA(BCVA)

- Debt valuation adjustment(DVA): is the mirror image of CVA(own credit risk).
 - * DVA is the expected cost to the counterparty
 - * DVA will oppose the CVA as a benefit.
 - * DVA term corresponds to fact that in cases where party themselves default, they will make a “gain” if a negative exposure.
 - * DVA creates “price symmetry”. Parties can agree on prices.
- 交易对手 A 和交易对手 B (My CVA is your DVA)

- * $V_A = -V_B$
- * $V_A + CVA_A + DVA_A$ (party A point of view)
- * $V_B + CVA_B + DVA_B$ (party B point of view)
- * $BCVA = CVA + DVA$
- * $CVA_A = -DVA_B$
- * $CVA_B = -DVA_A$
- * CVA 是一个负数
- * DVA 是一个正数
- CVA 和 DVA 计算

$$CVA = -LGD_C \times \sum_{i=1}^m d(t_i) \times EPE(t_i) \times PD_C(t_{i-1}, t_i) \times [1 - PD_P(0, t_{i-1})]$$

$$DVA = -LGD_P \times \sum_{i=1}^m d(t_i) \times ENE(t_i) \times PD_P(t_{i-1}, t_i) \times [1 - PD_C(0, t_{i-1})]$$

- * P is the party making the calculation
- * C is its counterparty
- * 注意加入考虑生存概率（有关协方差 PE 与原版生存概率处理的不同说明）
- As spread

$$BCVA \approx -averageEPE \times Spread_c - averageENE \times Spread_p$$

- * Assuming that average EPE=-average ENE

$$BCVA \approx -EPE \times (Spread_c - Spread_p)$$

- * Weaker counterparties would pay stronger counterparties in order to trade with them based on the differential in credit quality.

3 CVA allocation

- Netting CVA
 - **Netting CVA:** Risk mitigants such as netting reduce the value of CVA.
 - Under one netting agreement (netting set):

$$CVA_{NS} \leq \sum_{i=1}^n CVA_i$$

- where CVA_{NS} is the total CVA of all transactions under the netting agreement and CVA_i is the stand-alone CVA for transaction i .
- The above effect can be significant.

- Incremental CVA

- **Incremental CVA:** The CVA of a transaction i is calculated based on the incremental effect this transaction has on the netting set.

$$\begin{aligned} CVA^{NS \rightarrow NS^*} &= CVA^{NS^*} - CVA^{NS} \\ &= -LGD \times \sum_{i=1}^m EPE^{NS \rightarrow NS^*}(t, t_i) \times PD(t_{i-1}, t_i) \end{aligned}$$

- Incremental EPE can be negative, due to beneficial netting effects.
- The incremental CVA is never higher than the standalone CVA without netting.
- The benefit of netting seen in incremental CVA of a new transaction also depends on the relative size of the new transaction. 净额结算的好处还和交易本身规模有关
 - * As the transaction size increases, the netting benefit is lost, and CVA(or DVA) will approach the standalone value.
- Marginal CVA: break down a CVA for any number of netted transactions into transaction-level contributions that sum to total CVA.
- Impacts on CVA
 - Spread curve: if upwards-sloping, flat and inverted credit curves all have the same cumulative PD at the five-year point, the marginal default probabilities differ substantially.
 - * Upwards sloping: largest CVA
 - * Inverted sloping: smallest CVA
 - Credit spread: CVA increases with increasing credit spread.(In default, CVA may be zero.)
 - Impact of recovery rate:increase the recovery rate will reduce the value of CVA.
 - Initial margin and threshold: a high coverage of initial margin is held, CVA can be assumed to be zero.
 - Variation margin:
 - * MPoR/Cure period: It is typically 10 or 20 days.

4 Wrong-way risk

- 定义
 - Wrong-way risk (WWR):indicate an unfavorable dependence between exposure and counterparty credit quality.
 - * The exposure is high when counterparty is more likely to default and vice versa.
 - Right-way risk(RWR): cases where the dependence between exposure and credit quality is a favourable one.
 - * Right-way situations will reduce counterparty risk.
- Examples of WWR and RWR
 - 看跌期权
 - * A 公司向 B 公司购入以 B 公司的股票为标的的看跌期权, 当 B 公司违约概率上升时, 股票价格会下跌, 此时 A 所购入的看跌期权的盈利上升, 信用风险口上升。所以此时对于 A 来说是错路风险
 - 商品互换
 - * 以 A 公司与 B 公司的原油互换为例, A 公司支付固定的原油价格, 收到 B 公司所支付的浮动的原油价格, 此时 A 公司是 WWR 还是 RWR 要根据 B 公司的具体情况进行判断。
 - * 如果 B 公司是原油的经销商, 此时当原油价格上升, 有利于 B 公司的经营发展, 所以 B 公司的 PD 下降, 并且 A 公司的获利也更大, 信用口上升, 所以此时 A 公司面临 RWR。
 - * 如果 B 公司是航空公司, 当原油价格上升时, B 公司的经营成本上升, 不利于 B 公司的经营发展, 所以 B 公司的 PD 会上升, 此时 A 公司的信用口也上升, 面临 WWR
 - 信用违约互换
 - * 公司向 B 公司购入信用违约互换。此信用违约互换的标的资产和 B 公司的信用质量高度正相关, 当 B 公司信用质量恶化, B 公司的违约概率上升, 并且标的资产的信用质量也恶化, 此时会产生两种情况:
 - 当标的资产信用质量恶化, 但没有发生违约时, CDS 的保费会上升, A 公司之前购入的 CDS 相对于当前市场价格是较低的, A 公司是相对盈利的, 信用敞口上升, A 公司面临 WWR。
 - 当标的资产信用质量恶化且发生违约, 则会触发信用违约互换的赔付机制, A 公司的信用风险口也上升, 面临 WWR。

- 外汇产品
 - * A 公司与美国政府进行货币互换的交易, A 公司支付美元, 收英镑, 美国政府支付英镑, 收美元。当美国市场出现萧条情况时, 美元贬值, 此时美国政府较易违约, 并且美元贬值对于 A 公司来说是支出少, 收入多, 所以 A 公司的信用敞口上升, 面临 WWR
- 利率产品
 - * 假设 A 银行进入支付浮动利息, 收入固定利息的利率互换, B 公司支付固定利息, 收入浮动利息, 当经济恶化时, 市场利率下降, B 公司的违约概率上升, 并且 A 银行支付的浮动利息相对减少, A 银行在此笔互换中盈利上升, 信用敞口上升, 面临 WWR。
- WWR modelling
 - Hazard rate (intensity) approaches: introduce a stochastic process for the credit spread. Conditional EPE will be calculated. (easy to use)
 - * Correlation: directly via historical time series of credit spreads and other relevant market variables.
 - * Weak dependence between exposure and default
 - Structural approaches: mapping the default distribution and exposure distribution onto a bivariate distribution.
 - * WWR: Positive dependency will lead to an early default time being coupled with a higher exposure
 - Parametric approaches: linking the default probability parametrically to the exposure using a simple, functional relationship.
 - * If the portfolio has historically shown high values together with larger than average credit spreads, then this will indicate WWR
 - Jump approaches: jump at default.
 - * E.g.: counterparty defaults with FX rate
 - * Implied jump is larger for better-rated sovereigns
 - * A default of a large corporation should be expected to have quite a significant impact on the local currency
- Collateralization and WWR
 - WWR may also be present in terms of the relationship between the value of margin and the underlying exposure. 可以定义为敞口和抵质押物价值的关系
 - * 敞口上升, 抵质押物价值反倒下降
 - 例子:
 - * Payer interest rate swap collateralized with a high-quality government bond. (WWR: 利率上升, 敞口上升, 抵质押物价格下降)
 - * Cross-currency swap collateralized by cash in one of the two underlying currencies. Margin is held in the currency being paid.(如: 付美元利息, 收日元利息, 抵质押物为美元) (WWR: 美元贬值, 敞口上升, 抵质押物价值下降)
- CCP and WWR
 - CCPs tend to disassociate credit quality and exposure. CCPs are in danger of implicitly ignoring WWR.
 - WWR increases with increasing credit quality
 - * A large dealer represents more WWR than a smaller and/or weaker credit quality counterparty.
 - * CCPs should require greater initial margin and default fund contributions from better-credit-quality and more-systemically-important members. 信用质量较好的成员违约的可能性较小, 但如果违约, 影响可能会更严重。
 - CCPs face WWR on the margin they receive.
 - * Under pressure to accept a wide range of eligible securities for initial margin purposes.

- * Adverse selection: clearing members(and clients) will naturally choose to post margin that has the greatest risk (relative to its haircut) and may also present the greatest WWR to a CCP

考点 8 Counterparty Credit Risk 压力测试 CR-21

- 把 CCR 看成是 credit risk 和 market risk 的结合
- CCR 的压力测试主要包括 EL 和 CVA 两个方面。

(1) 贷款组合的压力测试

- stress test could take EAD and LGD as deterministic and focus on stresses where the PD is subject to a stress.
- 压力情况下的 PD，违约概率可以表示为汇率或者失业率等变量在压力情况下的函数

$$EL_s = \sum_{i=1}^N PD_i^s \times EAD_i \times LGD_i$$

EL_s : Stressed EL

- 压力损失 Stress loss 表示为压力下的 EL 和无压力下的 EL 的差额:

$$\text{Stress loss} = EL_s - EL$$

(2) 衍生产品组合的压力测试

- Average EPE multiplied by an alpha factor: allows CCR exposures to be placed in a portfolio credit model at a high percentile loss of the portfolio of exposures。【需要考虑压力下的 PD 和 EPE】

$$EL = \sum_{i=1}^N PD_i \times \alpha \times EPE_i \times LGD_i$$

$$EL_s = \sum_{i=1}^N PD_i^s \times \alpha \times EPE_i^s \times LGD_i$$

$$\text{Stress loss} = EL_s - EL$$

(3) CVA 的压力测试

$$CVA_n = \sum_{n=1}^N LGD_n \times \sum_{i=1}^m EE_n(t_i) \times PD_n(t_{i-1}, t_i)$$

$$CVA^s = \sum_{n=1}^N LGD_n \times \sum_{i=1}^m EPE_n^s(t_i) \times PD_n^s(t_{i-1}, t_i)$$

$$\text{Stress loss} = CVA^s - CVA。$$

(4) Pitfalls in stress testing CCR

- 现在的压力测试，仅限于当前敞口的压力测试
- CCR 很少与贷款组合或交易头寸的压力测试结果在一个一致的框架中。
- 对于 CCR 来说，使用 delta 敏感度来计算敞口变化是有问题的，因为 CCR 具有高度非线性特性。

5 信用风险管理

考点 1 Credit Risk Management CR-3

Chapter 16、17 和 18 都是关于信用风险的管理

* Chapter 16: 站在银行贷款的角度管理风险，更多的涉及的是规章制度。

* Chapter 17: 站在持有 bond 的角度，用信用衍生品，管理 bond 投资带来的信用风险。

* Chapter 18: 站在银行资产 [各种 loan、mortgage 等] 角度，采取证券化手段把这些有风险资产转嫁出去。

【本节知识点，站在银行贷款的角度管理风险。】

- Key Elements of Effective Lending/Financing Policy

Credit facility scope and allocation: 明确规定银行的信贷业务范围，包括哪些类型的贷款和融资产品可以提供，涵盖的客户群体，以及允许的最大敞口限额。

Credit portfolio management 信贷组合管理

信贷风险管理的全过程，包括贷款的 origin、评估 (appraisal)、内部监督 (internal supervision) 和收款程序 (collection procedures)。

强调在每个阶段如何识别、评估和管理风险，确保在整个贷款生命周期中对风险的持续控制和监管。

Emphasize non-restrictiveness and flexibility 强调非限制性和灵活性

强调政策的灵活性，使其能够适应市场和经济环境的变化。银行应能够根据新的风险和市场条件及时调整其信贷策略。符合监管机构对信贷政策的最低标准要求。

董事会和高级管理层的参与

风险分类和损失准备金

包括将资产按信贷风险敞口和可收回性分类，以及为潜在损失建立拨备的具体方法和标准。

Setting Exposure and Concentration Limits

监管机构通常关注三个领域：

- 对单一客户的敞口 exposure to a single customer
- 关联方融资 related-party financing

3 对某一经济部门或地理区域的过度敞口 over exposure to an economic sector or geographic area。

/ **Concentration limits** 集中度限制: 指的是对单一客户、地区或经济部门的最大敞口。银行通常根据准备金或资本的百分比设定限制

Related parties 关联方可能会对银行的信贷决策产生控制作用。关联方包括母公司、子公司、主要银行股东、关联公司、董事和高级管理人员。

- **Credit Facility Allocation and Credit Risk Reduction:** 银行在制定和执行贷款政策时需要考虑的关键因素

A Lending authority 贷款授权: 贷款政策应设定贷款官员的授权限制。有经验或任期较长的贷款官员可能会有更高的限额。

在小型银行通常是集中化的，而在大型银行则可能根据地理区域、贷款产品或客户类型进行分散化授权。

Types and distribution of loans 贷款类型和分布

通常对商业、房地产、消费者等贷款类别设定限制，并且任何超出限制的行为应得到董事会的批准。

Appraisal process 评估过程

贷款定价 **Loan pricing:** 贷款利率应根据资金成本、贷款监督和行政开支的结构。应考虑损失概率和银行的合理利润率。

Maturities 到期期限: 贷款政策应为各种贷款类型设定明确的最大到期期限。

- **Exposure to geographic or sector-level concentration** 地理或行业集中度风险

Availability of current financial information 当前财务信息的可用性: 贷款政策应明确规定商业和消费者贷款所需的财务信息。

- Monitoring of collections 催收监控

Limits on total loans outstanding 总贷款余额限制: 贷款政策应设定关于总贷款余额的限制。

A Maximum ratio of loans to market value of pledged securities 质押证券市值与贷款最大比率: 应为所有抵押品类型明确设定保证金要求。

Impairment recognition 贷款减值确认: 应定期系统地识别贷款减值

Renegotiated debt treatment 重组债务处理

V Written internal guidelines 书面内部指导方针: 所有贷款政策都应有适当的文件记录, 不能仅靠口头理解。

- Credit Asset Classification 信贷资产分类

资产分类是根据偿还可能性将银行资产分配到信用风险评级的过程, 是信贷风险管理的关键工具。类别:

Standard, or pass 标准/及格: 无逾期问题, 有充分现金或等值现金担保

□ Specially mentioned, or watch 特别提及/观察: 存在影响借款人偿还能力的弱点

Substandard 次级: 具有明确的还款弱点, 可能危及偿还。

Doubtful 可疑: 与次级相似但预期损失更为明显

Collection in full is questionable

Loss: 偿还迟滞相对确定, 准备核销

Nonperforming Loans analysis 不良贷款分析

Aging of past-due loans 逾期时间

- Reasons for the deterioration of loan portfolio quality

Case-by-case assessment of a list of nonperforming loans 单独审查每个逾期案例

Provision levels 准备金水平

Loan Loss Provisions and Loan Loss Reserves

Loan Loss Provisions 贷款损失准备金

Specific amounts set aside by a bank for potential losses on loans.

考虑因素: like credit policies, prior loss experiences, and economic conditions

Loan Loss Reserves 贷款损失储备

Not assigned to specific assets

General reserves counted as Tier 2 capital, used to absorb unexpected losses. 作为二级资本的一部分, 用于吸收意外损失

Loan loss provisions cover specific amounts, while reserves are general and form part of the bank's capital structure.

考点 2 Credit Derivatives CR-13

金融机构可以使用信用衍生品管理信用风险。本章节主要涉及 CDS、TRS。【关于 CDS 相关知识点, 在 FRM 一级以及二级其他科目都或多或少的有所涉及, 这里做一个相对全面的介绍。CDS 是本科目重要知识点。】

1. Credit default swap 信用违约互换

(1) 基本概念

一个典型的例子:

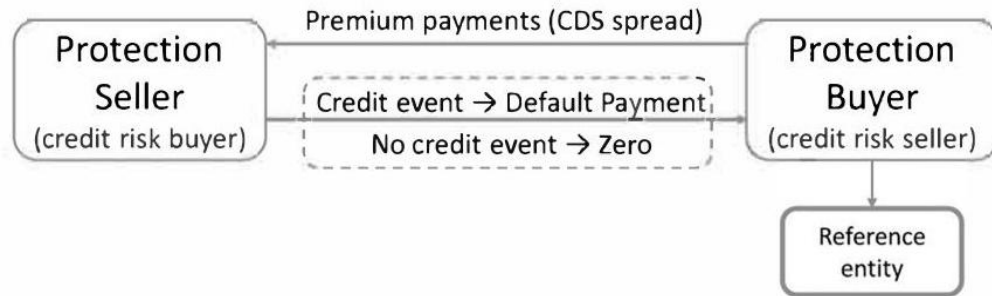
一个投资者 A 买了一个 C 公司发行的债券, 但是担心 C 公司未来违约 (信用风险), 然后向一家 B 金融机构购买了一个“保险”, 提供的保护就是关于公司 C 是否违约。

零时刻或者特定时刻: 投资者 A 向金融机构 B 支付保费

未来: 如果公司 C 违约, 投资者 A 虽然不能收到来自公司 C 的资金, 但是因为购买了“保险”, 金融机构 B 将会对此进行“保险”赔付。

这里的“保险”就是我们将要介绍的 CDS。

- A credit default swap 信用违约互换 (CDS) is a derivative contract between two parties, a credit protection buyer and credit protection seller, in which the buyer makes a series of cash payments to the seller and receives a promise of compensation for credit losses resulting from a credit event in an underlying.
- Credit default swaps (CDS) is essentially an insurance contract for the reference, the reference obligation is the fixed income security on which the swap is written-usually a bond but potentially also a loan.
- 与保险很像，与 put option 类似



- CDS 用作风险管理

如果持有一个有风险的债券，买一个 CDS，相当于将债券的风险转移出去，进而相当于购买了无风险债券进行风险管理。

一个有风险的公司债的多头相当于无风险债券的多头再卖出一份 CDS。

$$\text{Bond}_{\text{risky}} + \text{CDS} = \text{Bond}_{\text{risk free}}$$

$$\text{Bond}_{\text{risky}} = \text{Bond}_{\text{risk free}} - \text{CDS}$$

【这是个理念性公式。这个公式在以前的原版书上是有的，现在没了。我感觉很好，笔记就保留了下来】

Y 另外有 CDS 可以只管理信用风险，并不能管理市场风险，这不同于后面学习的总收益互换。但同时 CDS 还会引入新的信用风险，即 CDS 的卖方可能违约。

- 交易方:

credit protection buyer=the buyer of CDS(CDS 买方，保护的买方，风险的卖方)=short position of CDS

例如上述例子中的投资者 A。

支付一定的保费 fee/premium(可能是期初一次支付，也可以是后期定期支付)。如果标的资产违约，CDS 卖出方给予买方一定的赔付。如果标的资产没有违约，白白支付了 premium。

注意: 风险的卖方，因此也称 short position of CDS (受益于受益于标的资产信用质量的恶化，也就是风险变大)。

credit protection seller=the seller of CDS(CDS 卖方，保护的卖方，风险的买方)=long position of CDS

例如上述例子中的金融机构 B。

收取一定保费。如果标的资产违约，给予买方一定的赔付。如果标的资产不违约，白白获得 premium。

注意: 风险的买方，因此也称 long position of CDS (受益于标的资产信用质量的好转，也就是风险变小)。

注意: 在 CDS 这里，long CDS 意味着是 credit protection seller; short CDS 意味着 credit protection buyer。这与其他衍生品不同，这里可做特殊记忆。

- Reference entity 参考实体:

也就是债券的发行人。类似例子中的 C 公司

reference obligation 参考债务/参照资产: 发行人发行的 (一个或多个) 债务工具 (bond/loan)

- Notional amount/principal:

类似保险的保额

- The total face value of the bonds that can be sold is known as the credit default swap's notional principal. 【total face value of the bonds=CDS 的 notional principal】

The amount of protection being purchased. 【最后的赔付是按照这个 Notional amount 来的】

- CDS spread 保费: the periodic premium that the buyer of a CDS pays to the seller for protection against credit risk.

The protection buyer pays the seller a premium that is either paid upfront or over a period of time (一次性预交或定期支付).

类似上述例子中的: 投资者 A 向金融机构 B 支付的保费

CDS spread 也称 credit spread 信用利差。从概念上讲, 它与债券的信用利差相同, 是承担信用风险的补偿。

CDS spread 反映 reference obligation 的信用风险。

- CDS coupon rate (%):

【在 CDS 的发展历史上, 最初 CDS 是非常定制化的, 因此, CDS spread 各异。随着 CDS 的发展, CDS 慢慢标准化。CDS coupon 是标准化的 CDS spread】

标准化的 CDS spread, 也称 fixed coupon on CDS.

1% for a CDS on an investment-grade company or index;

5% for a CDS on a high-yield company or index.

投资级债券的 CDS coupon 为 1%, 非投资级债券的 CDS coupon 为 5%。【实际市场交易的 CDS 保费在此基础上进行调整】

CDS spread: 市场定价

CDS coupon: 标准化合约下的标准化保费支付

- Upfront premium → The difference between the standardized coupon rate and the credit spread is paid upfront by one of the parties to the contract. 【处理原则: 期初一次性; 多退少补】

若交易的 CDS spread 不是标准化的, 比如 CDS spread=3%。

如果交易的是投资级的 CDS, 少交 2% 的 spread (本身需要支付 3% 的保费, 但是按照标准化合约, 只支付 1%, 那么还少交了 2%)。那么保护的买方在期初一次性把所有差额补给保护的卖方。

如果交易的是投机级的 CDS, 多交 2% 的 spread。那么保护的卖方在期初一次性退回所有差额给保护的买方。

Upfront premium 预付保费 (%) $\approx$ (Credit spread - Fixed coupon) $\times$ Duration

Duration: CDS 的有效久期。题目通常会给定

Fixed coupon: 投资级是 1%; 投机级是 5%

例题: Assume a high-yield company's 10-year credit spread is 600 bps and the duration of the CDS is 8 years. What is the approximate upfront premium required to buy 10-year CDS protection? Assume high-yield companies have 5% coupons on their CDS.

* Solution: upfront premium = (6% - 5%) $\times$ 8 = 8%

- Credit Events 信用事件

这些信用事件发生, 意味着债券发行人违约, 也就是保护的卖方要赔付了。credit events 类型: 【关于类型, 了解即可】

Bankruptcy 债券发行人破产

Failure to pay. 还没有到破产程度, 但是未按计划支付

Restructuring 重组: 指一系列可能的事件, 包括减少或推迟本金或利息、债务优先级或顺位的变更等

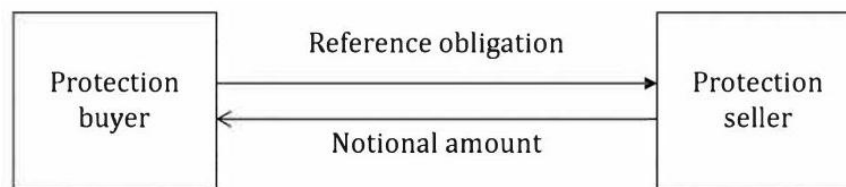
在美国, 重组通常不被视为信用事件, 因为破产通常是美国公司的首选路径。

在美国以外的地区, 重组被视为信用事件。

(2) CDS 的结算

physical settlement 实物结算 【一手交钱 (赔付), 一手交货 (违约的债券)】 involves actual delivery of the debt instrument in exchange for a payment by the credit protection seller of the notional amount of the contract

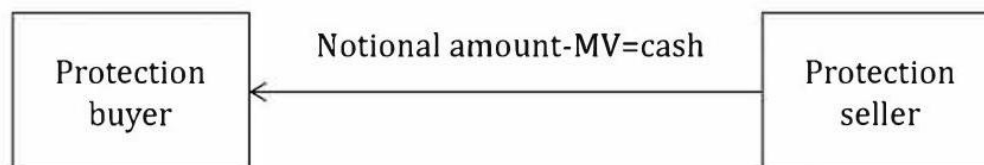
也就是信用事件发生后，CDS buyer 把这个违约的债务工具 (Reference obligation)，递交给 CDS seller。CDS seller 进行赔付 (Notional amount)。



实际市场中，CDS buyer 递交的这个债务工具是有一定选择范围的，只要满足标准即可。这个选择叫做 **cheapest deliverable bond**。(如同一级学习的长期国债期货合约那样，可以选择)

Cash settlement 现金结算

如果标的资产发生违约，信用违约互换的卖方会按照违约差价赔付给买方，违约差价指的是资产的面值和违约发生后的价值之差。



$\text{Payout amount} = \text{Notional amount} - \text{债券回收的市场价值}$

(3) Pricing of CDS 【这个知识学习性价比比较低，会涉及大量计算。这里借由原版书例子给出】

- Pricing means determining the CDS spread or upfront payment given a particular coupon rate for a contract. 【定价定的是 CDS spread 或者 upfront payment。而我们这里主要指的是确定 CDS spread】
- accrual payment 应计支付: 在 CDS 定价中考虑应计支付

因为违约可能在一年中的任何时间发生，而保费支付通常年末。因此，应计支付考虑了在支付期中途违约的情况下，投资者应获得的部分保护。【类似于债券中的累计利息概念】

- 定价逻辑:

✓ CDS 买方支付的保费现值之和 (没有违约和违约情况) = 其收到的赔付现值之和

CDS 买方支付的保费现值之和: $\text{Present Value of Expected Payments (没有违约)} + \text{the Present Value of Accrual Payment (违约情况下)}$

CDS 买方收到的赔付之和: $\text{the Present Value of Expected Payoff}$

在这个原则下，计算 CDS spread。【这种分析方式类似，一级中的保险保费的计算】

- 说明: 如果已知 CDS spread，可以求得违约概率。【见“Estimating Default Probabilities from Credit Spreads”】
- 原版书例子: 一个 5-year CDS
- Hazard rate of the reference entity: 2% per annum
- Risk-free interest rate is 5% per annum with continuous compounding

Payments are made once a year at the end of each year. 【保费在每年的期末支付】

- Notional principal of \$1

Defaults happen halfway through a year 【如果参考实体违约，假设发生中一年的年中】

Recovery rate is 40% 【参考实体发行的债券回收率】

计算 CDS spread。(CDS spread 用 s 表示)

- 分析:

第一步: 计算所需概率【利用违约强度模型】

Year	Probability of surviving to year end	Probability of default during year
1	0.9802	0.0198
2	0.9608	0.0194
3	0.9418	0.0190
4	0.9231	0.0186
5	0.9048	0.0183

$$\text{Cumulative PD} = 1 - e^{-\bar{\lambda}t}$$

$$\text{Cumulative survival rate} = e^{-\bar{\lambda}t}$$

例如: 第一年的生存概率 $= e^{-0.02 \times 1} = 0.9802$

第二年的生存概率 $= e^{-0.02 \times 2} = 0.9608$

第三年的生存概率 $= e^{-0.02 \times 3} = 0.9418$

□ 以此类推

第一年的违约概率 $= 1 - 0.9802 = 0.0198$

第二年的违约概率 $= 0.9802 - 0.9608 = 0.0194$

/ 第二步, Calculation of the Present Value of Expected Payments [计算没有违约的情况下, 支付的保费之和]

只有在前面时间没有违约的情况下, 才会支付下一期的保费

Payment = s per annum.				
Time (years)	Probability of survival	Expected payment	Discount factor	PV of expected payment
1	0.9802	0.9802s	0.9512	0.9324s
2	0.9608	0.9608s	0.9048	0.8694s
3	0.9418	0.9418s	0.8607	0.8106s
4	0.9231	0.9231 s	0.8187	0.7558s
5	0.9048	0.9048s	0.7788	0.7047 s
Total				4.0728s

例如: 第一年的预期支付现值 = 生存概率 × 保费 × 贴现因子 $= 0.9802 \times s \times e^{-0.05 \times 1} = 0.9802 \times s \times 0.9512 = 0.9324 s$

总现值: 所有年份的预期支付现值之和 = 4.0728s。

第三步, Calculation of the Present Value of Accrual Payment 【计算有违约的情况下的支付】

违约发生在每年年中, 应计支付 = 0.5s

Time (years)	Probability of default	Expected accrual payment	Discount factor	PV of expected accrual payment
0.5	0.0198	0.0099s	0.9753	0.0097 s
1.5	0.0194	0.0097s	0.9277	0.0090s
2.5	0.0190	0.0095s	0.8825	0.0084s
3.5	0.0186	0.0093s	0.8395	0.0078s
4.5	0.0183	0.0091 s	0.7985	0.0073s
Total				0.0422s

第一年的应计支付现值 = 违约概率 × 应计支付 × 贴现因子 = $0.0198 \times 0.5 S \times 0.9753 = 0.0097 S$

总现值: 所有年份的应计支付现值之和 = 0.0422 s。

/ 第四步: Calculation of the Present Value of Expected Payoff 【赔付现值之和】

Notional principal = +1 .					
Time (years)	Probability of default	Recovery rate	Expected payoff (\$)	Discount factor	PV of expected payoff (\$)
0.5	0.0198	0.4	0.0119	0.9753	0.0116
1.5	0.0194	0.4	0.0116	0.9277	0.0108
2.5	0.0190	0.4	0.0114	0.8825	0.0101
3.5	0.0186	0.4	0.0112	0.8395	0.0094
4.5	0.0183	0.4	0.0110	0.7985	0.0088
Total					0.0506

计算每年的预期回报现值: 【考虑每个违约情况】

例如: 第一年的预期回报现值 = expected payoff × 贴现因子 = $0.0119 \times 0.9753 = 0.0116$ 【这里在折现的时候, 注意折现时间, 是每年的年中, 折现因子这里已经体现。】

□ 总现值 = 0.0506。

最后: 联立等式

$$4.0728 s + 0.0422 s = 0.0506 \rightarrow s = 0.0123$$

The mid-market CDS spread for the 5-year deal we have considered should be 0.0123 times the principal or 123 basis points per year.

(4) 其他

A. CDS indices(CDS 指数)

- participants in credit markets have developed indices to track credit default swap spreads.
- 市场中比较著名的两个 CDS 指数:

CDX NA IG, a portfolio of 125 investment grade companies in North America

- iTraxx Europe, a portfolio of 125 investment grade names in Europe
- 这些投资组合每年三月和九月进行更新, 如果不再是投资级的公司会被移除, 并加入新的投资级公司。

B. CDS Forward 和 CDS Options

- CDS Forward
- Obligation to buy or sell a particular credit default swap on a particular reference entity at a particular future time T 【远期的标的资产是 CDS】
- CDS Options

Option to buy or sell a particular credit default swap on a particular reference entity at a particular future time T. 【期权的标的资产是 CDS】

C. Types of CDS

- Single-name CDS

A CDS on one specific borrower is called a single-name CDS. 【CDS 只覆盖一个借款人】

这是最常见的 CDS

- Basket Credit Default Swap: there are a number of reference entities

累加型篮子 CDS(Add-up Basket CDS): 任意一个参考实体违约时, 提供赔付。

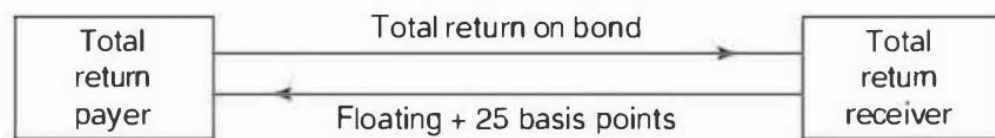
首次违约 CDS(First-to-Default CDS): 仅在第一个违约发生时提供赔付。

第二次违约 CDS(Second-to-Default CDS): 仅在第二个违约发生时提供赔付。

第 k 次违约 CDS(kth-to-Default CDS): 仅在第 k 个违约发生时提供赔付。

- It is an agreement to exchange the total return on a bond (or any portfolio of assets) for a floating rate plus a spread. 【TRS 指的是用债券的总收益换取 floating rate+spread】

The total return includes coupons, interest, and the gain or loss on the asset over the life of the swap.



The payer: 转嫁市场风险和信用风险

The receiver: 获得融资, 融资成本是 floating rate+ spread

- 原版书例题: a total return swap is a 5-year agreement with a notional principal of \$100 million to exchange the total return on a corporate bond for the floating rate plus 25 basis points.

期间:

The payer: 支付 coupons

The receiver: 支付 floating rate+ 25 bps

期末:

如果债券价值增长 10%: The payer pays $100\text{ m} \times 10\% = 10\text{ m}$

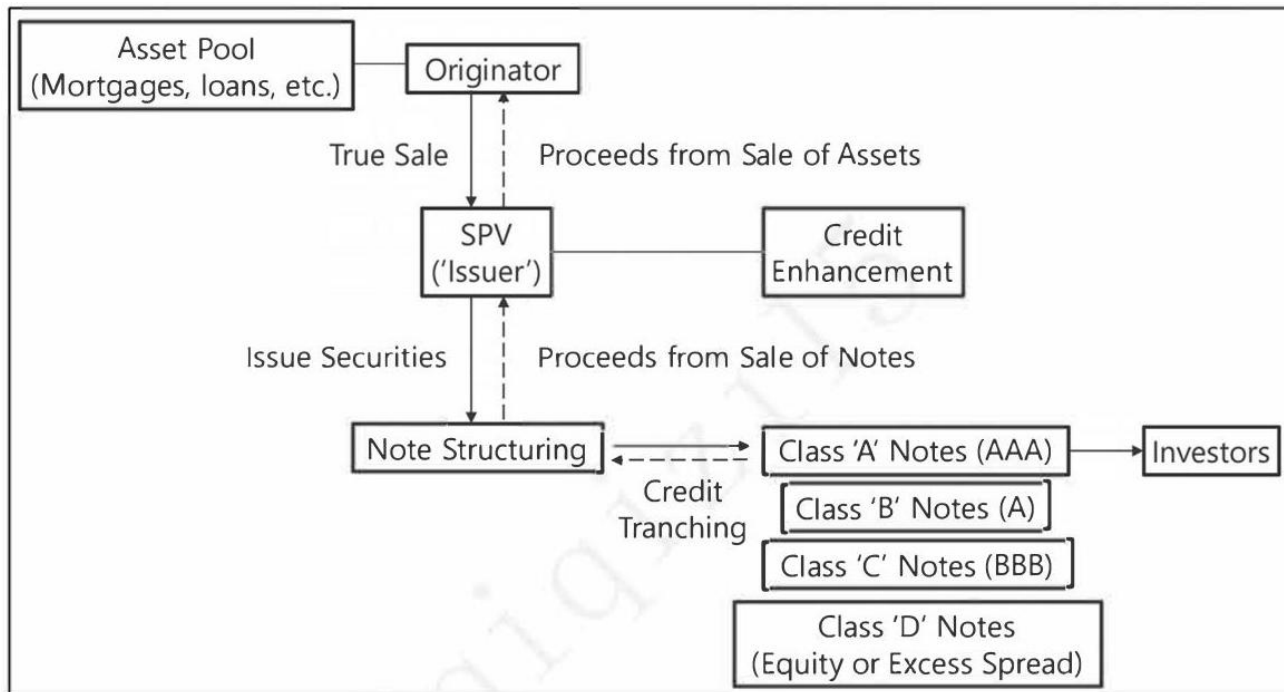
如果债券价值降低 15%: The receiver pays $100\text{ m} \times 15\% = \15 m

债券违约: TRS 停止, receiver 支付债券市场价格和 \$100 million 的差额

考点 3 Structured credit products CR-22

1. 资产证券概述

- securitisation 证券化过程将贷款或应收款等资产 (叫做 collateral or securitized assets) 的所有权从原始所有者转移到一个特殊法律实体 (SPV/SPE)。然后, SPV 发行证券, 利用资产产生的现金流向投资者支付利息和偿还本金。这些证券统称为资产支持证券 (asset-backed securities, ABS) - 资产证券化过程



- 证券化的参与者

发起人 (sponsor, seller)。

the entity whose assets are being securitized.

The loan originator is the original lender who creates the debt obligations in the collateral pool, often a bank.

一般是银行

I Issuer (underwriter, arranger, SPV): is set up specially for the purpose of securitization.

Aggregates the underlying loans, designs the securitization structure and markets the liabilities.

Warehousing risk 仓储风险: risk that deal will not be completed and value of accumulated collateral falls. 【产品无法完成或没卖出去, 且已积累的抵押品价值下跌的风险。】

评级机构 (rating agencies), 为证券化产品评定信用评级。

Rating agencies are compensated by issuers, creating a potential conflict

服务机构和管理机构 (servicers and managers): 服务和管理资产池的机构。

比较复杂的证券化产品, 需要专人进行管理, 比如 CDO

托管人和监管机构 (trustee and custodian), 对账户或者标的资产进行管理的机构。

- 资产证券化的动机 【主要从银行和投资者角度分析】

融资 Funding

Banks can use securitisation to:

- (i) support rapid asset growth 支持资产快速增长;
- (ii) diversify their funding mix and reduce cost of funding 多样化融资来源并降低融资成本;
- (iii) reduce maturity mismatches 减少期限错配

Balance Sheet Capital Management 资产负债资本管理

Banks use securitisation to improve balance sheet capital management. This provides:

regulatory capital relief 监管资本减免: 通过证券化减轻银行的资本要求。

economic capital relief 经济资本减免;

diversified sources of capital 多样化资本来源: 证券化作为传统资本筹集的替代

Risk Management 风险管理

减少信用风险暴露: 资产出售给 SPV, 降低或完全移除银行的信用风险。

剥离不良资产: 移除不良资产和负面情绪, 同时释放监管资本。对投资者的好处

diversify sectors of interest;

access different (and sometimes superior) risk-reward profiles;

access sectors that are otherwise not open to them.

- SPV Structures

按照证券化产品的结构可将其分为不同的类别。主要有以下分类:

摊销结构 Amortizing structures(pass-through structure): pay principal and interest to investors on a coupon-by-coupon basis throughout the life of the security. 在证券的整个存续期间按期支付本金和利息给投资者。

比如常见的 MBS

They are priced and traded based on expected maturity and weighted-average life (WAL) 【基于 WAL 定价与交易】

* weighted-average life (WAL): 即本金未偿还的时间加权时间

$$\bullet \text{ WAL} = \sum \frac{\text{Actual Days}}{365} * \text{Pool factor}$$

Pool factor: 池因子, is the repayment weighting adjustment to the notional value outstanding (O/S) 【会给出】

A WAL approach incorporates various pre-payment assumptions, and any change in this pre-payment speed will increase or decrease the rate at which principal is repaid to investors. 【WAL 方法包含各种提前还款假设】

循环结构 (revolving structures): 使资产本金循环流转。这种结构适用于像信用卡贷款或者汽车贷款。这类贷款的期限通常期限较短且提前还款速度较快, 其现金流划分为两个期限, 分别是循环期和摊销期。

During the revolving period 循环期, principal collections are used to purchase new receivables that fulfil the necessary criteria.

【收到的本金用于购买符合条件的新应收账款】

During the amortisation period 摊销期, principal payments are paid to investors either in a series of equal installments (controlled amortisation) or the principal is "trapped" in a separate account until the expected maturity date and then paid in a single lump sum to investors (soft bullet). 【在摊销期, 本金可以通过等额分期付款或在到期日一次性支付的方式支付给投资者。】

综合信托 (master trust): 指证券化产品是根据投资者需求灵活设计的, 既可以构建摊销结构, 也可以构建循环结构。

Frequent issuers under US and UK law use master trust structures, which allow multiple securitisations to be issued from the same SPV.

Under such schemes, the originator transfers assets to the master trust SPV. Notes are then issued out of the asset pool based on investor demand. 【发起人将资产转移到 master trust SPV, 然后根据投资者的需求从资产池中发行票据。】

Master trusts are used by MBS and credit card ABS originators.

2. asset pools

- Classifications of asset pools
- 根据资产类别:

债权类资产

Auto loans 汽车贷款

Credit card 信用卡

Mortgage 住房贷款

Debt obligation

Loan pool: 比如某个游乐场的门票收益，打包发行证券化产品

- By type of pools
- Static pools 静态池: are amortizing pools in which a fixed loans is placed in the trust. 【资产池的资产是固定的，比如 100 个贷款，假设 4 个违约，那么资产池还剩 96 个贷款】

E.g.: auto loans and residential mortgages.

Revolving pools 循环池: specify an overall level of assets to be maintained during a revolving period. As underlying loans are repaid, the size is maintained by introducing additional loans from the balance sheet of the originator.

E.g.: credit card debt.

Managed pool 管理池 s : manager of the structured product has discretion to remove individual loans from the pool, and replace them with others. 【移除池中的个别贷款，并用其他贷款替换】

E.g.: CDOs.

- 证券化产品主要受资产池的影响。因此在评估证券化产品的风险时，我们要更加关注标的资产池的 performance measure。【这个表格虽然是截图的，但是有几个指标需要重点掌握: 比如 CPR、WAL、DSCR。下面给了几个例题可以做一做感受一下】

Performance Measure	Calculation	Typical Asset Class
Public Securities Association (PSA)	$PSA = [CPR / (.2)(months)] * 100$	mortgages, home-equity, student loans
Constant prepayment rate (CPR)	$1 - (1 - SMM)^{12}$	mortgages, home-equity, student loans
Single monthly mortality (SMM)	Prepayment/Outstanding pool balance	mortgages, home-equity, student loans
Weighted average life (WAL)	$\sum (a/365).PF(s)$ Where PF (s)	mortgages
Weighted average maturity (WAM)	Weighted maturity of the pool	mortgages
Weighted average coupon (WAC)	Weighted coupon of the pool	mortgages
Debt service coverage ratio (DSCR)	Net operating income/Debt payments	commercial mortgages
Monthly payment rate (MPR)	Collections/Outstanding pool balance	all non-amortising asset classes
Default ratio	Defaults/Outstanding pool balance	credit cards
Delinquency ratio	Delinquents/Outstanding pool balance	credit cards
Absolute prepayment speed (ABS)	Prepayments/Outstanding pool balance	auto loans, truck loans
Loss curves	Show expected cumulative loss	auto loans, truck loans

Example 1 : Given an ABS with total outstanding balance of credit card receivables of \$57, 800, 000. \$900, 000 are over 90 days past due. Additionally, \$1□200□000 of receivables were written off. Total monthly principal and interest payments per month are \$1□580□000 . Calculate the delinquency ratio, default ratio, and monthly payment rate for this ABS.

$$\text{Delinquency Ratio} = \$900 \square 000 / \$57 \square 800 \square 000 = 1.577\%$$

$$\text{Default Ratio} = \$1 \square 200 \square 000 / \$57 \square 800 \square 000 = 2.076\%$$

$$\text{Monthly Payment Rate (MPR)} = \$1 \square 580 \square 000 / \$57 \square 800 \square 000 = 2.734\%$$

Example 2: Suppose an MBS has net operating income from commercial mortgaged properties equal to \$89□572□500 . The total debt payments for notes issued against these mortgages is equal to \$87□958□000 . Calculate the debt service coverage ratio (DSCR).

$$DSCR = \$89 \square 572 \square 500 / \$87 \square 958 \square 000 = 1.02$$

Example 3 : Suppose an MBS is composed of three different pools of mortgages: \$6 million of mortgages that yield 7.8%, \$10 million of mortgages that yield 6.0% , and \$4 million of mortgages that yield 5%. Calculate the weighted average coupon (WAC).

$$WAC = 6 / (6 + 10 + 4) \times 7.8\% + 10 / (6 + 10 + 4) \times 6\% + 4 / (6 + 10 + 4) \times 5\% = 6.34\%$$

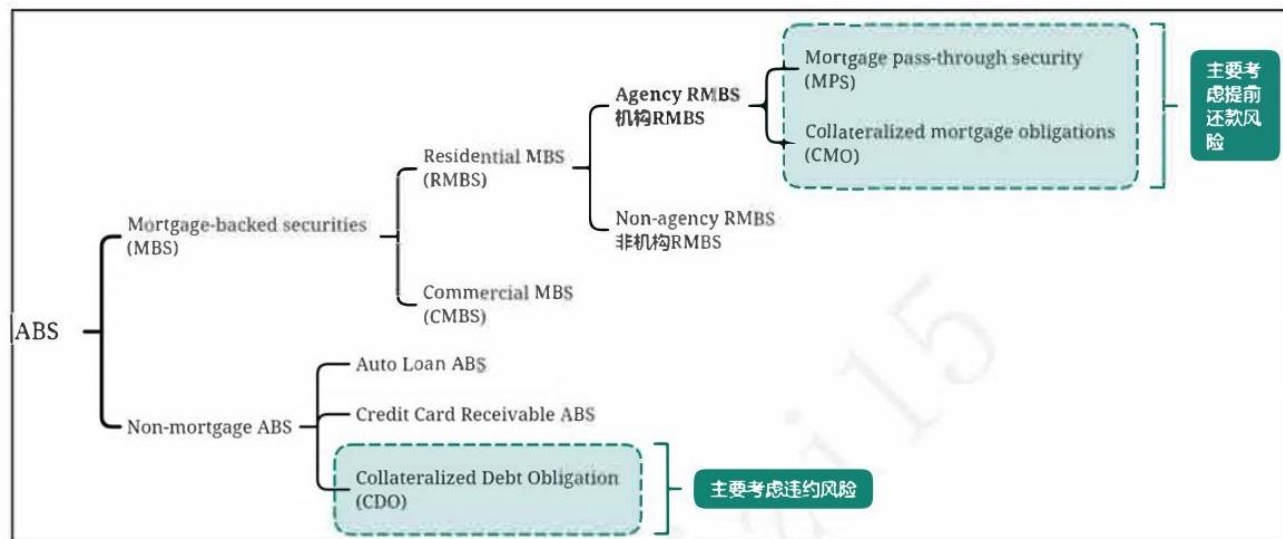
Example 4 : Given an MBS composed of 3 different pools of mortgages: \$2 million of mortgages with a maturity of 180 days, \$6 million of mortgages with a maturity of 360 days, and \$12 million of mortgages with a maturity of 90 days. Calculate the WAM.

$$WAM = 2 / (2 + 6 + 12) \times 180 + 6 / (2 + 6 + 12) \times 360 + 12 / (2 + 6 + 12) \times 90 = 180$$

Example 5: Given an ABS with an SMM of 1.2%. This implies that the approximate prepayment for the month is equal to 1.2% of the remaining mortgage balance for the month less the scheduled principal repayment. Calculate the CPR for this MBS.

$$CPR = 1 - (1 - 1.2\%)^{12} = 0.1349$$

3. Structured products



(1) mortgage pass-through security (MPS)

- The simplest agency MBSs are pass-throughs。
- 所有的投资者拿到 same return. 投资者每个月拿到的现金流来源于借款人每月归还的房贷。其中包括利息、本金以及提前归还 (prepayment) 的本金。

(2) Collateralized mortgage obligations (CMOs) 担保抵押债券

- 对于抵押资产池或者 MPS 的现金流, 进行重新分配, 以形成不同 tranche。不同的 tranche 会有不同的风险和收益, 以满足不同投资者的需求。这样形成的证券, 叫做 CMO。
- CMO 的产生并不能消除而只是转移提前还款风险。【我们这里介绍的是机构 MBS, 信用风险比较小, 提前还款风险比较大。所以, 这里的分层主要针对的是提前还款风险的再分配】

(3) 担保债务凭证 (Collateralized Debt Obligation, CDO)

- 抵押资产: a diversified pool of one or more debt obligations

种类:

名称	抵押资产
collateralized bond obligations (CBOs)	corporate and emerging market bonds
collateralized loan obligations	leveraged bank loans

(CLOs)	
structured finance CDOs	ABS, RMBS, CMBS, and other CDOs
synthetic CDOs	a portfolio of credit default swaps for other structured securities

- 根据信用风险可以分为: senior, mezzanine, and equity tranches.

高级层 (senior tranche): 违约风险最小, 收益最小-最安全

夹层 (mezzanine tranche): 违约风险和收益居中

股权层 (equity tranche): 违约风险最大, 收益最大-一般发行人自留;

如果贷款人违约, 首先损失的 equity tranche, 其次是 mezzanine tranche, 最后是 senior tranche

- Credit enhancement 信用增强

有些证券化产品信用风险比较大, 会采用一些手段让这些产品看起来更好看 (风险小)。这些手段中, 其中常用的是 Credit enhancement 信用增强。

The lower the quality of the assets being securitized, the greater the need for credit enhancement. 具体而言有: 超额抵押 (overcollateralization)

post more collateral than is needed to obtain or secure financing. 实际发行的证券化产品的规模小于资产的规模。

比如贷款资产池规模是 100million, 发行面值 90million 的证券化产品

超额利差 (excess spread)

贷款资产池收到的的现金流大于支付的证券化产品的利息。【后面会更详细的介绍】

资产池保险 (pool insurance) 购买保险, 如果贷款资产池违约, 保险公司给与赔付

分层 (subordination): 划分偿付的优先顺序, 对最高层级产品进行信用增级。

逐渐上升的利息 (margin step-up)

- PD and Default Correlation

【底层资产池的违约, 以及违约相关性, 是影响证券化产品风险的关键因素。我们这里从资产池角度讨论】

Increasing Default Probability (Correlation Constant)

	Mean Value	Credit VaR
Equity tranche	↓	↓
Mezzanine tranche	↓	↑ then ↓
Senior tranche	↓	↑

Increasing Correlations (Default Probability Constant)

	Mean Value	Credit VaR
Equity tranche	↑	↑
Mezzanine tranche	(at low default rates) ↑ (at high default rates)	↑
Senior tranche	↓	↑

Default'01

Default'01 用于衡量当 PD 变动一个基点时, 对于证券化产品价值的影响。通常以违约概率每上升或下降 10 个基点为整体衡量对于证券化产品价值的影响:

$$\text{Default'01} = \frac{1}{20} [(\text{mean value/loss based on } \pi + 0.001) -$$

(mean value/loss based on $\pi - 0.001$)]

- CDO 的种类

现金型 CDO(cash CDO): 最传统的 CDO

合成型 CDO(synthetic CDO): 卖出 CDS 和买入无风险资产共同构成资产证券化产品

单一级 CDO(single-tranche CDO): 所发行的证券化产品只有一个层级 【为了满足投资者的定制化需求】。

通常不会直接收购资产池，而是类似合成 CDO 一样，通过 CDS 构造。4. Cash waterfall 现金流瀑布

- 现金流瀑布分析: all the cash that is generated by the asset pool is paid in order of payment priority.

The senior debt receives all of its promised payments before any lower tranche receives any monies.

我们这里分析的不是摊销结构 (amortizing structure)，也就是前期只有利息，最后一期还本付息。【和普通债券一样的现金流结构】

资产池现金流流入 cash inflow=Aggregate loan interest received+Recovery amount

Aggregate loan interest received 利息部分

Recovery amount 有些资产会违约，违约时的回收部分

最后一期的本金

- 资产池现金流流出 cash outflow=Bond coupon interest due to both the junior and senior bonds.
- Overcollateralization account (OC) 超额抵押账户: 像一个蓄水池，把多余的资金存起来，以备不时之需。

包含三部分资金:

Excess spread: Excess spread= 资产池 Cash Inflows-Cash outflows

每年存入的 Excess spread 会有上限，多余的就给到 equity 层

有些资产池违约时的 Recovery amount

之前年份的 OC earn the financing money market rate of r 。即之前年份的 OC 的利息

- 原版书例题:

Imagine a CLO, the underlying assets of which are 100 identical leveraged loans, with a par value of \$1,000,000 each, and priced at par. The loans are floating rate obligations that pay an annual coupon of fixed spread of 3.5 percent over Libor. Assume there are no upfront, management, or trustee fees. The maturity of underlying assets is five-year and the maturity of CLO is five-year.

- Capital structure displayed in this schematic balance sheet:

Senior tranche: 本金 \$85 million; coupon rate= $L+50$ bps

Mezz Tranche: 本金 \$10 million; coupon rate= $L+500$ bps

Equity Tranche: 本金 \$5 million;

问题 1. If defaults happen, $PD=2\%$, $RR=40\%$. Given $N=100$, K (存入 Excess spread 的上限) = \$1.75 million, Libor = 5%. please define the coupon cash flow of CLO for the five years.

1 t	2 Def	3 Cum	4 Srv	5 Loan int	6 Exc spr	700	8 Recov	90C+ Recov	10 Eq flow	11 OC a/c
1	2	2	98	8, 330, 000	2, 655, 000	1, 750, 000	800, 000	2, 550, 000	905, 000	2, 550, 000
2	2	4	96	8, 160, 000	2, 485, 000	1, 750, 000	800, 000	2, 550, 000	735, 000	5, 227, 500

3	2	6	94	7, 990, 000	2, 315, 000	1, 750, 000	800, 000	2, 550, 000	565, 000	8, 038, 875
4	2	8	92	7, 820, 000	2, 145, 000	1, 750, 000	800, 000	2, 550, 000	395, 000	10, 990, 819
5	2	10	90	97, 650, 000			800, 000	800, 000	-	11, 540, 360

Key to columns:

(1)Year index

(2)Number of defaults during year t

(3) Cumulative number of defaults, at the end of year t

(4) Number of surviving loans, at the end of year t

(5) Loan interest

(6)Excess spread

(7)Overcollateralization increment

(8)Recovery amount R_t

(9) Aggregate flow into overcollateralization account $OC_t + R_t$

(10) Interim cash flow to the equity at the end of year t .

(11) The value of the overcollateralization account at time t .

分析:

第一年: 资产池中的 100 个贷款有 2 个违约, 剩余 98 个未违约

$$\text{Loan int} = 98 \times 1000 \times 8.5\% = 8330000$$

3 支付给最高层级和中间层级支付的利息总额为 5, 675, 000

$$\text{Senior tranche: } 85 \text{ million} \times (5\% + 0.5\%) = 1 \text{ million}$$

$$\text{Mezz Tranche: } 10 \text{ million} \times (5\% + 5\%) = 4.675 \text{ million}$$

$$\text{Total cash outflow} = 1 \text{ million} + 4.675 \text{ million} = 5675000$$

$$\text{Exc spr} = 8330000 - 5675000 = 2655000$$

其中 1, 750, 000 美元放入 OC

剩余的 905, 000 美元给 equity

$$OC + \text{Recov} = 1, 750, 000 + 2 \times 1000000 \times 40\% = 2, 550, 000$$

$$OC_a / c = 2655000$$

第二年: 剩余 96 个未违约

$$\text{Loan int} = 96 \times 1000 \times 8.5\% = 8160000$$

3 支付给最高层级和中间层级支付的利息总额为 5, 675, 000

$$\text{Senior tranche: } 85 \text{ million} \times (5\% + 0.5\%) = 1 \text{ million}$$

$$\text{Mezz Tranche: } 10 \text{ million} \times (5\% + 5\%) = 4.675 \text{ million}$$

$$\text{Total cash outflow} = 1 \text{ million} + 4.675 \text{ million} = 5675000$$

$$\text{Exc spr} = 8160000 - 5675000 = 2485000$$

其中 1, 750, 000 美元放入 OC

剩余的 735, 000 美元给 equity

$$* OC + \text{Recov} = 1, 750, 000 + 2 \times 1000000 \times 40\% = 2, 550, 000$$

$$* OC_a / c = 2, 550, 000 \times (1 + 5\%) + 2, 550, 000 = 5, 227, 500$$

- 依次可得第三年和第四年。

第五年: 利息和本金

4 剩余 90 个未违约

* Loan 本息和 (cash inflow)=90, 000, 000*(1+8.5%)=97, 650, 000

总计: $97 \square 650 \square 000 + 10 \square 990 \square 819 * (1 + 5\%) + 800 \square 000 = 109 \square 990 \square 360$

outflow=85 million*(1+5%+0.5%)+10

million* (1 + 5% + 5%) = 100 □ 675 □ 000

* 109, 990, 360 > 100, 675, 000 , 因此在最后一年 Senior 和 Mezz 的投资者可以得到自己应得的资金

* 而 equity 投资者可得 109, 990, 360-100, 675, 000=9315360

问题 2: If defaults happen, PD=10%, RR=40%. Given N= 100, K(存入 Excess spread 的上限)=\$1.75million, Libor=5%. please define the coupon cash flow of CLO for the five years.

1 t	2 Def	3 Cum	4 Srv	5 Loan int	6 Exc spr	700	8 Recov	9OC+ Recov	10 Eq flow	11OCa/c
1	10	10	90	7, 650, 000	1, 975, 000	1, 750, 000	4, 000, 000	5, 750, 000	225, 000	5, 750, 000
2	9	19	81	6, 885, 000	1, 210, 000	1210, 000	3, 600, 000	4, 810, 000	0	10, 847, 500
3	8	27	73	6, 205, 000	530, 000	530, 000	3, 200, 000	3, 730, 000	0	15, 119, 875
4	7	34	66	5, 610, 000	-65, 000	-65, 000	2, 800, 000	2, 735, 000	0	18, 610, 869
5	7	41	59	64, 015, 000			2, 800, 000	2, 800, 000		19, 541, 412

第一年: 资产池中的 100 个贷款有 10 个违约, 剩余 90 个未违约

Loan int = $90 \square 000 \square 000 \times 8.5\% = 7 \square 650 \square 000$

支付给最高层级和中间层级支付的利息总额为 5, 675, 000

Senior tranche: $85 \text{million} * (5\% + 0.5\%) = 1 \text{million}$

Mezz Tranche: $10 \text{million} * (5\% + 5\%) = 4.675 \text{million}$

Total cash outflow = $1 \text{million} + 4.675 \text{million} = 5 \square 675 \square 000$

Exc spr=7, 650, 000-5, 675, 000=1, 975, 000

其中 1, 750, 000 美元放入 OC

剩余的 225, 000 美元给 equity

OC+ Recov=1, 750, 000+10*1000000*40%=5, 750, 000

• OCa/c = $5 \square 750 \square 00$

第二、三年分析方式一样

第四年, cash outflow 大于 cash inflow, 因此 Exc spr 小于零, 从 OC 中拿出一分部给到 Senior 和 Mezz 的投资者。

第五年: 还利息和本金

4 剩余 59 个未违约

* Loan 本息和 (cash inflow)=59, 000, 000*(1+8.5%)=64, 015, 000

* $18, 610, 869 * (1 + 5\%) + 2, 800, 000 = 19, 541, 412$

* 总计: $64 \square 015 \square 000 + 18 \square 610 \square 869 * (1 + 5\%) + 2 \square 800 \square 000 = 86356412$

* Total cash outflow = $85 \text{million} * (1 + 5\% + 0.5\%) + 10 \text{million} * (1 + 5\% + 5\%) = 89675000 + 11000000$

* $86, 356, 412 < 89675000$, 因此在最后一年 Senior、Mezz 和 equity 都遭受损失